

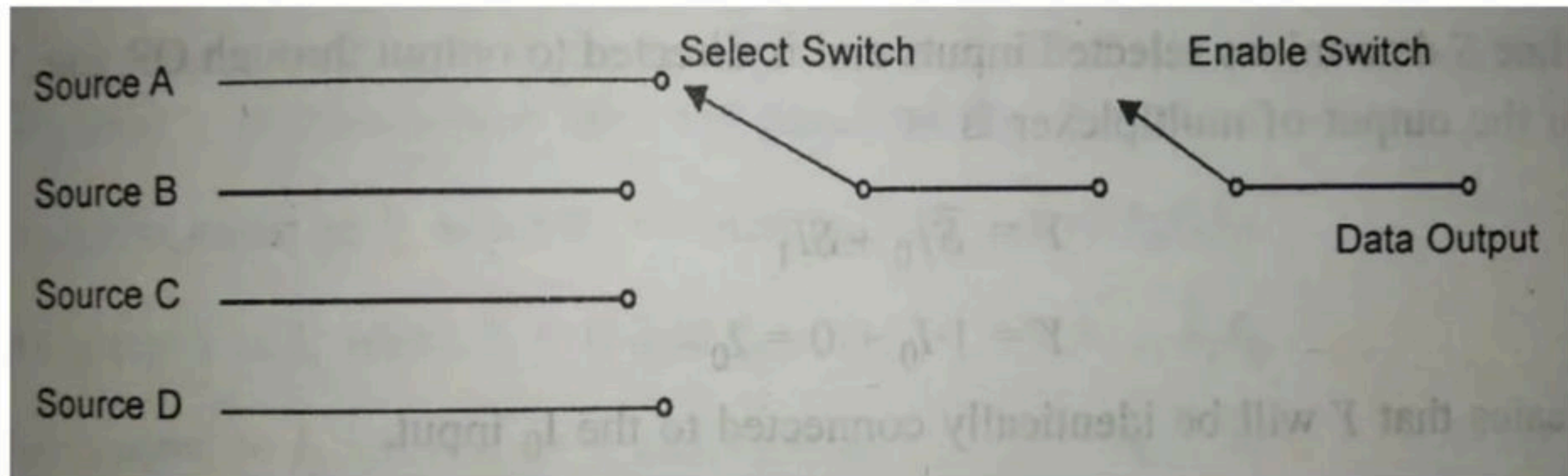


Doubt Clearing Session

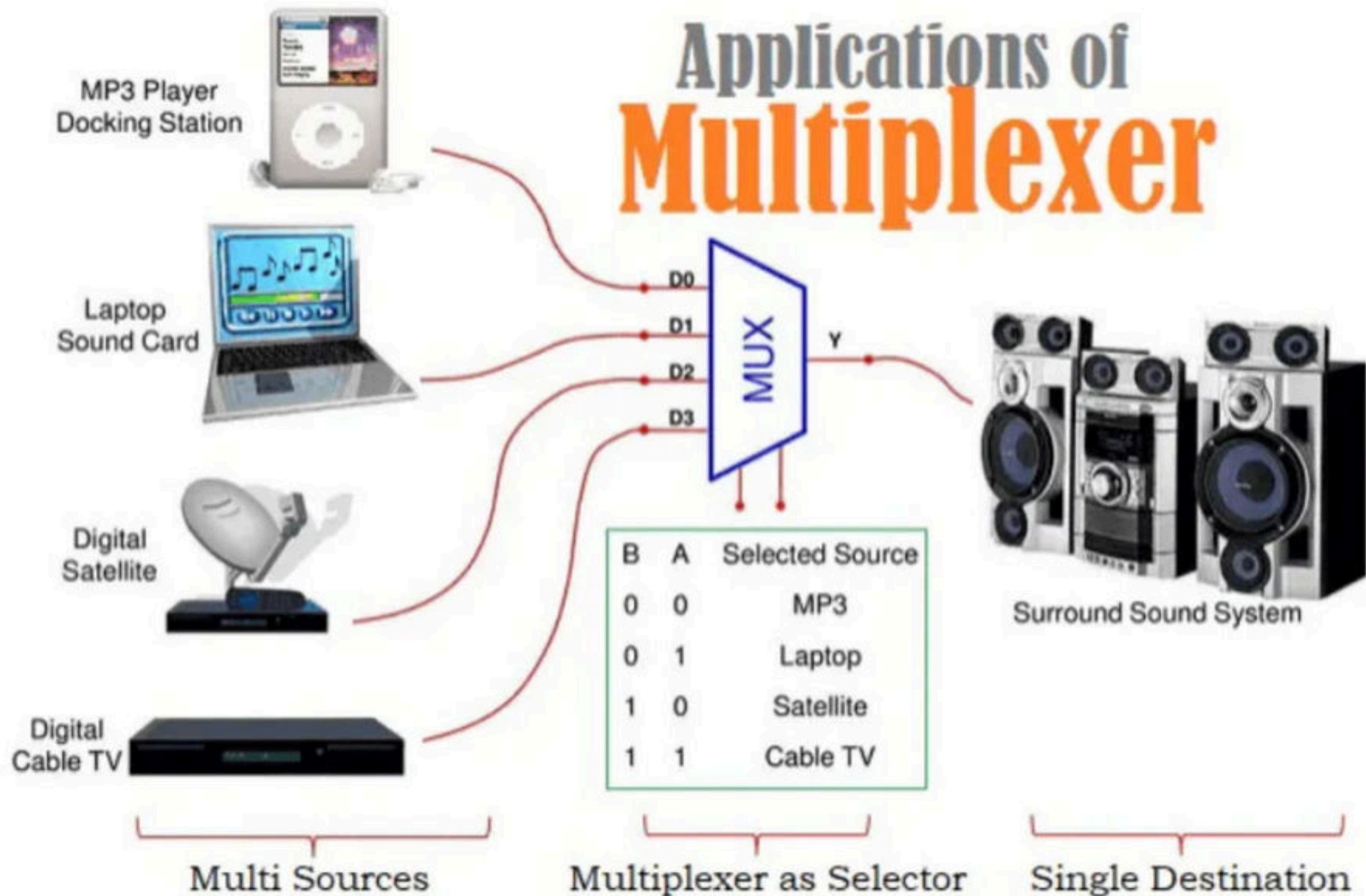
Foundation Course on Digital Electronics

Multiplexer (Selector)

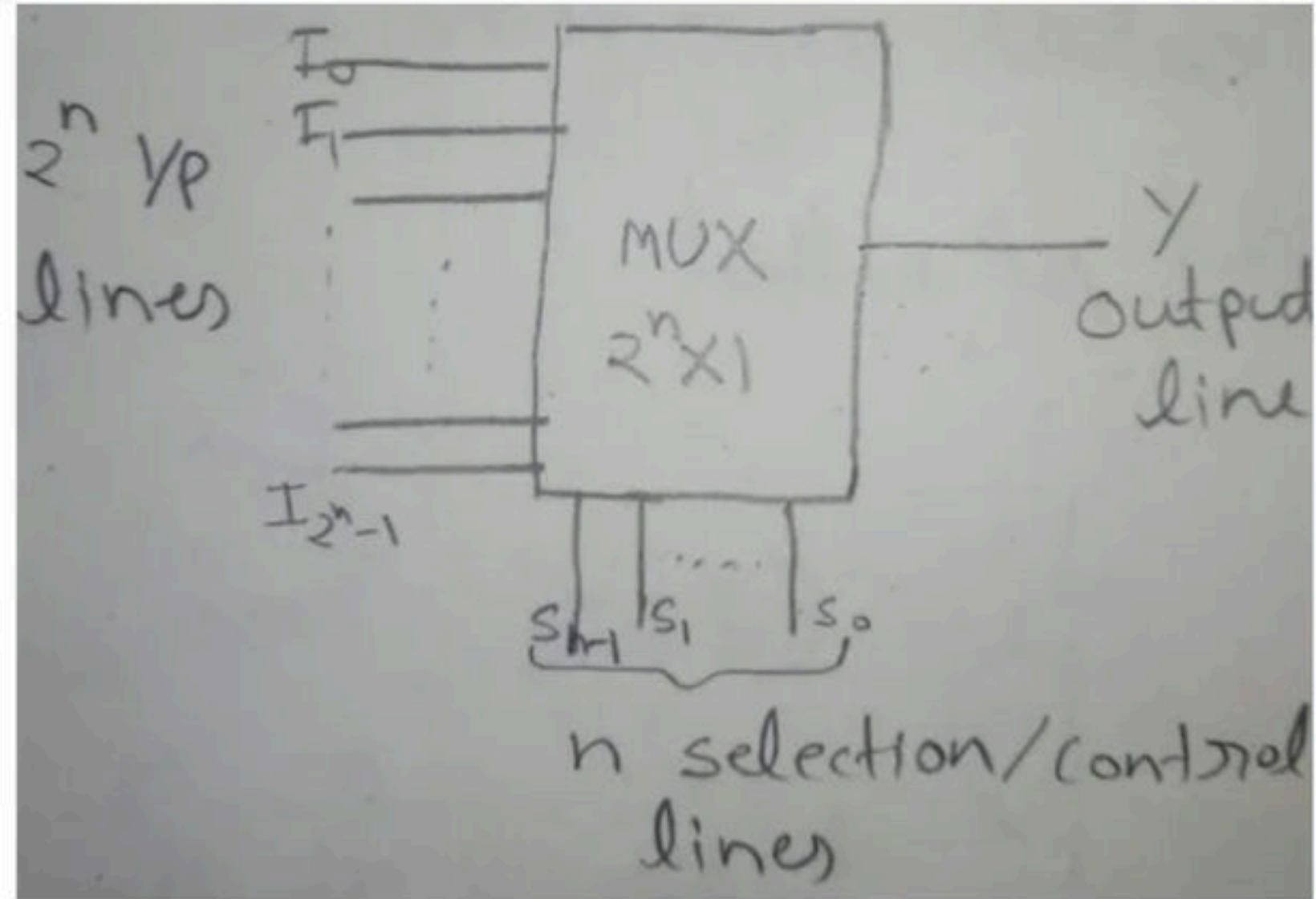
- Multiplexer are special and one of the most widely used combinational circuits.
- Main requirement is out of many inputs we have to select one for e.g. telephone or train leaving the station. So multiplexer do not perform any logical operation or comparison, it just acts as a switch or relay.



Applications of Multiplexer



1. A multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line.
2. A multiplexer is also called a data selector, since it selects one of many inputs and steers the binary information to the output line.
3. The selection of a particular input line is controlled by a set of selection lines.
4. There are 2^n input lines and n selection lines whose bit combinations determine which input is to be selected.
5. No of Input line $= 2^n$



Note:

- Can never have two i/p connected to out at any time

Application

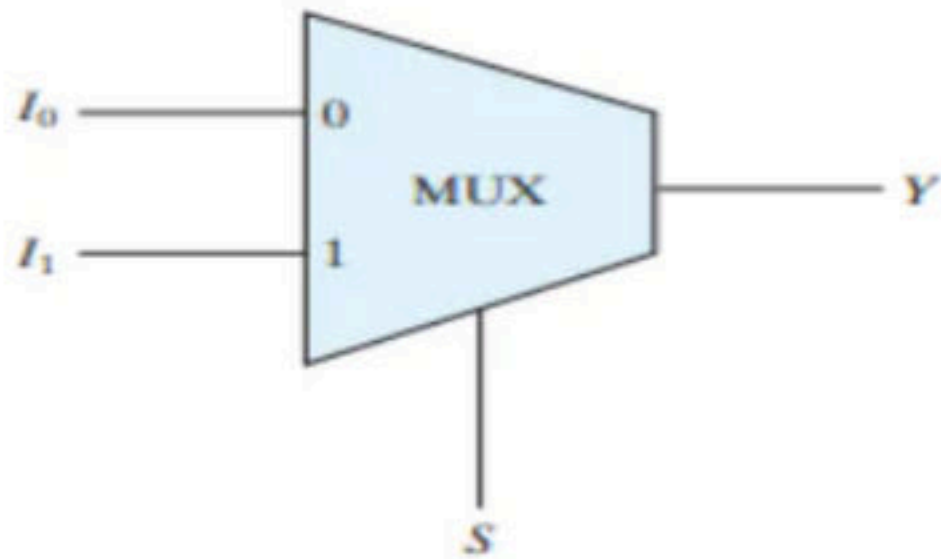
- Parallel data to serial data conversion
- Used as data selector, as the output of a multiplexer is directed from one of various inputs
- Used in implementation of Boolean functions
- Used in communication systems, **Computer Memory, Telephone Network, Transmission from the Computer System of a Satellite**

A two-to-one-line multiplexer

- This multiplexer acts like an electronic switch that selects one of two sources.
- The circuit has two data input lines, one output line, and one selection line S.

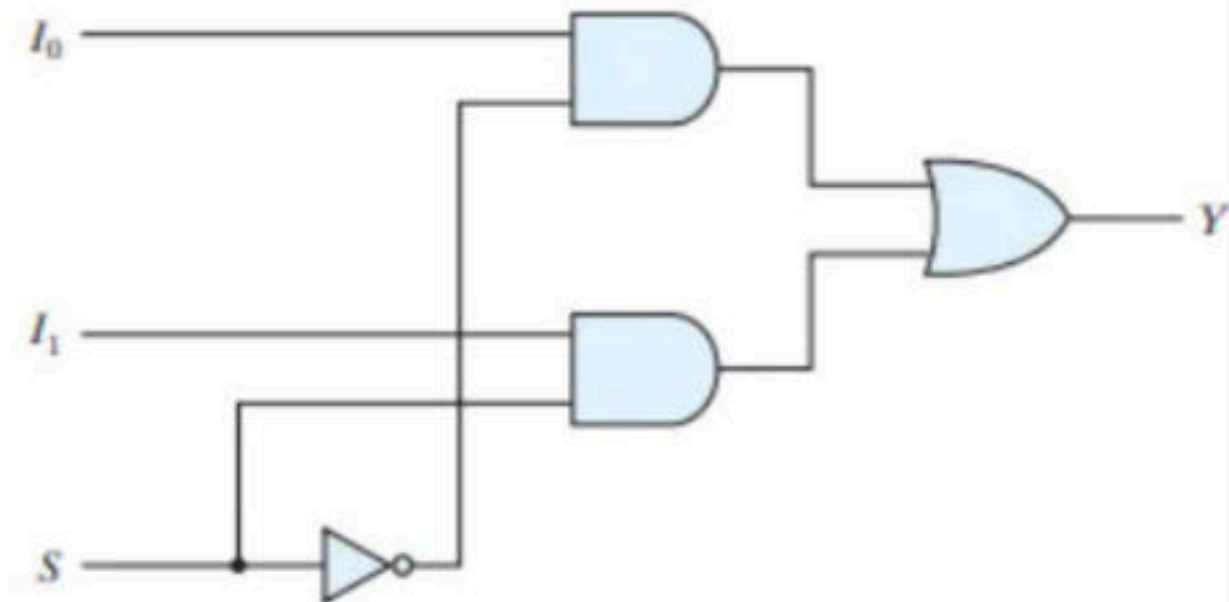
A two-to-one-line multiplexer

- When $S = 0$, the upper AND gate is enabled and I_0 has a path to the output.
- When $S = 1$, the lower AND gate is enabled and I_1 has a path to the output.



(b) Block diagram

Input	Output
S	Y
0	I_0
1	I_1



(a) Logic diagram

Characteristic equation: $Y = S_0' I_0 + S_0 I_1$

Q A multiplexer is a logic circuit that **(NET-JUNE-2011)**

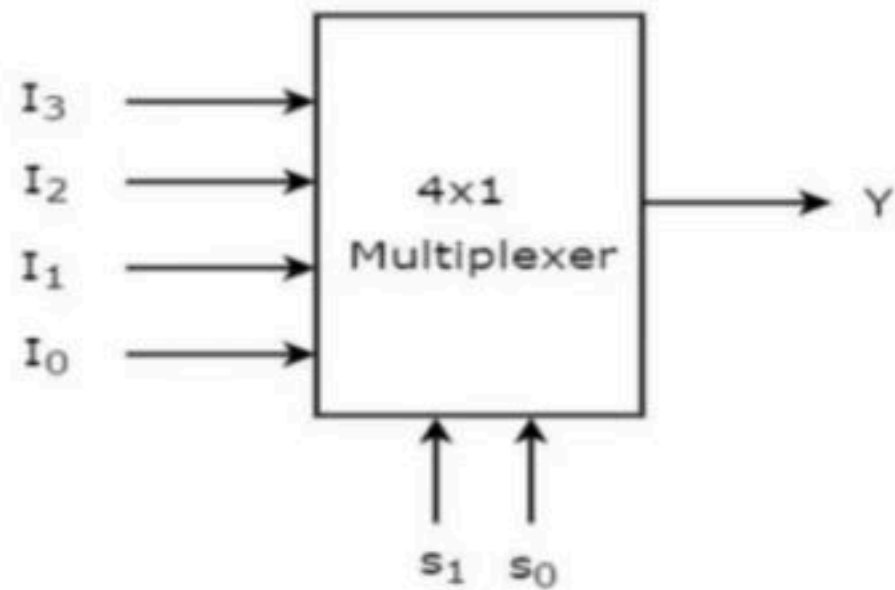
- (A)** accepts one input and gives several outputs
- (B)** accepts many inputs and gives many output
- (C)** accepts many inputs and gives one output
- (D)** accepts one input and gives one output

Case study of 4 to 1

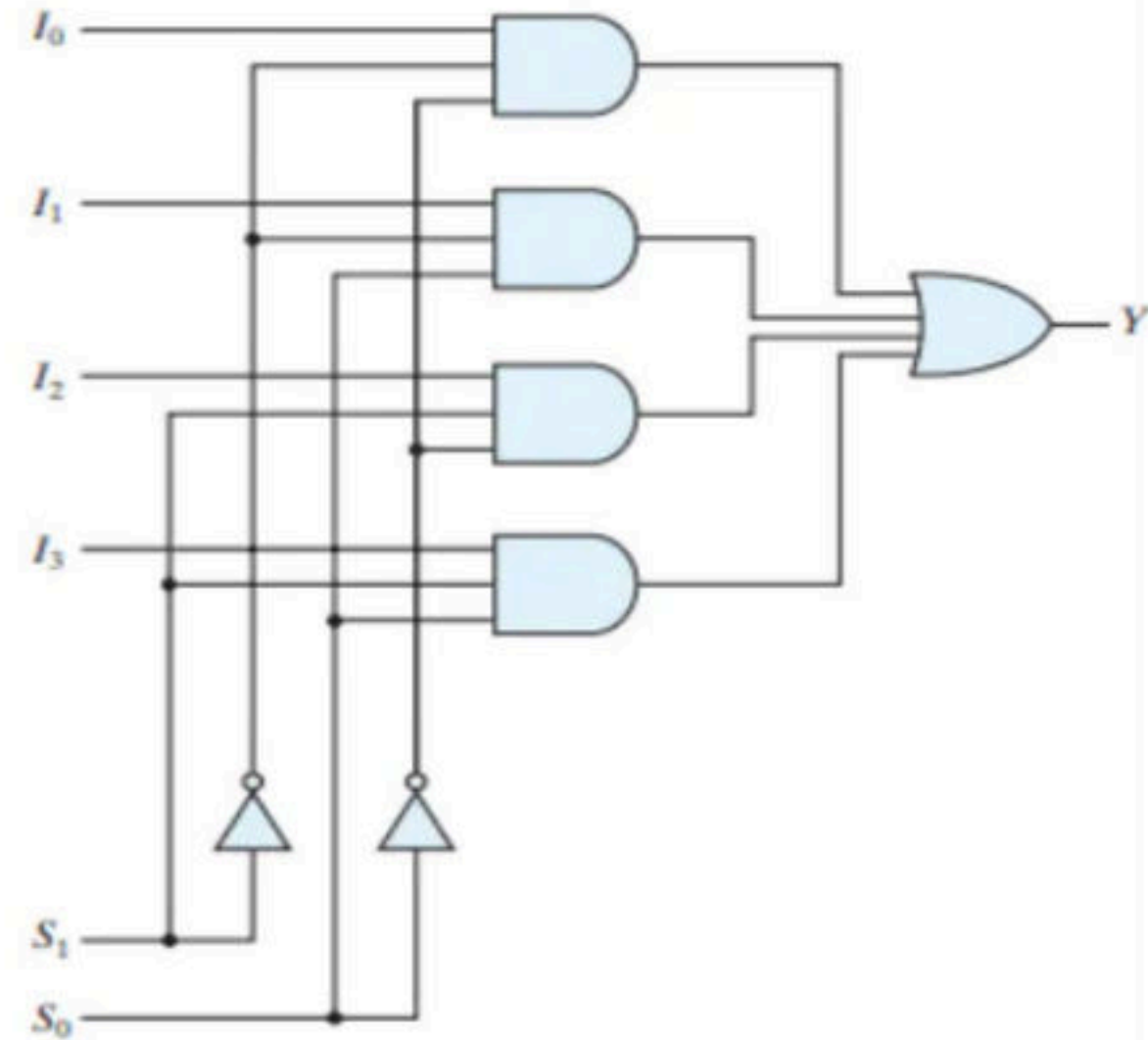
- The circuit has four data input lines I_0, I_1, I_2, I_3 one output line Y , and two selection line S_0 and S_1 .

Case study of 4 to 1

- Each of the four inputs, I_0 through I_3 , is applied to one input of an AND gate.
- Selection lines S_1 and S_0 are decoded to select a particular AND gate.
- The outputs of the AND gates are applied to a single OR gate that provides the one-line output.



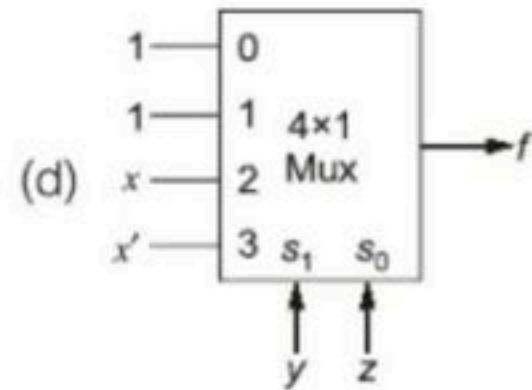
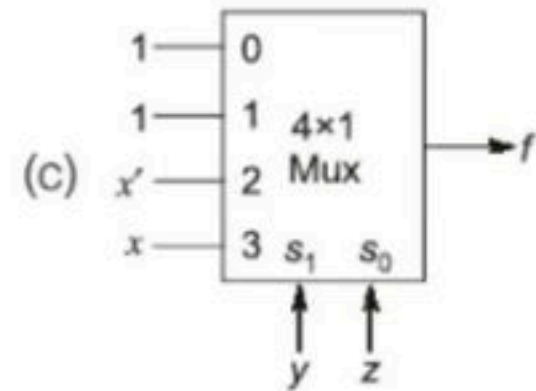
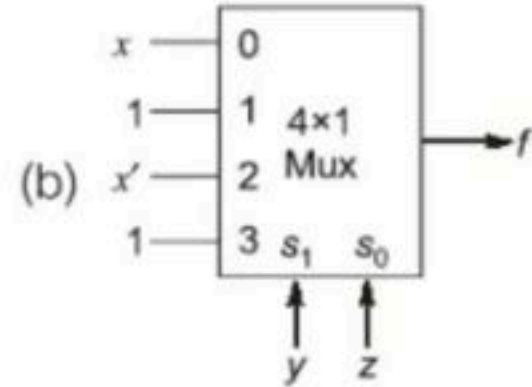
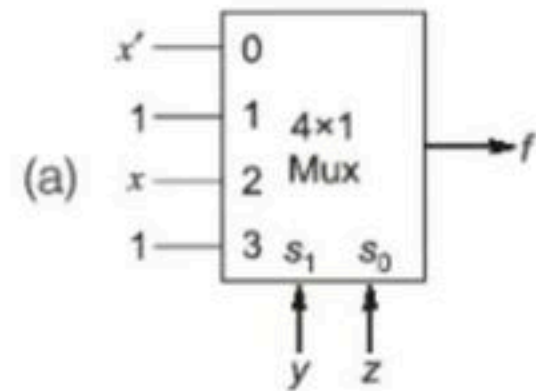
Input		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



Characteristic equation

$$Y = E [(S'_1 S'_0 I_0) + (S'_1 S_0 I_1) + (S_1 S'_0 I_2) + (S_1 S_0 I_3)]$$

Q.46 Which one of the following circuits implements the Boolean function given below?
 $f(x, y, z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm.



Ans. (d)

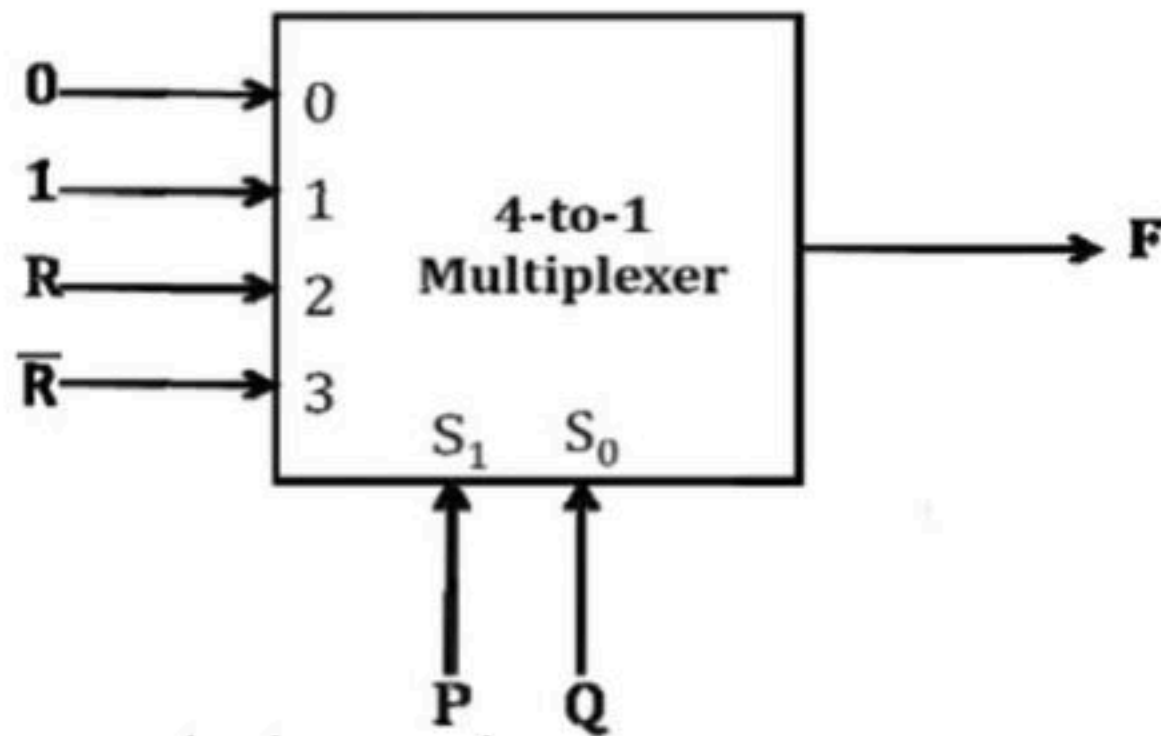
Q Consider a 4-to-1 multiplexer with two select lines S_1 and S_0 , given below
The minimal sum-of-products form of the Boolean expression for the output F of the multiplexer is **(Gate-CS-2014-Set-1) (2 Marks)**

(A) $P'Q + QR' + PQ'R$

(C) $P'QR + P'QR' + QR' + PQ'R$

(B) $P'Q + P'QR' + PQR' + PQ'R$

(D) PQR'



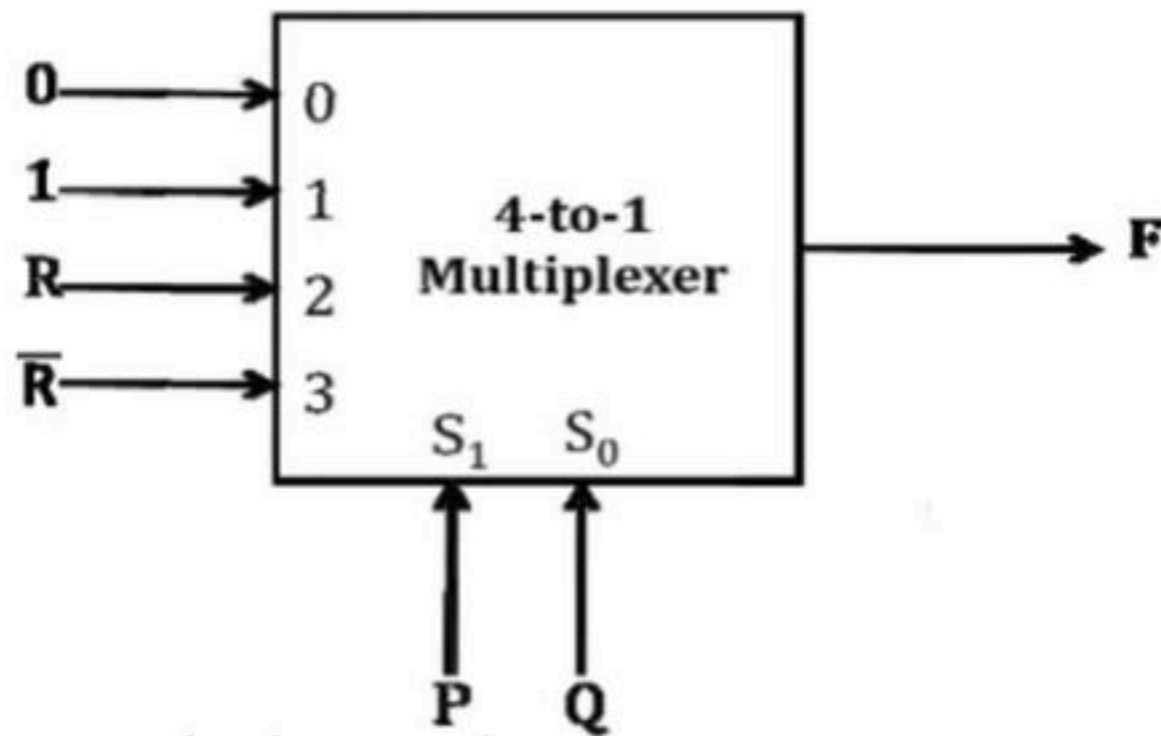
Q Consider a 4-to-1 multiplexer with two select lines S_1 and S_0 , given below
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(A) $P'Q + QR' + PQ'R$

(B) $P'Q + P'QR' + PQR' + PQ'R$

(C) $P'QR + P'QR' + QR' + PQ'R$

(D) PQR'



$$Y = [(S'_1 S'_0 I_0) + (S'_1 S_0 I_1) + (S_1 S'_0 I_2) + (S_1 S_0 I_3)]$$

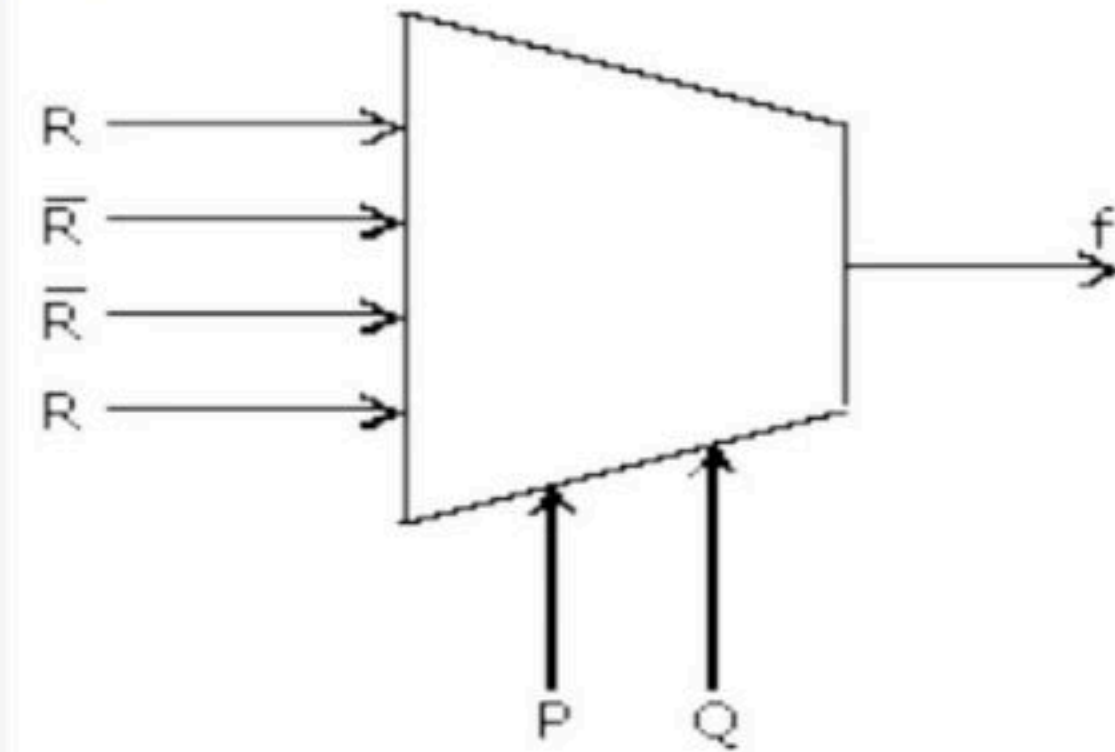
Q The Boolean expression for the output 'f' of the multiplexer shown below is **(GATE-2010) (2 Marks)**

(A) $(P \oplus Q \oplus R)'$

(B) $P \oplus Q \oplus R$

(C) $(P + Q + R)'$

(D) $P + Q + R$



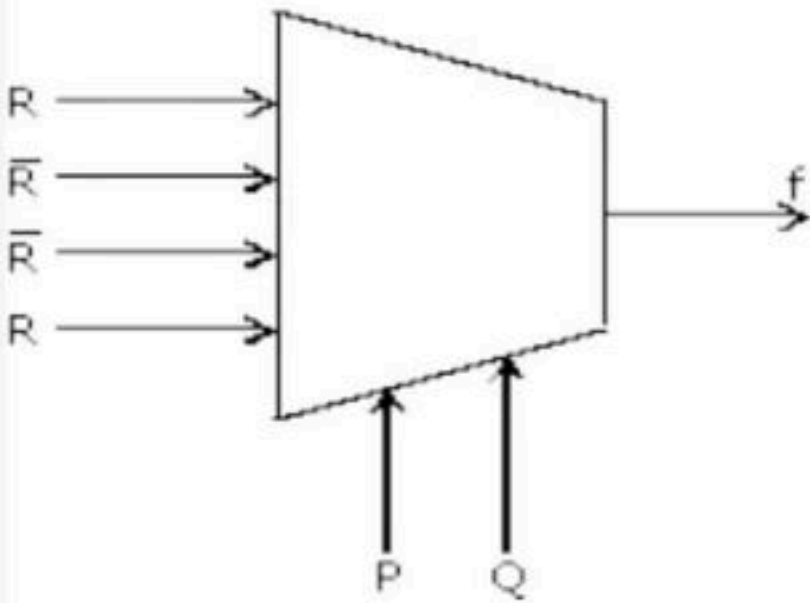
Q The Boolean expression for the output 'f' of the multiplexer shown below is **(GATE-2010) (2 Marks)**

(A) $(P \oplus Q \oplus R)'$

(B) $P \oplus Q \oplus R$

(C) $(P + Q + R)'$

(D) $P + Q + R$

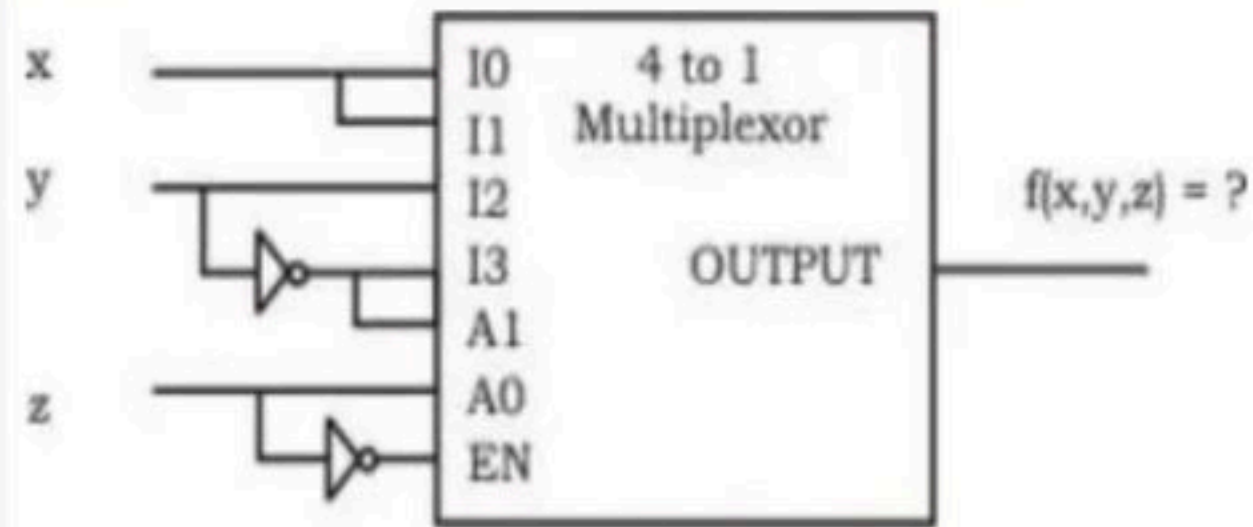


$$Y = [(S'_1 S'_0 I_0) + (S'_1 S_0 I_1) + (S_1 S'_0 I_2) + (S_1 S_0 I_3)]$$

Q Consider the following multiplexor where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$ respectively and f is “the output of the multiplexor. EN is the enable input. **(Gate-CS-2002) (2 Marks)**

The function $f(x, y, z)$ implemented by the above circuit is :

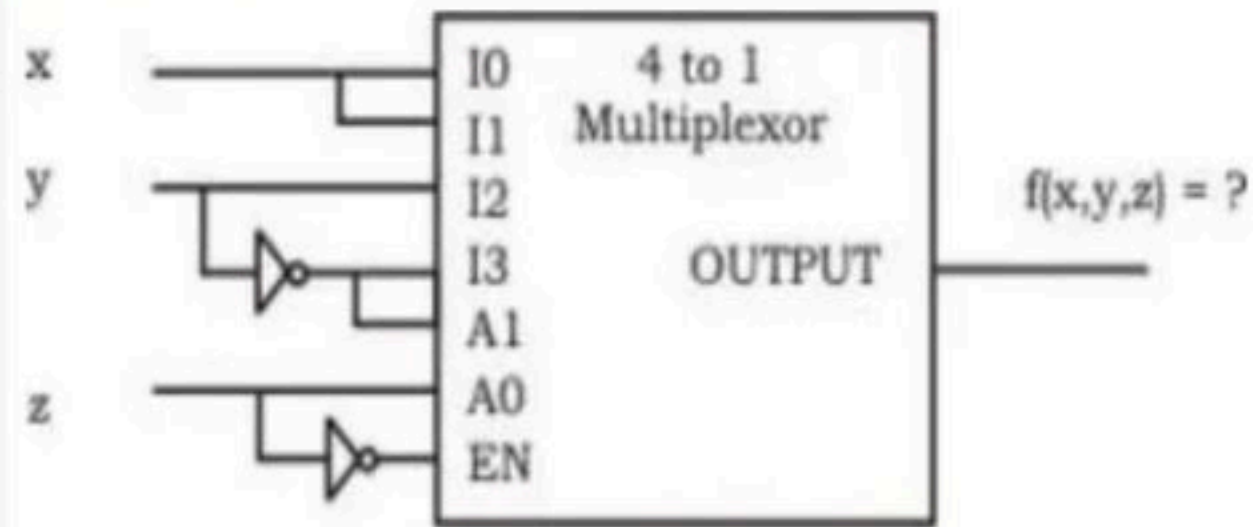
- (A)** xyz' **(B)** $xy + z$ **(C)** $x + z$ **(D)** None of these



Q Consider the following multiplexor where I_0, I_1, I_2, I_3 are four data input lines selected by two address line combinations $A_1A_0 = 00, 01, 10, 11$ respectively and f is “the output of the multiplexor. EN is the enable input. **(Gate-CS-2002) (2 Marks)**

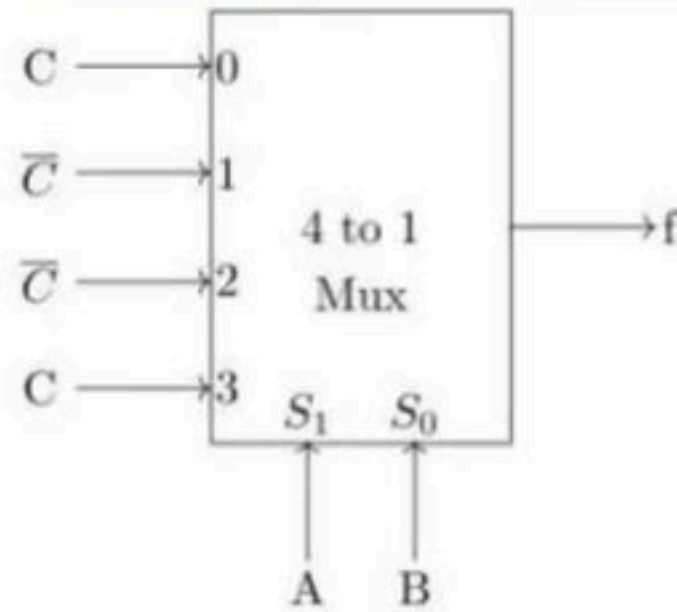
The function $f(x, y, z)$ implemented by the above circuit is :

- (A)** xyz' **(B)** $xy + z$ **(C)** $x + z$ **(D)** None of these



$$Y = E [(S'_1 S'_0 I_0) + (S'_1 S_0 I_1) + (S_1 S'_0 I_2) + (S_1 S_0 I_3)]$$

Q Consider the circuit in below figure. f implements (GATE-1996) (2 Marks)



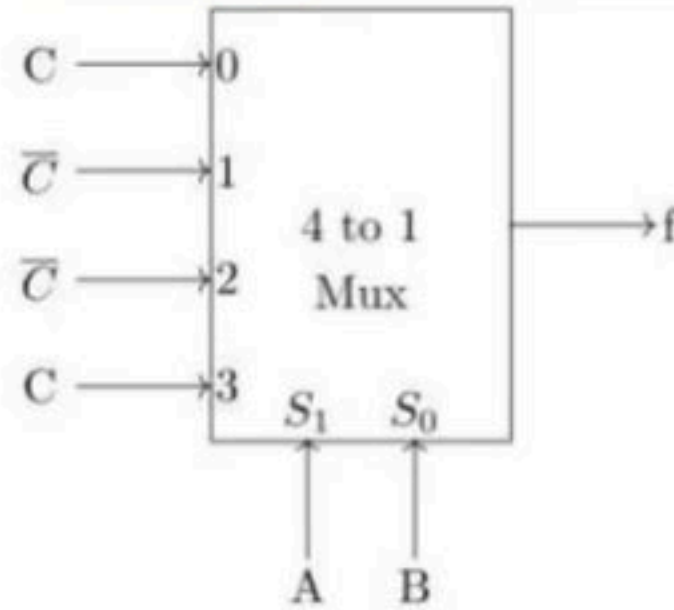
(A) $(ABC)' + A'BC' + ABC$

(B) $A + B + C$

(C) $A \oplus B \oplus C$

(D) $AB + BC + CA$

Q Consider the circuit in below figure. f implements **(GATE-1996) (2 Marks)**



(A) $(ABC)' + A'BC' + ABC$

(B) $A + B + C$

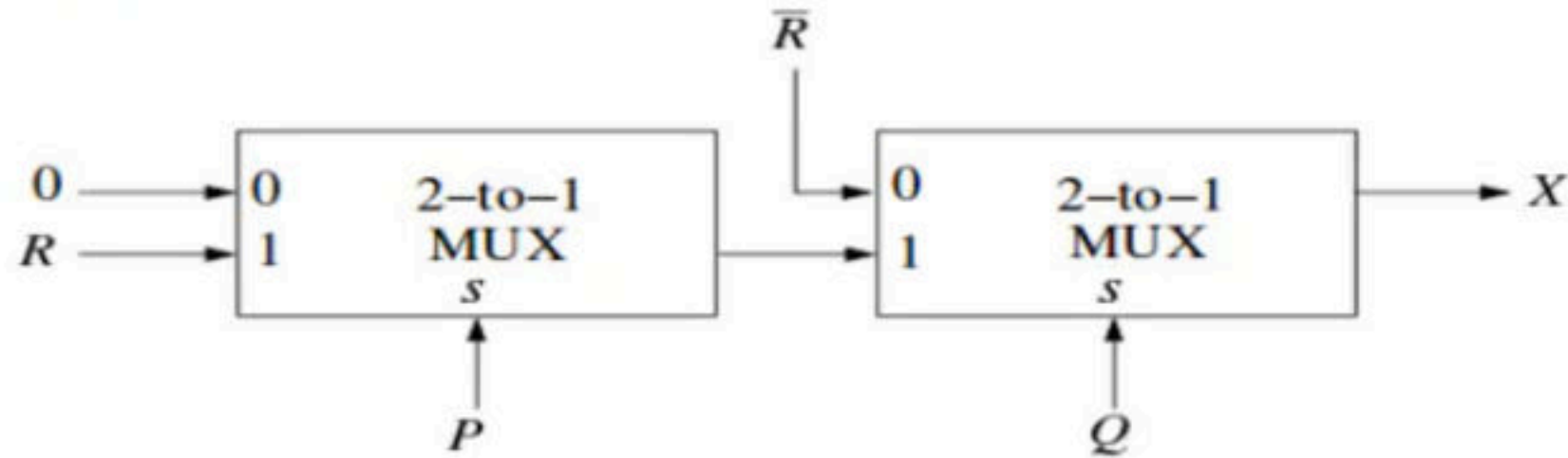
$$Y = [(S'_1 S'_0 I_0) + (S'_1 S_0 I_1) + (S_1 S'_0 I_2) + (S_1 S_0 I_3)]$$

(C) $A \oplus B \oplus C$

(D) $AB + BC + CA$

Many multiplexer

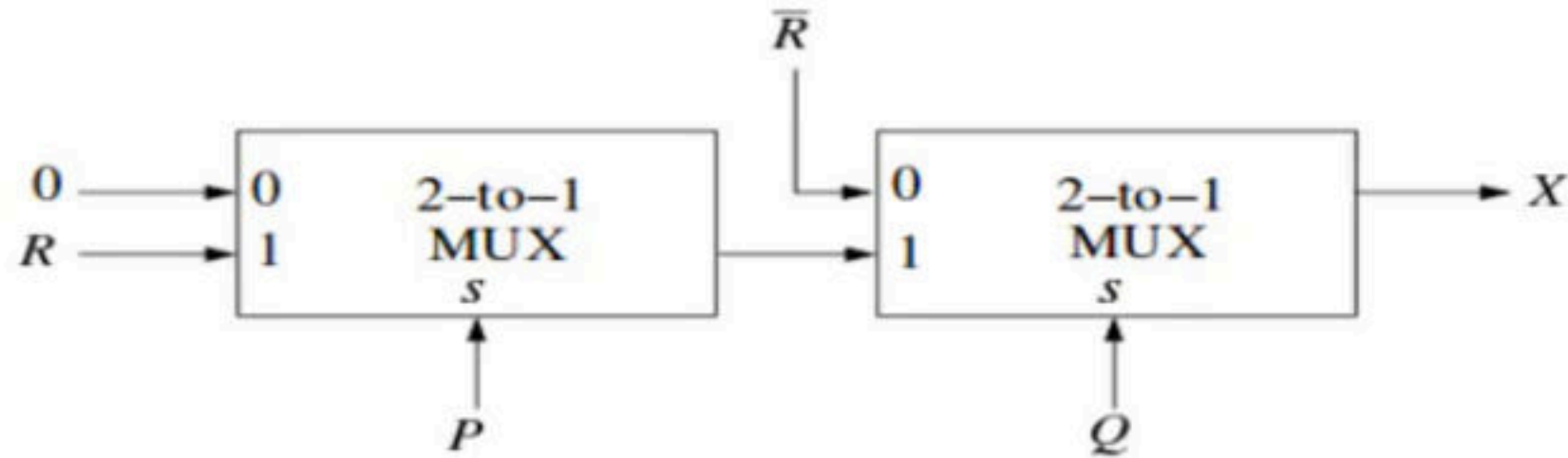
Q Consider the two cascaded 2-to-1 multiplexers as shown in the figure. (GATE-2016) (2 Marks)



The minimal sum of products form of the output X is

- (A) $\bar{P}\bar{Q} + PQR$
- (B) $\bar{P}Q + QR$
- (C) $PQ + \bar{P}\bar{Q}R$
- (D) $\bar{Q}\bar{R} + PQR$

Q Consider the two cascaded 2-to-1 multiplexers as shown in the figure. (GATE-2016) (2 Marks)



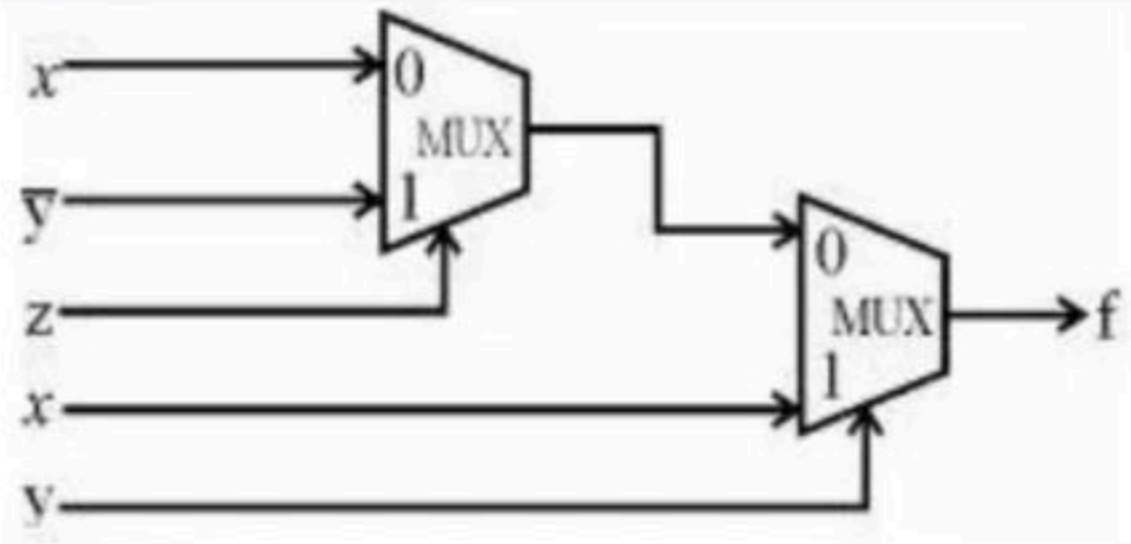
$$Y = S_0' I_0 + S_0 I_1$$

The minimal sum of products form of the output X is

- (A) $\bar{P}\bar{Q} + PQR$
- (B) $\bar{P}Q + QR$
- (C) $PQ + \bar{P}\bar{Q}R$
- (D) $\bar{Q}\bar{R} + PQR$

$$Y = S_0' I_0 + S_0 I_1$$

Q Consider the below circuit and find the output function $f(x, y, z)$. (NET-JUNE-2012)



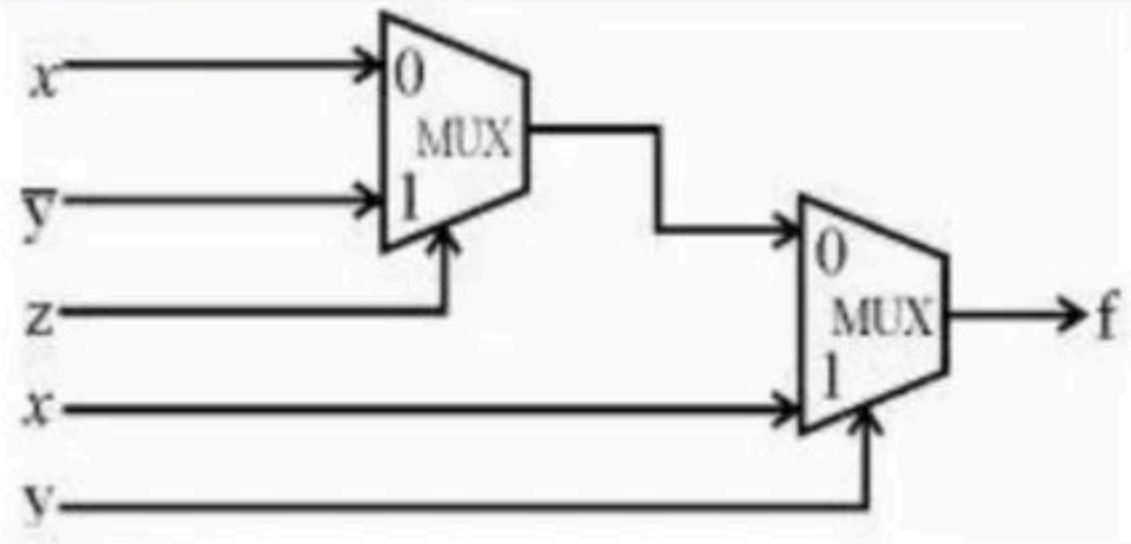
(A) $xz' + xy + y'z$

(B) $xz' + xy + y'z'$

(C) $xz + xy + y'z'$

(D) $xz + xy' + y'z$

Q Consider the below circuit and find the output function $f(x, y, z)$. (NET-JUNE-2012)



$$Y = S_0' I_0 + S_0 I_1$$

(A) $xz' + xy + y'z$

(B) $xz' + xy + y'z'$

(C) $xz + xy + y'z'$

(D) $xz + xy' + y'z$

$$Y = S_0' I_0 + S_0 I_1$$

Q The following circuit implements a two-input AND gate using two 2-1 multiplexers. **(GATE-2007) (1 Marks)**

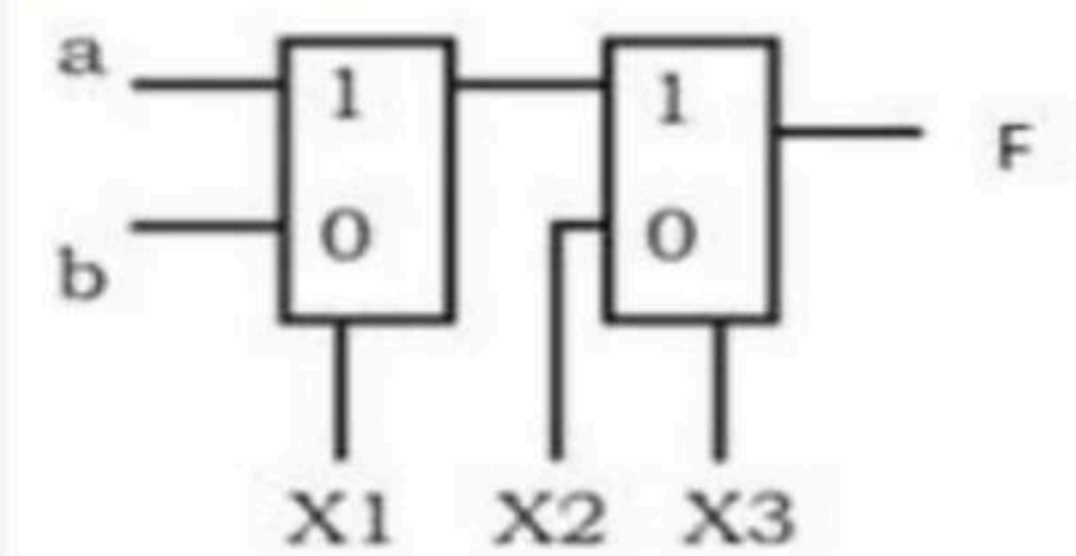
What are the values of X_1 , X_2 , X_3 ?

(A) $X_1=b$, $X_2=0$, $X_3=a$

(C) $X_1=a$, $X_2=b$, $X_3=1$

(B) $X_1=b$, $X_2=1$, $X_3=b$

(D) $X_1=a$, $X_2=0$, $X_3=b$



Q The following circuit implements a two-input AND gate using two 2-1 multiplexers. **(GATE-2007) (1 Marks)**

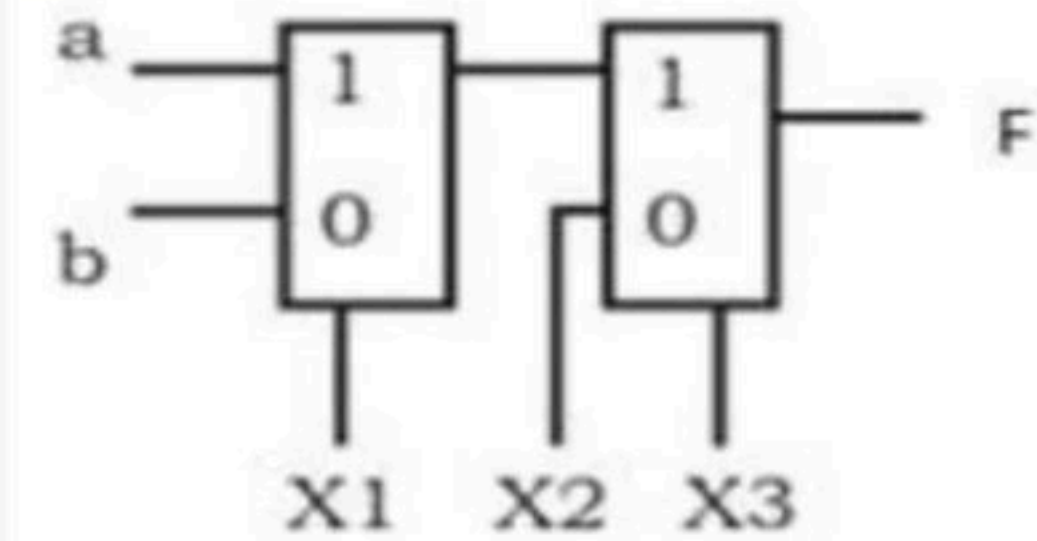
What are the values of X_1, X_2, X_3 ?

(A) $X_1=b, X_2=0, X_3=a$

(C) $X_1=a, X_2=b, X_3=1$

(B) $X_1=b, X_2=1, X_3=b$

(D) $X_1=a, X_2=0, X_3=b$

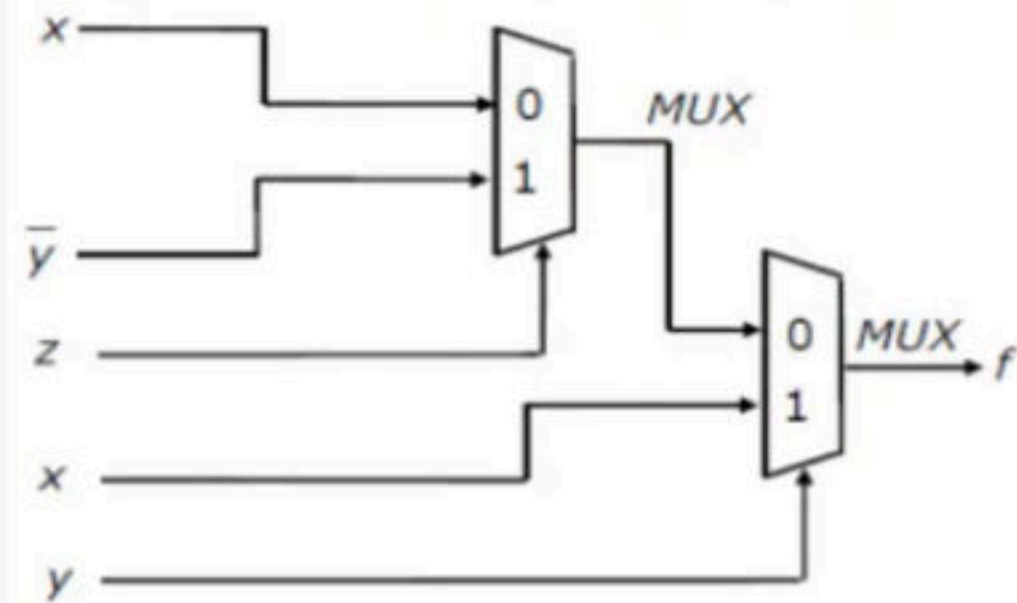


$$Y = S_0' I_0 + S_0 I_1$$

$$Y = S_0' I_0 + S_0 I_1$$

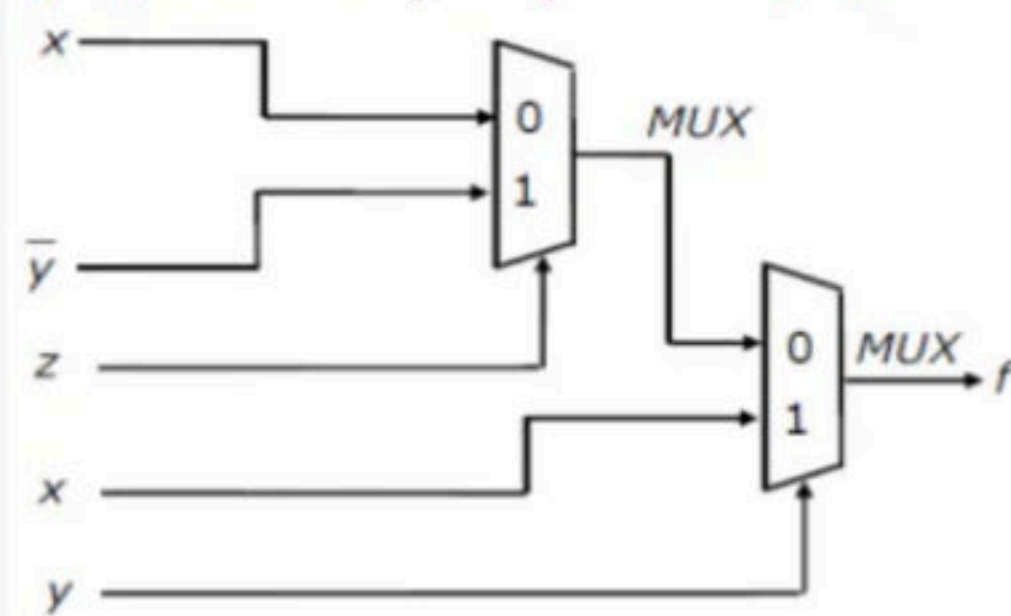
Q Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$? (**Gate-CS-2006**) (2 Marks)

(A) $xz' + xy + y'z$ **(B)** $xz' + xy + (yz)'$ **(C)** $xz + xy + (yz)'$ **(D)** $xz + xy' + y'z$



Q Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$? (**Gate-CS-2006**) (2 Marks)

(A) $xz' + xy + y'z$ (B) $xz' + xy + (yz)'$ (C) $xz + xy + (yz)'$ (D) $xz + xy' + y'z$

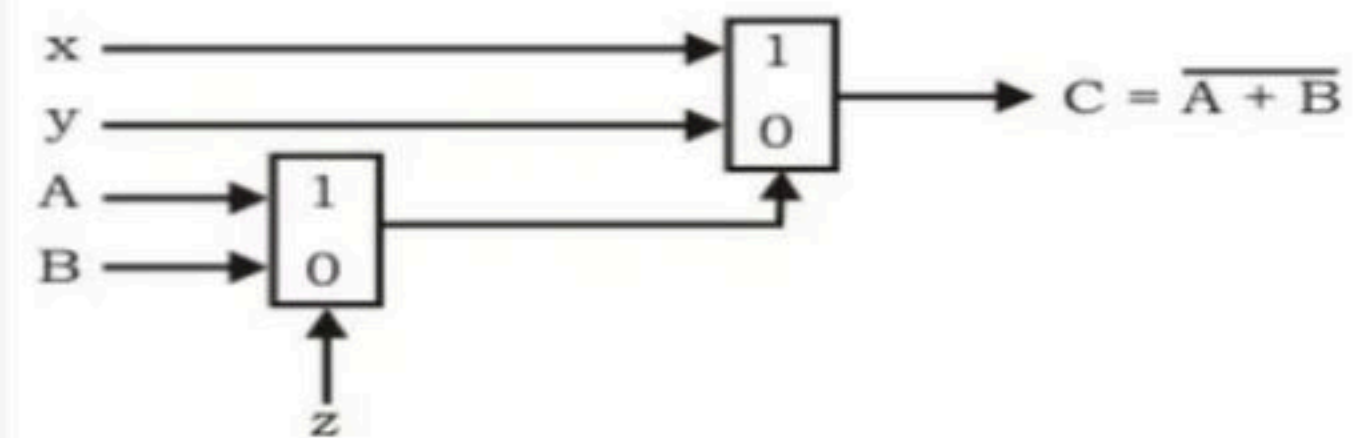


$$Y = S_0' I_0 + S_0 I_1$$

$$Y = S_0' I_0 + S_0 I_1$$

Q The circuit shown below implements a 2-input NOR gate using two 2-4 MUX (control signal 1 selects the upper input). What are the values of signals x, y and z? **(GATE-2005) (2 Marks)**

(A) 1, 0, B **(B)** 1, 0, A **(C)** 0, 1, B **(D)** 0, 1, A



(D) 0, 1, A



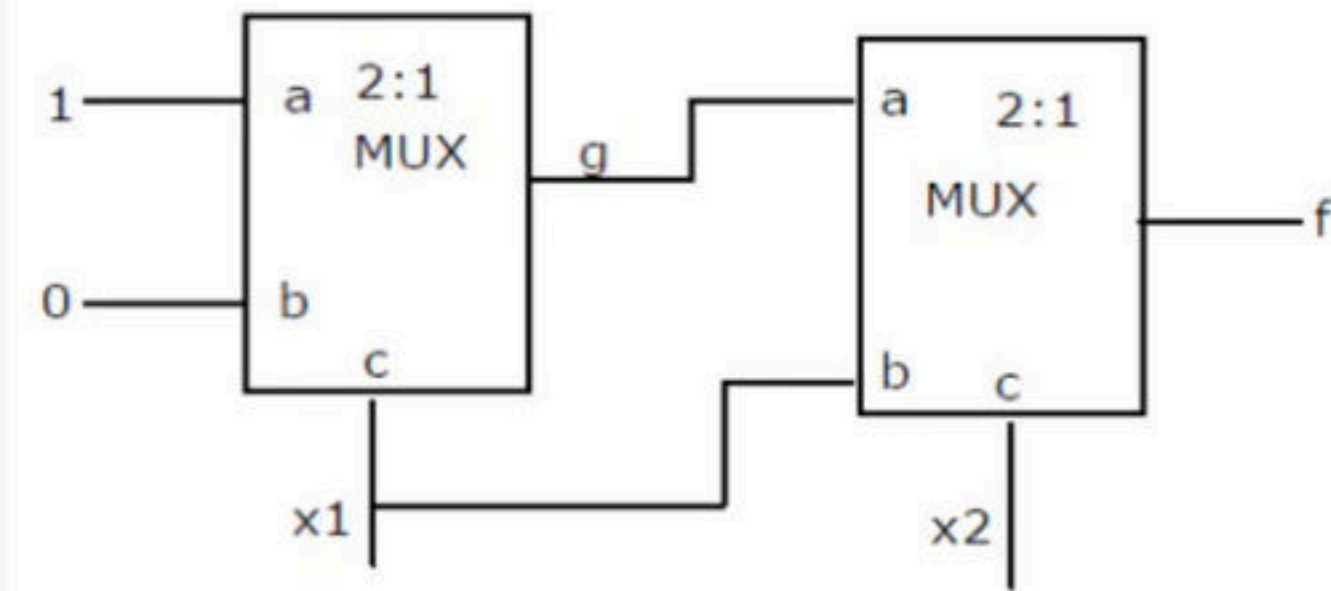
Q Consider the circuit shown below. The output of a 2:1 Mux is given by the function $(ac' + bc)$, Which of the following is true? **(GATE-2001) (2 Marks)**

(A) $f = x_1' + x_2$

(B) $f = x_1'x_2 + x_1x_2'$

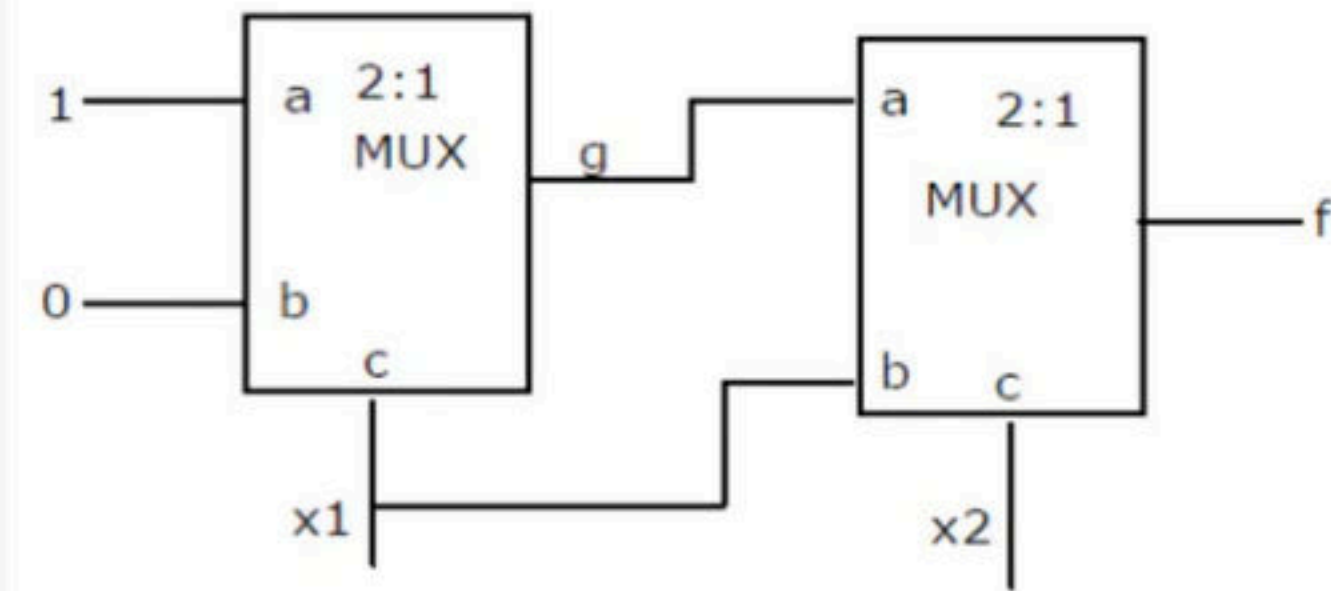
(C) $f = x_1x_2 + x_1'x_2'$

(D) $f = x_1 + x_2'$



Q Consider the circuit shown below. The output of a 2:1 Mux is given by the function $(ac' + bc)$, Which of the following is true? **(GATE-2001) (2 Marks)**

- (A)** $f = x_1' + x_2$ **(B)** $f = x_1'x_2 + x_1x_2'$ **(C)** $f = x_1x_2 + x_1'x_2'$ **(D)** $f = x_1 + x_2'$



$$Y = S_0' I_0 + S_0 I_1$$

$$Y = S_0' I_0 + S_0 I_1$$

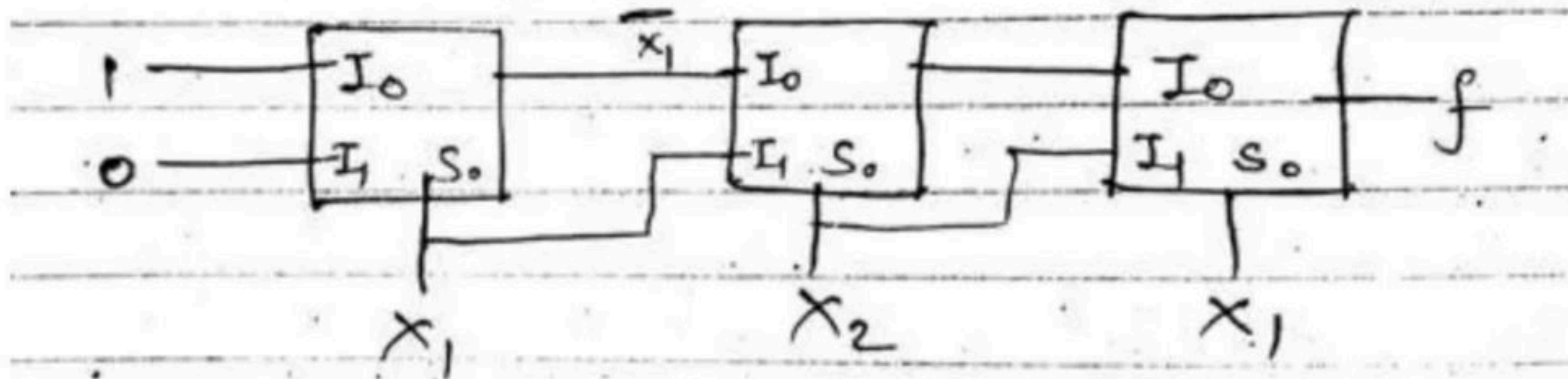
Q Consider the circuit shown below. Which of the following is true?

(A) $f = x_1' + x_2$

(B) $f = x_1'x_2 + x_1x_2'$

(C) $f = x_1x_2 + x_1'x_2'$

(D) none



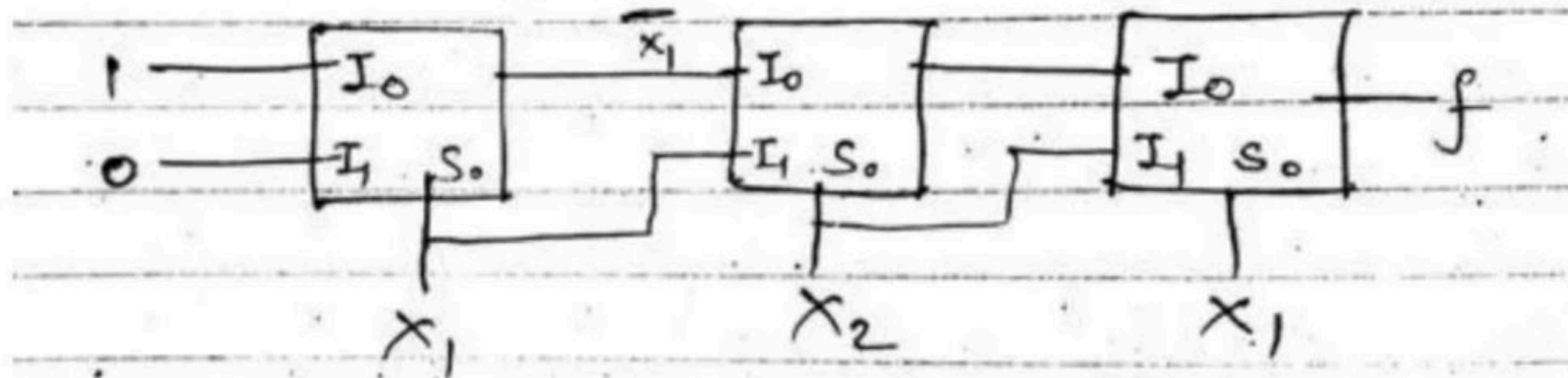
Q Consider the circuit shown below. Which of the following is true?

(A) $f = x_1' + x_2$

(B) $f = x_1'x_2 + x_1x_2'$

(C) $f = x_1x_2 + x_1'x_2'$

(D) none



$$Y = S_0' I_0 + S_0 I_1$$

$$Y = S_0' I_0 + S_0 I_1$$

$$Y = S_0' I_0 + S_0 I_1$$

Q Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed? **(Gate-CS-2007) (2 Marks)**

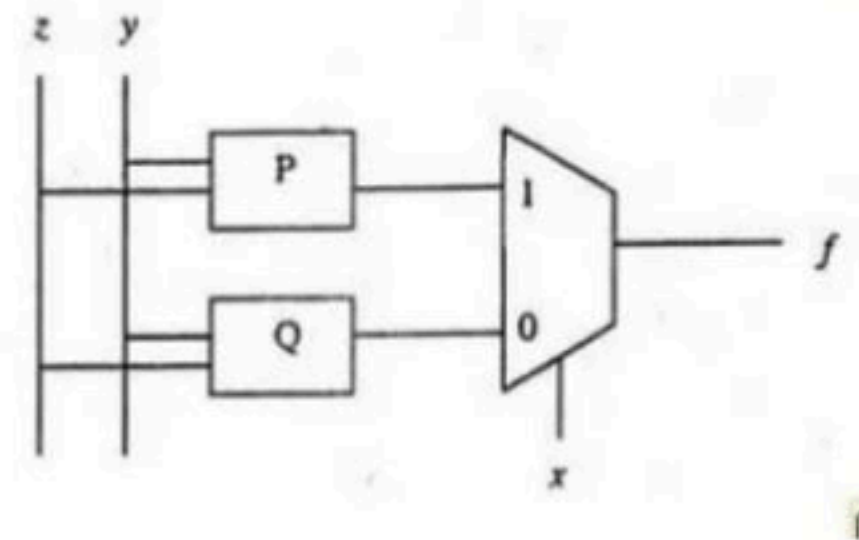
- (A)** 2^n line to 1-line
(C) 2^{n-1} line to 1-line

- (B)** 2^{n+1} line to 1 line
(D) 2^{n-2} line to 1 line

In order to implement a n variable switching function, a MUX must have :

- (A) 2^n inputs (B) $2^n + 1$ inputs (C) 2^{n-1} inputs (D) $2^n - 1$ inputs

Q The majority function is a Boolean function $f(x, y, z)$ that takes the value 1 whenever a majority of the variables x, y, z is 1. In the circuit diagram for the majority function shown below, the logic gates for the boxes labelled P and Q are, respectively? **(GATE-2006) (2 Marks)**



(A) XOR, AND

(B) XOR, XOR

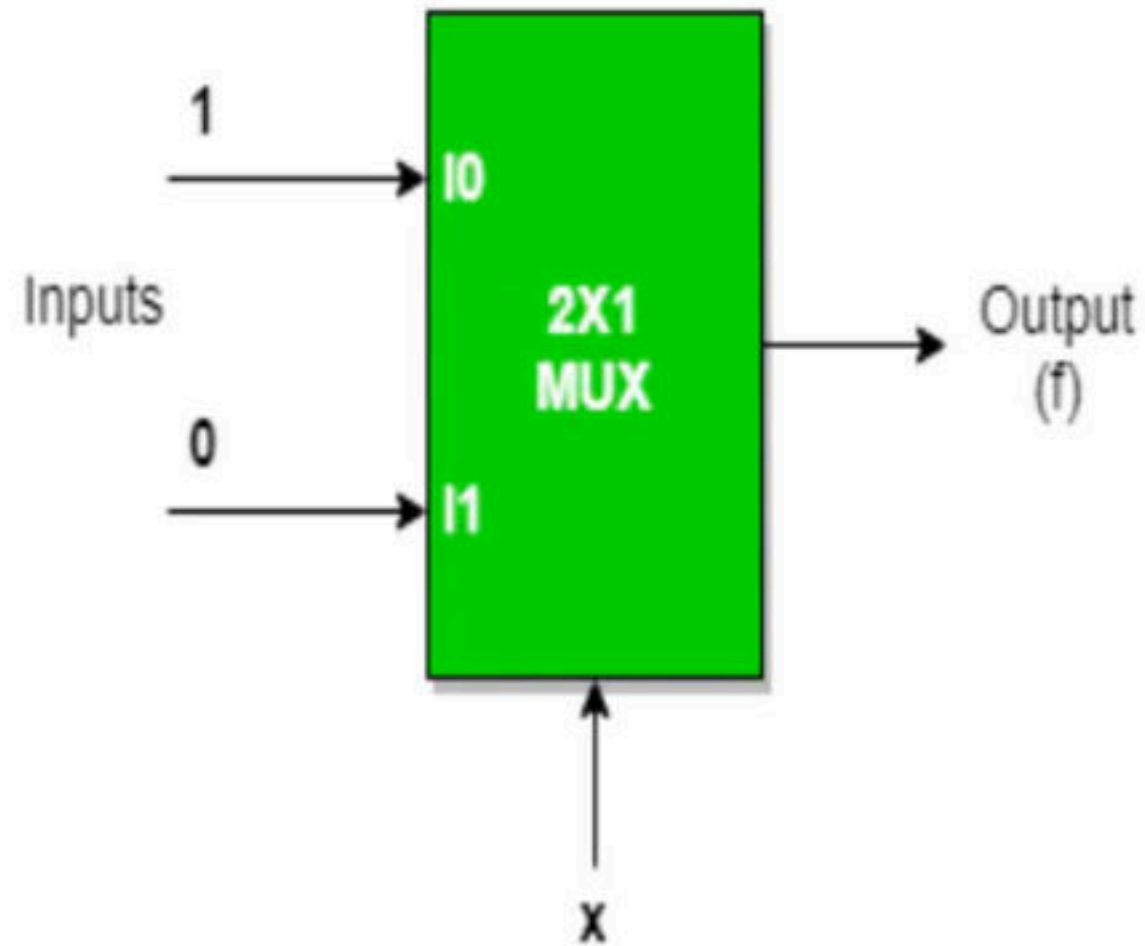
(C) OR, OR

(D) OR, AND

Break

NOT GATE

NOT GATE



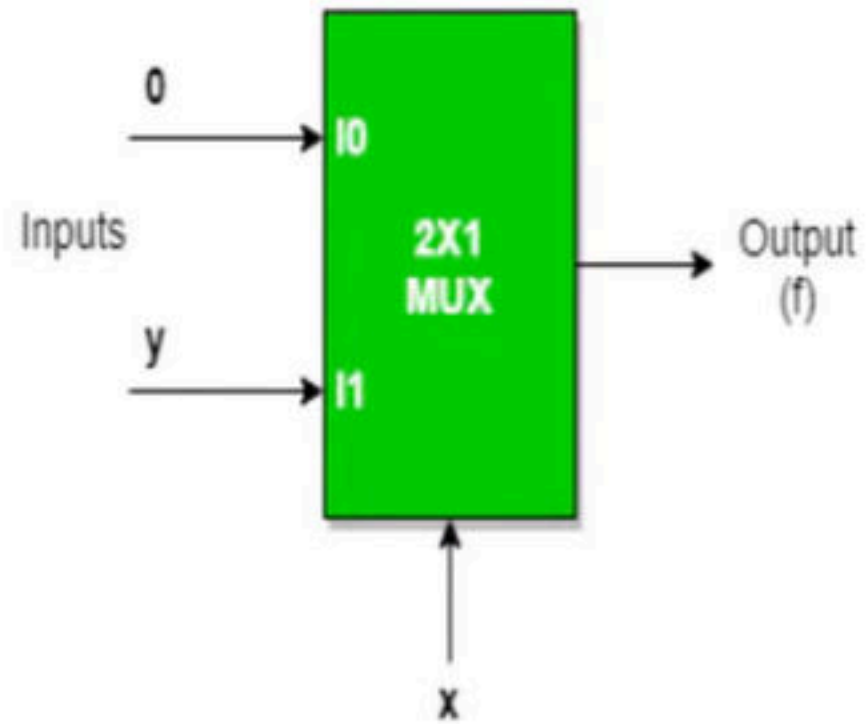
Truth Table

x	f
0	1
1	0

AND GATE

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AND GATE



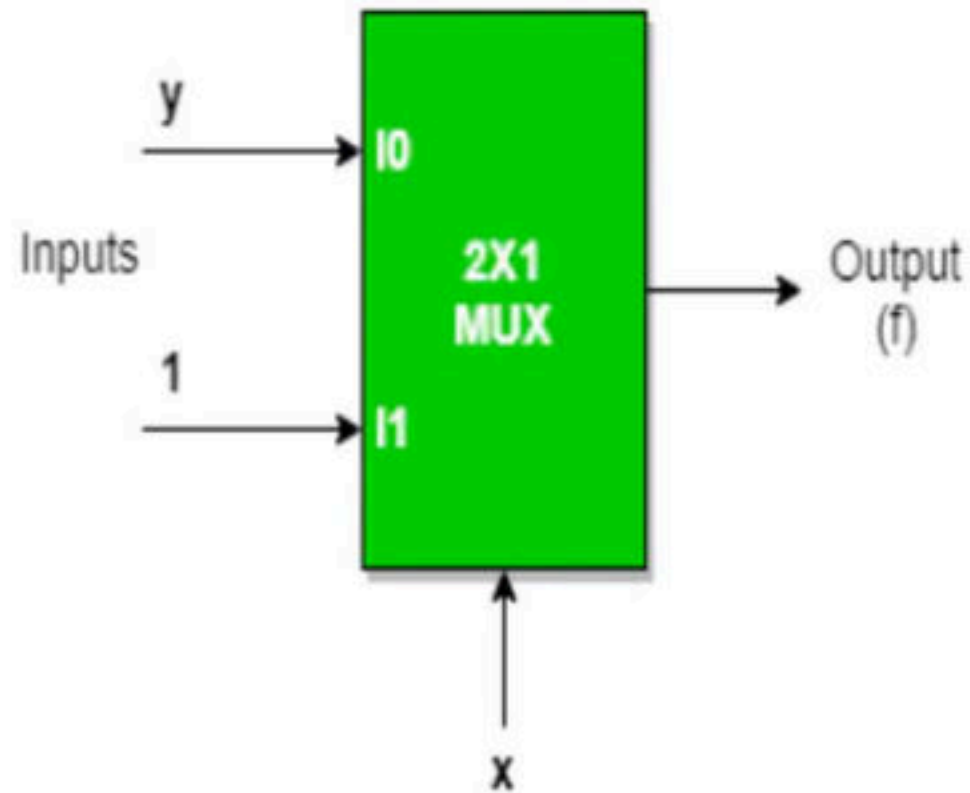
Truth Table

x	y	f	$f \rightarrow y$
0	0	0	f = 0
0	1	0	
1	0	0	f = y
1	1	1	

OR GATE

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OR GATE



Truth Table

x	y	f	$f \rightarrow y$
0	0	0	f = y
0	1	1	
1	0	1	f = 1
1	1	1	

Break

Multiplexer Expansion

(Implementation of Higher order MUX using lower order MUX)

Q 2 : 1 MUX are required to implement 4 : 1 MUX ?

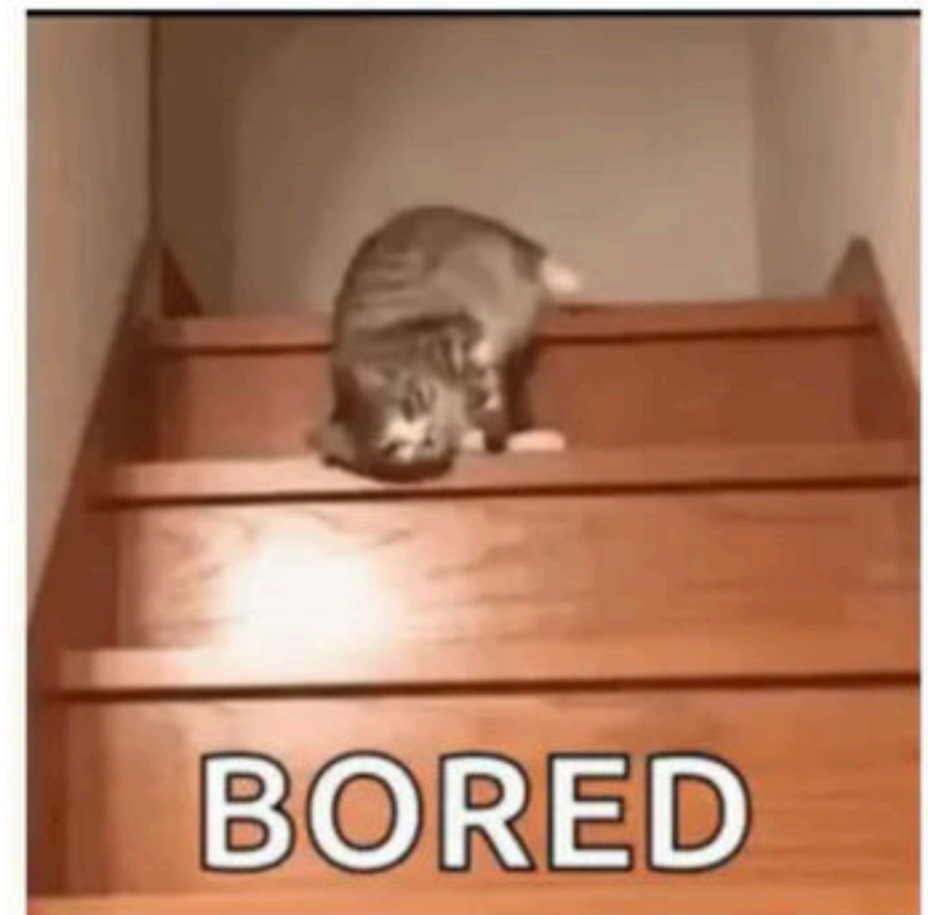
Input		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Q 16 : 1 MUX using 4 : 1 MUX ?

Q how many 4X1 Mux are required in order to construct 128X1?

Q how many 8X1 Mux are required in order to construct 4096X1?

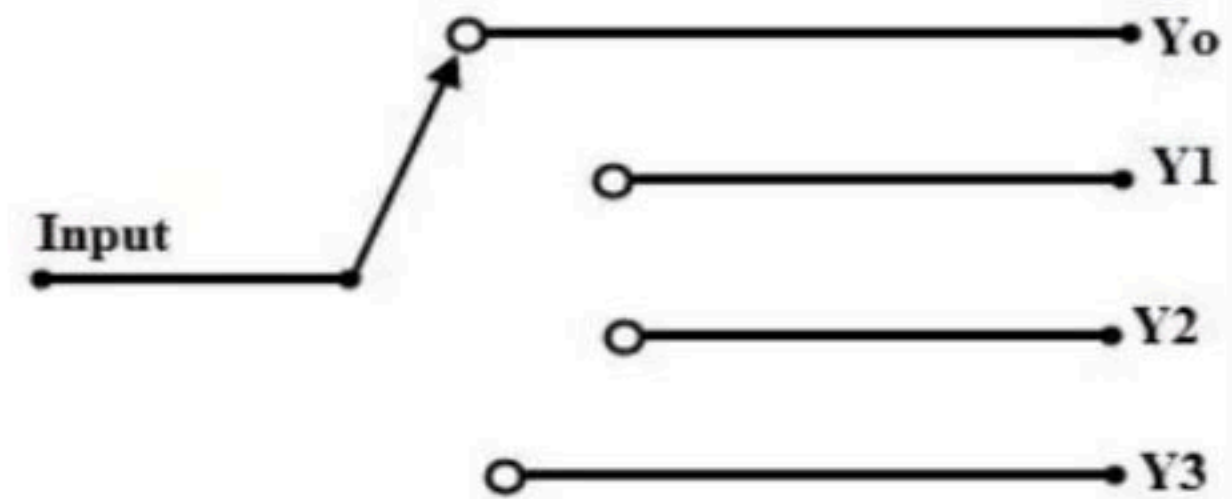
Given:	$m \times 1$
Target:	$n \times 1$
No of levels(K)	$\log_m n$
No of Mux at i^{th} level (x_i)	(n/m_i)
Total Mux required	$\sum_{i=1}^{i=k} x_i$
Maximum capacity	$m^k \times 1$



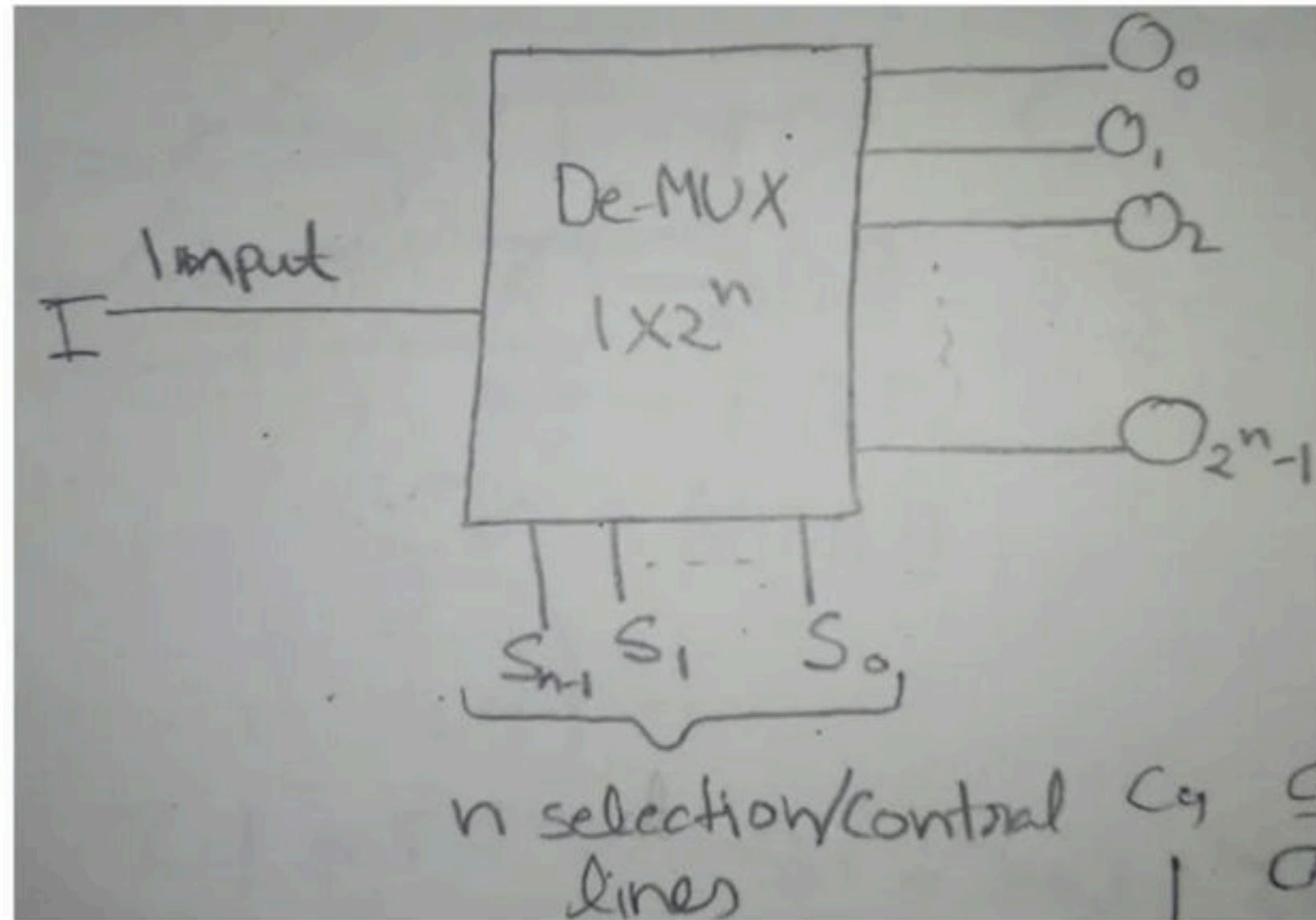
Break

Demultiplexer

1. A demultiplexer (or DeMux) is a device that takes a single input line and routes it to one of several digital output lines.
2. A demultiplexer is also called a data distributor.
3. It is conceptually same as Mux just with reverse logic.



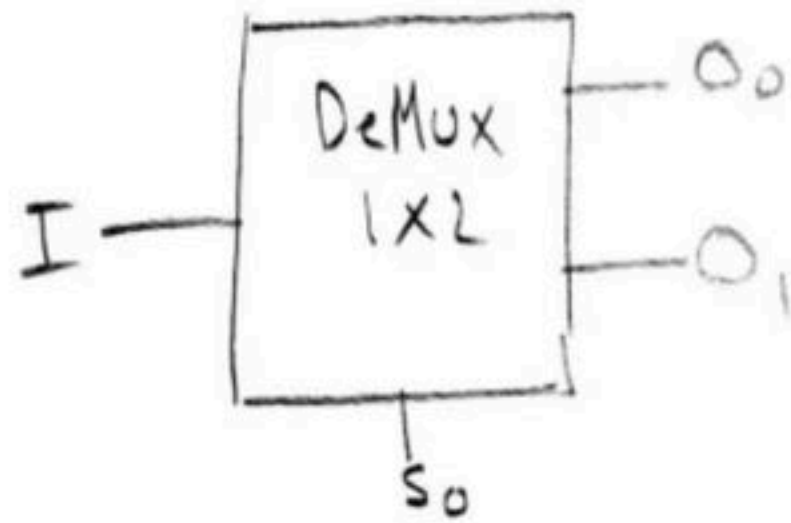
- A demultiplexer of 2^n outputs has n select lines, which are used to select which output line to send the input.



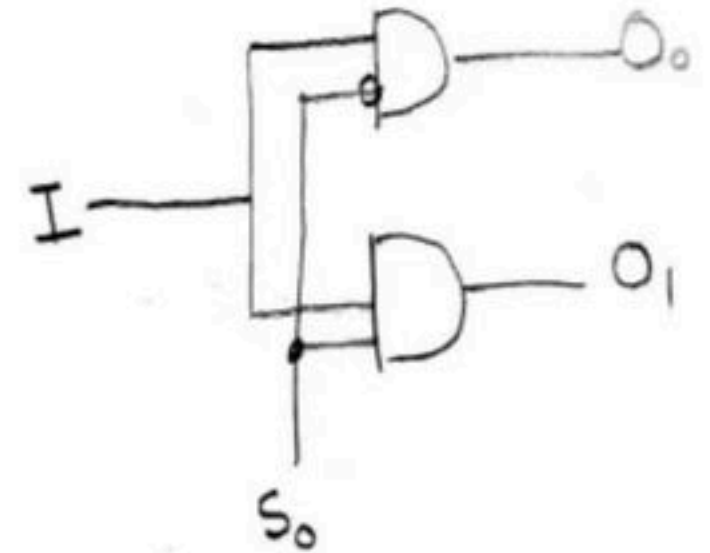
1 to 2 Demultiplexer

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1 to 2 Demultiplexer



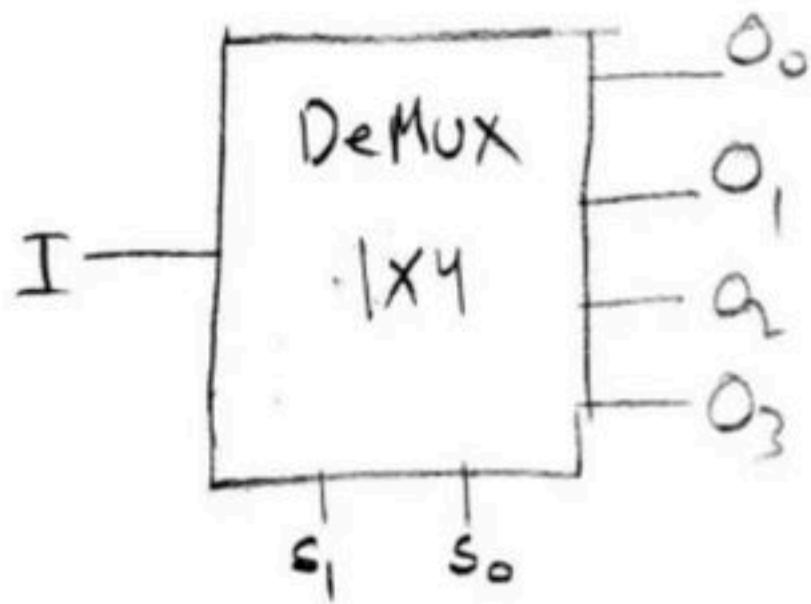
S_0	O_1	O_0
0	0	1
1	1	0



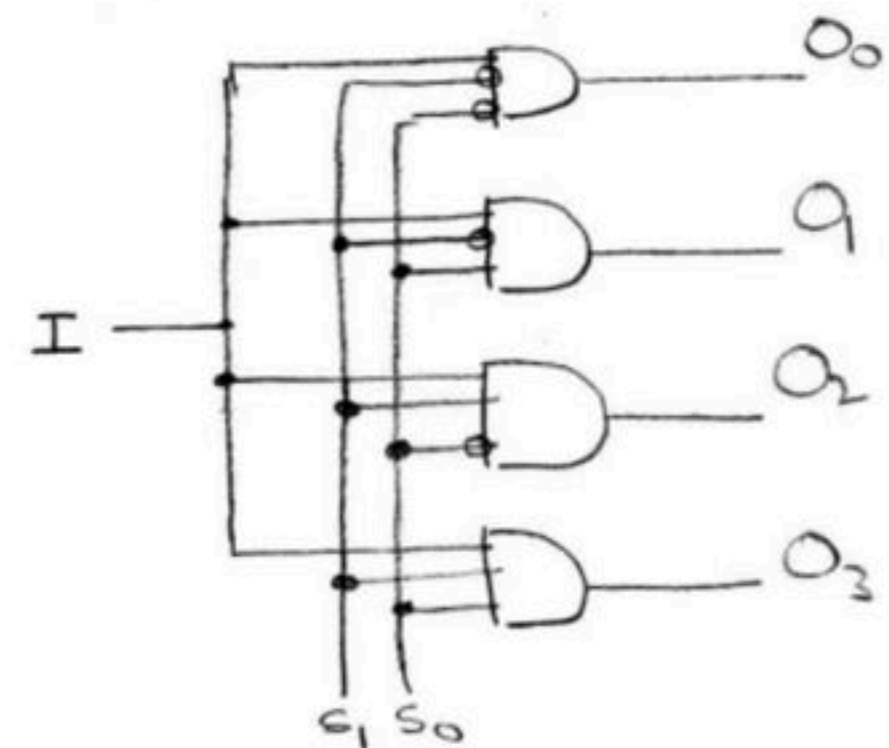
1 to 4 Demultiplexer

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1 to 4 Demultiplexer



S_1	S_0	O_3	O_2	O_1	O_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Break

Demultiplexer Expansion

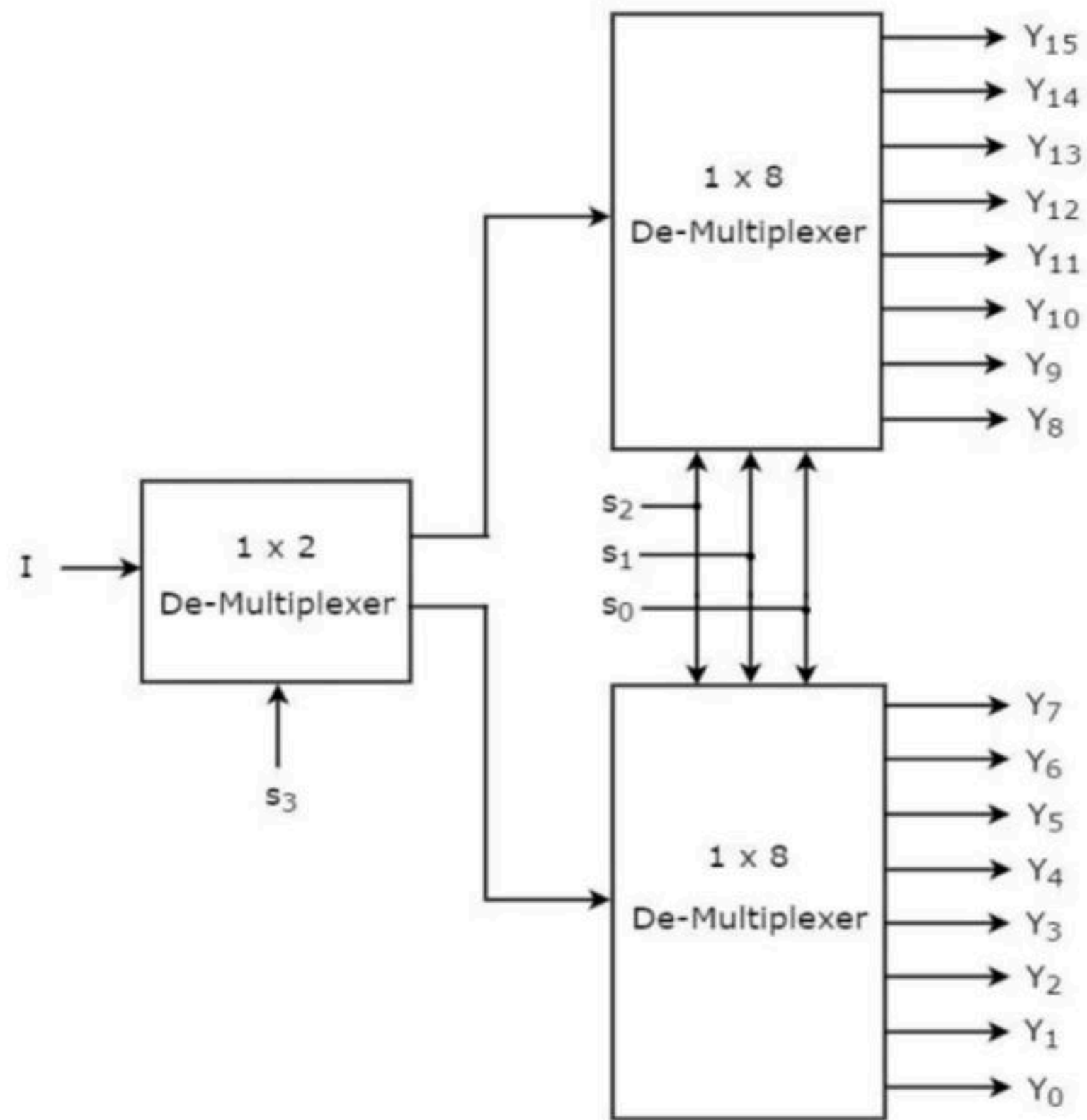
(Implementation of Higher order DeMux using lower order DeMux)

Q 1 : 2 DeMux are required to implement 1 : 4 DeMux ?

Demultiplexer Expansion

(Implementation of Higher order DeMux using lower order DeMux)

Q 1 : 2 DeMux are required to implement 1 : 8 DeMux ?

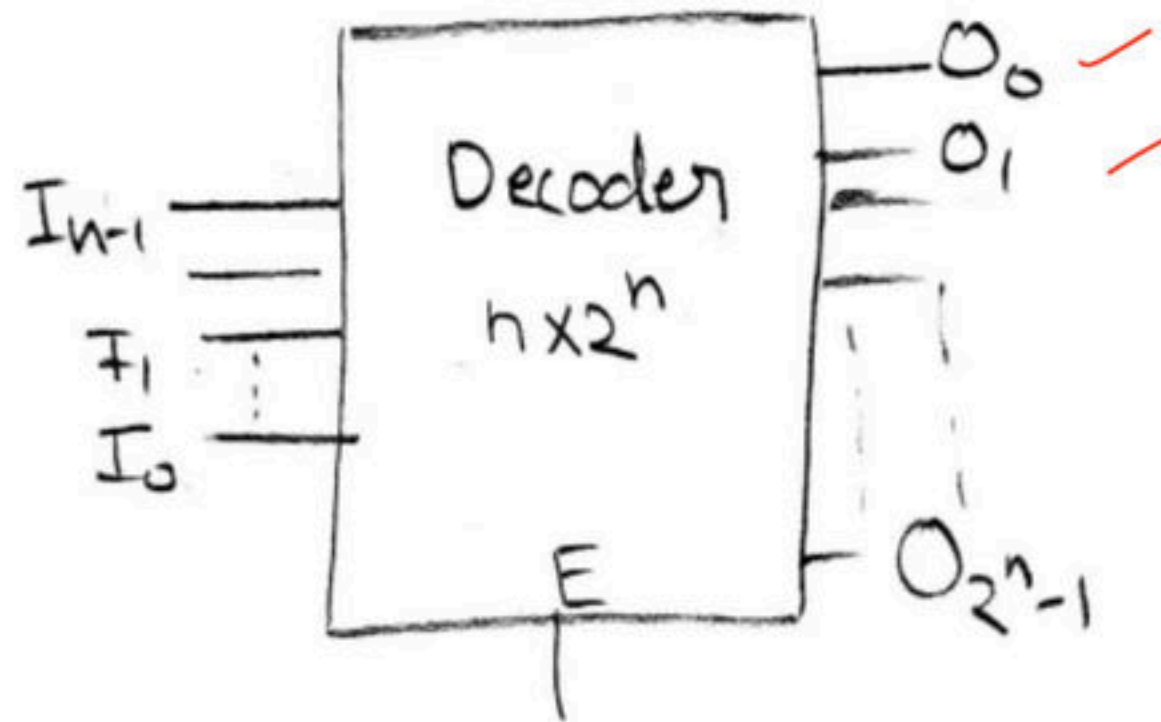


- A DeMux is a combinational circuit, which is used in data communication
- Serial to parallel conversation
- Different input/output configuration demultiplexers are available in the form of single integrated circuits (ICs).
- Demultiplexers are mainly used in Boolean function generators and decoder circuits.

Break

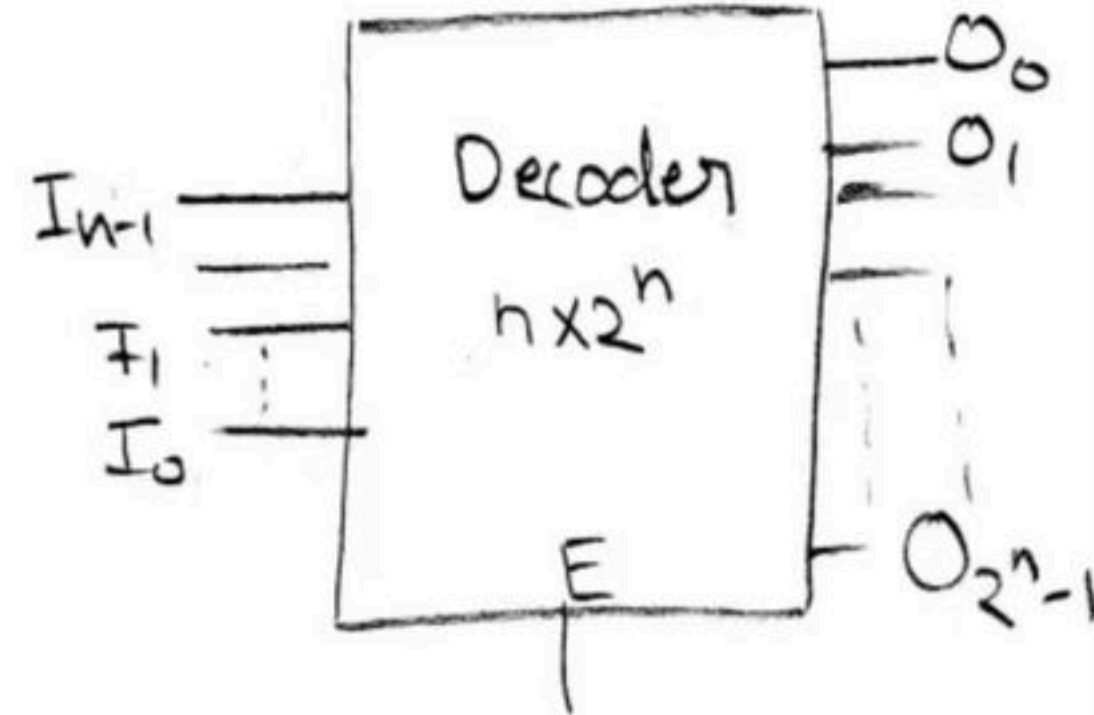
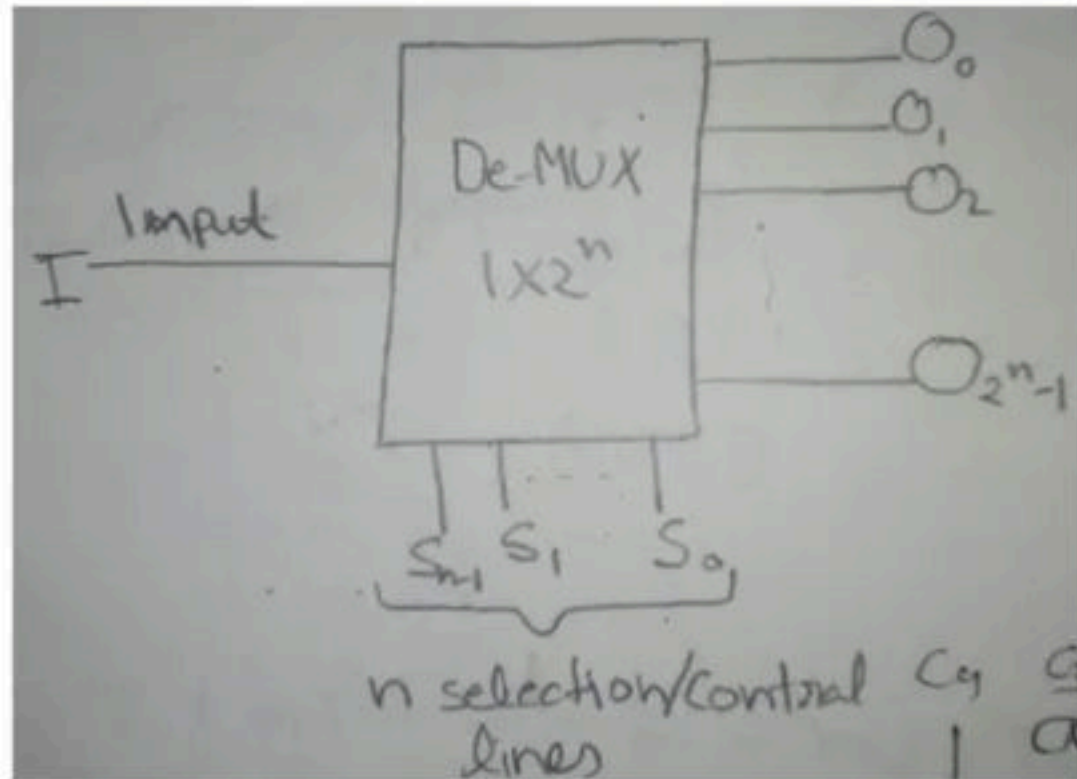
Decoder

- A decoder is a combinational circuit that decodes binary information from n input lines to a maximum of 2^n unique output lines.
- The decoders are called n -to- m -line decoders, where $m \leq 2^n$.
- Their purpose is to generate the 2^n (or fewer) minterms of n input variables. Each combination of inputs will assert a unique output.
- If the n -bit coded information has unused combinations, the decoder may have fewer than 2^n outputs.



Decoder

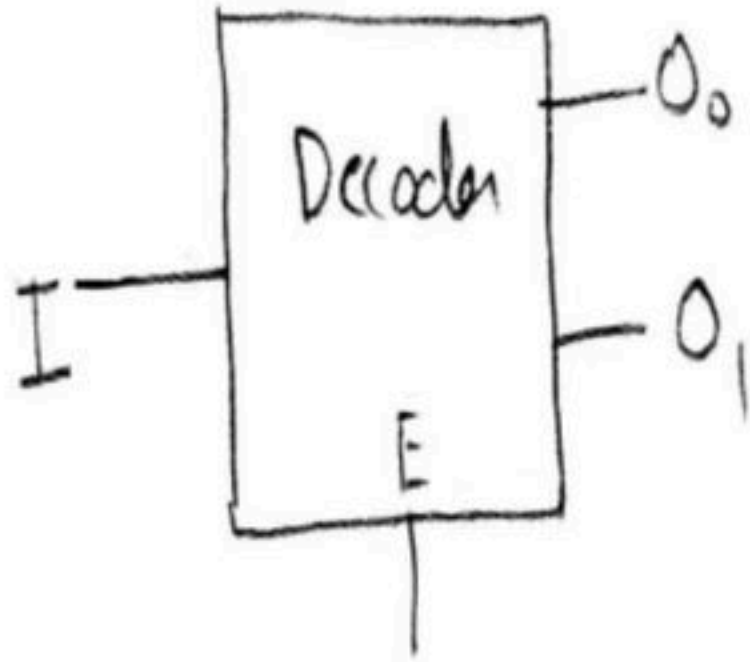
- Decoders are also combinational circuits logically we can say a DeMux can be converted into a decoder by setting input line as enable line and selection line as input lines.



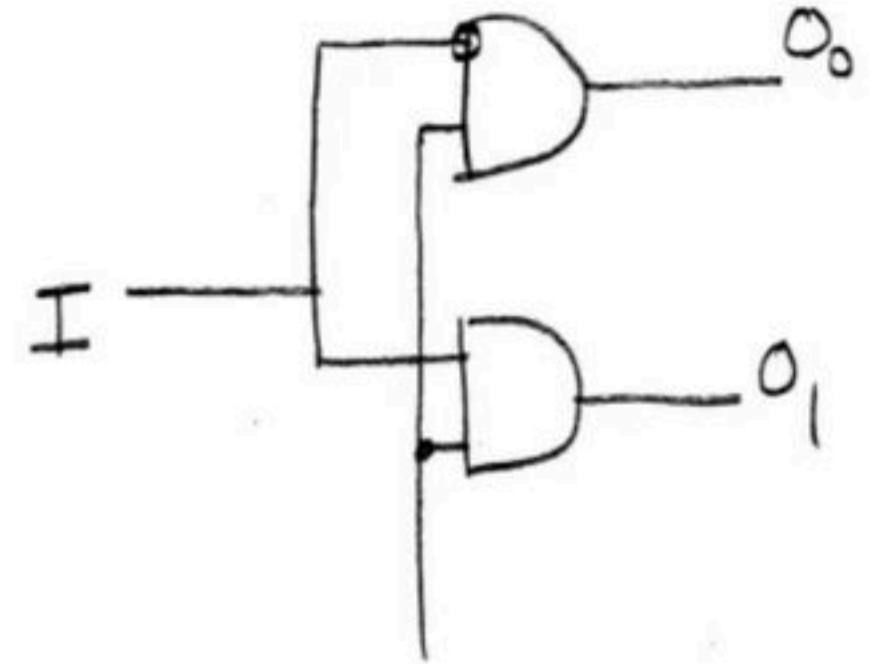
1-to-2 Decoder

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1-to-2 Decoder



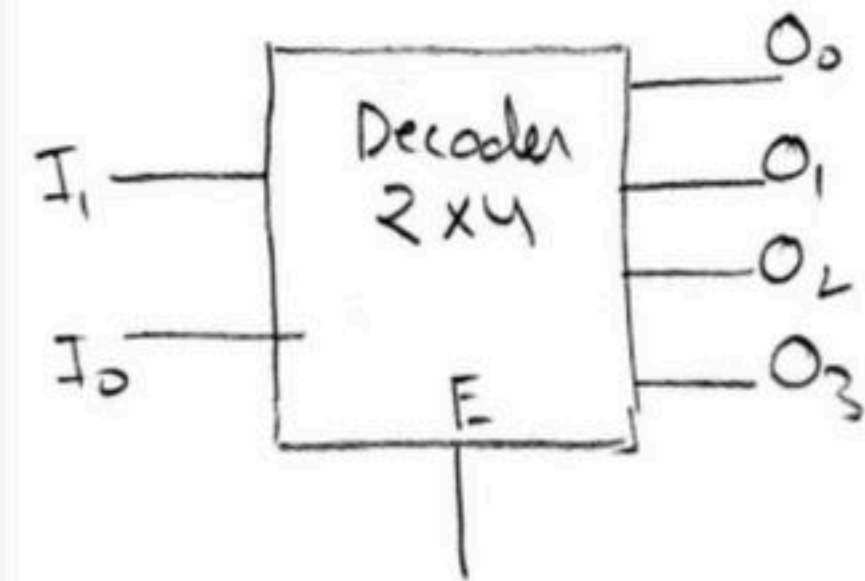
I	O ₁	O ₀
0	0	1
1	1	0



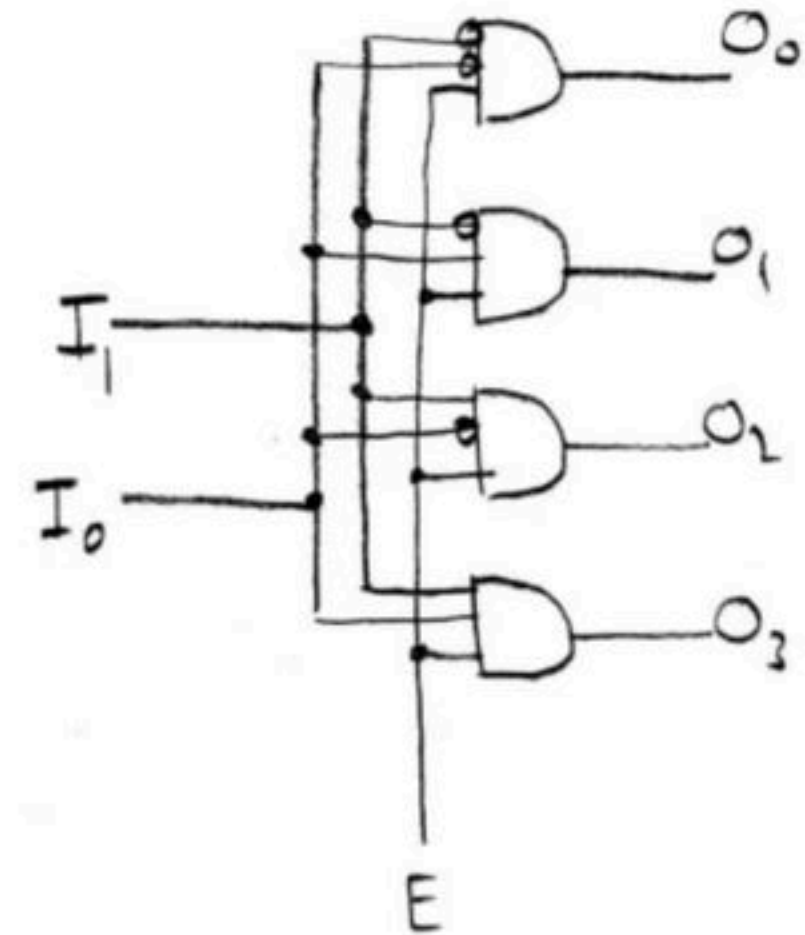
2-to-4 Decoder

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2-to-4 Decoder



I_1	I_0	O_3	O_2	O_1	O_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

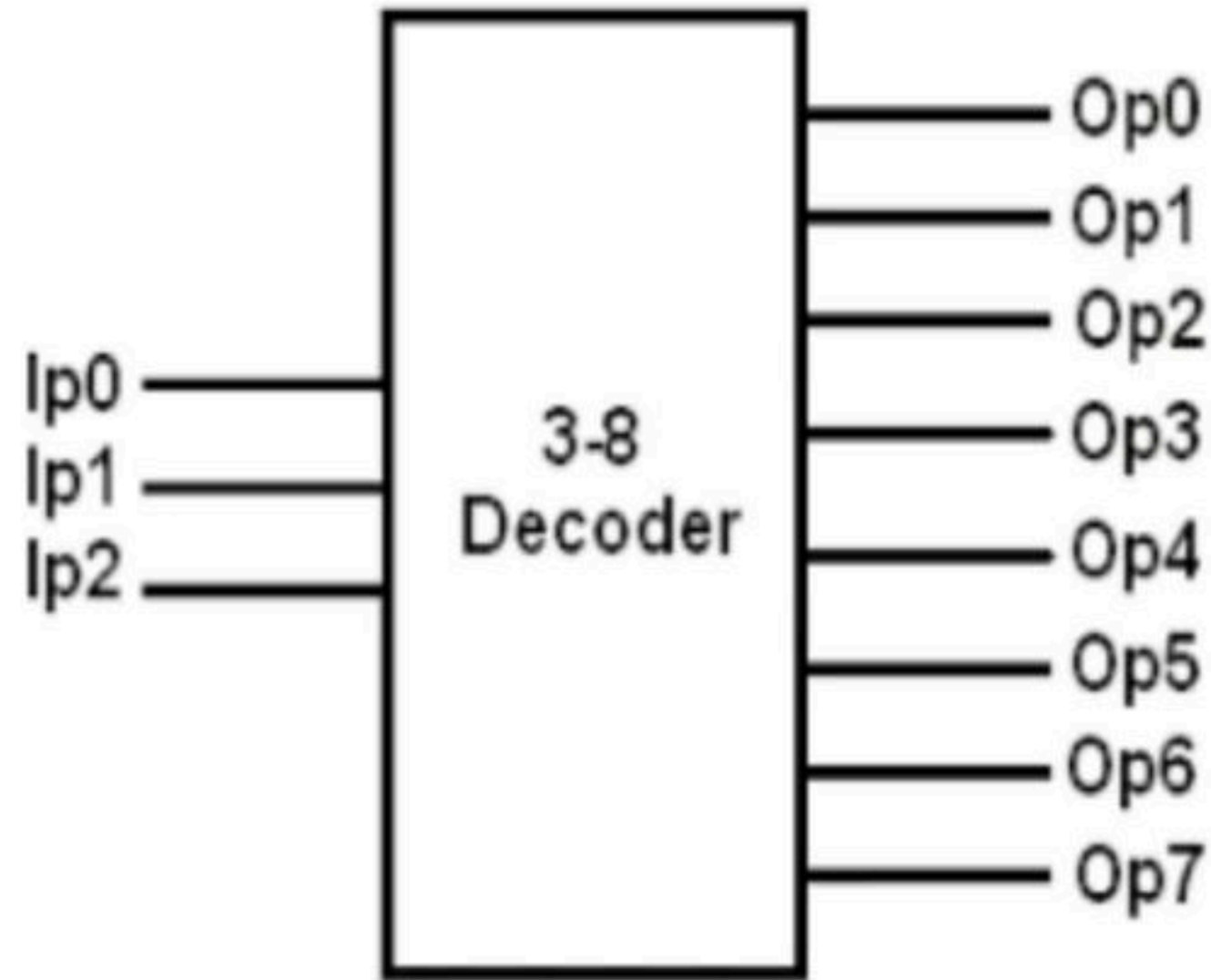


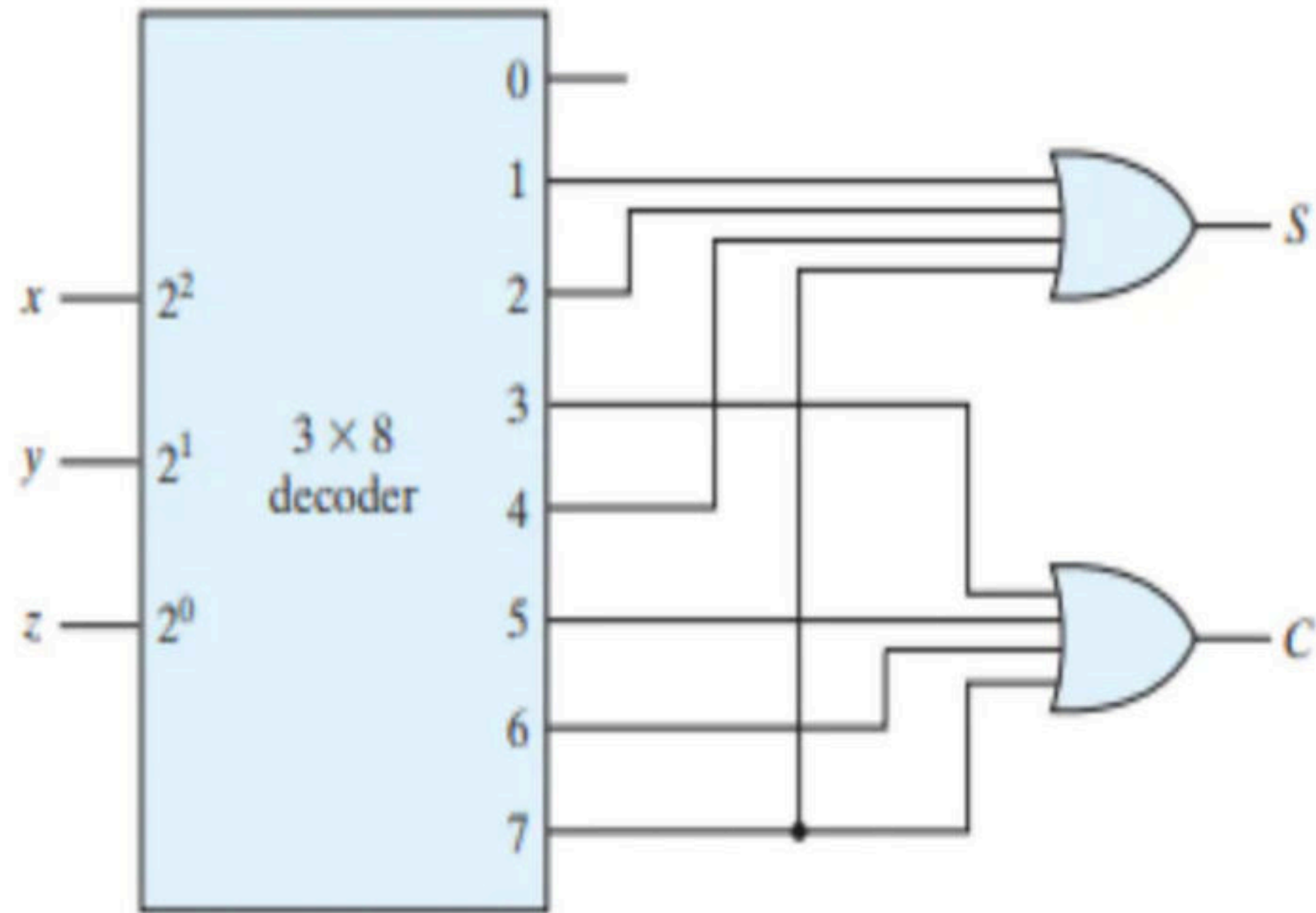
Q Implementation of a full adder with a decoder

From the truth table of the full adder, we obtain the functions for the combinational circuit in sum-of-minterms form:

$$S(x, y, z) = \sum (1, 2, 4, 7)$$

$$C(x, y, z) = \sum (3, 5, 6, 7)$$





Combinational Logic Implementation

- Any combinational circuit with n inputs and m outputs can be implemented with an n -to- 2^n -line decoder and OR gates.
- The procedure for implementing a combinational circuit by means of a decoder and OR gates requires that the Boolean function for the circuit be expressed as a sum of minterms.
- A decoder is then chosen that generates all the minterms of the input variables. The inputs to each OR gate are selected from the decoder outputs according to the list of minterms of each function.

- **Active High Decoder:-** When output is directly from AND gate, we get exact minterms then, it is called active high decoder. In the 2nd level here, we use OR-gate to find the function.
- **Active Low Decoder:-** In active low decoders, output will be from NAND gates, on the 2nd level we again use NAND gate, as we know NAND-NAND implementation is SAME as AND-OR.

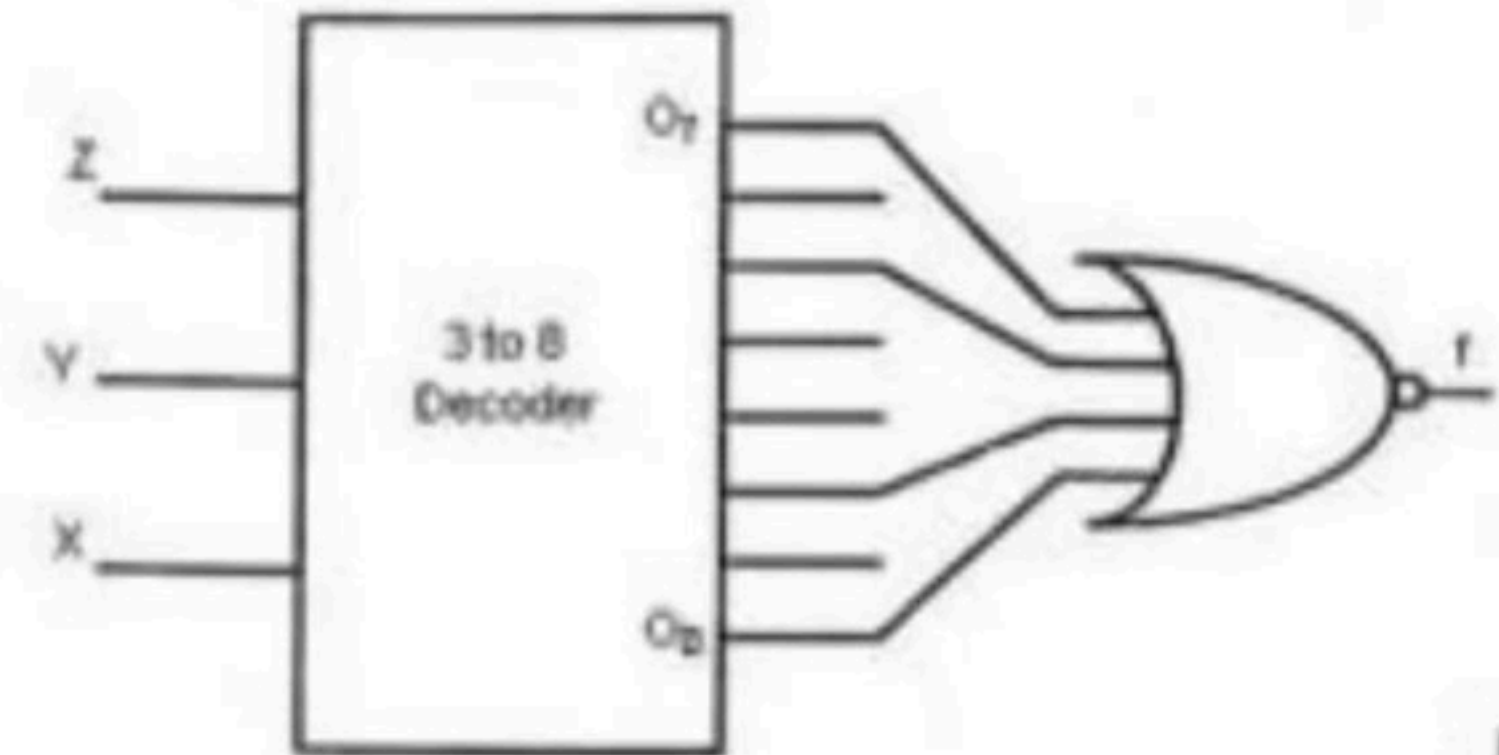
Q What Boolean function does the circuit below realize? (GATE-2006) (2 Marks)

(A) $xz + x'z'$

(B) $xz' + x'z$

(C) $x'y' + yz$

(D) $xy + y'z'$



Break

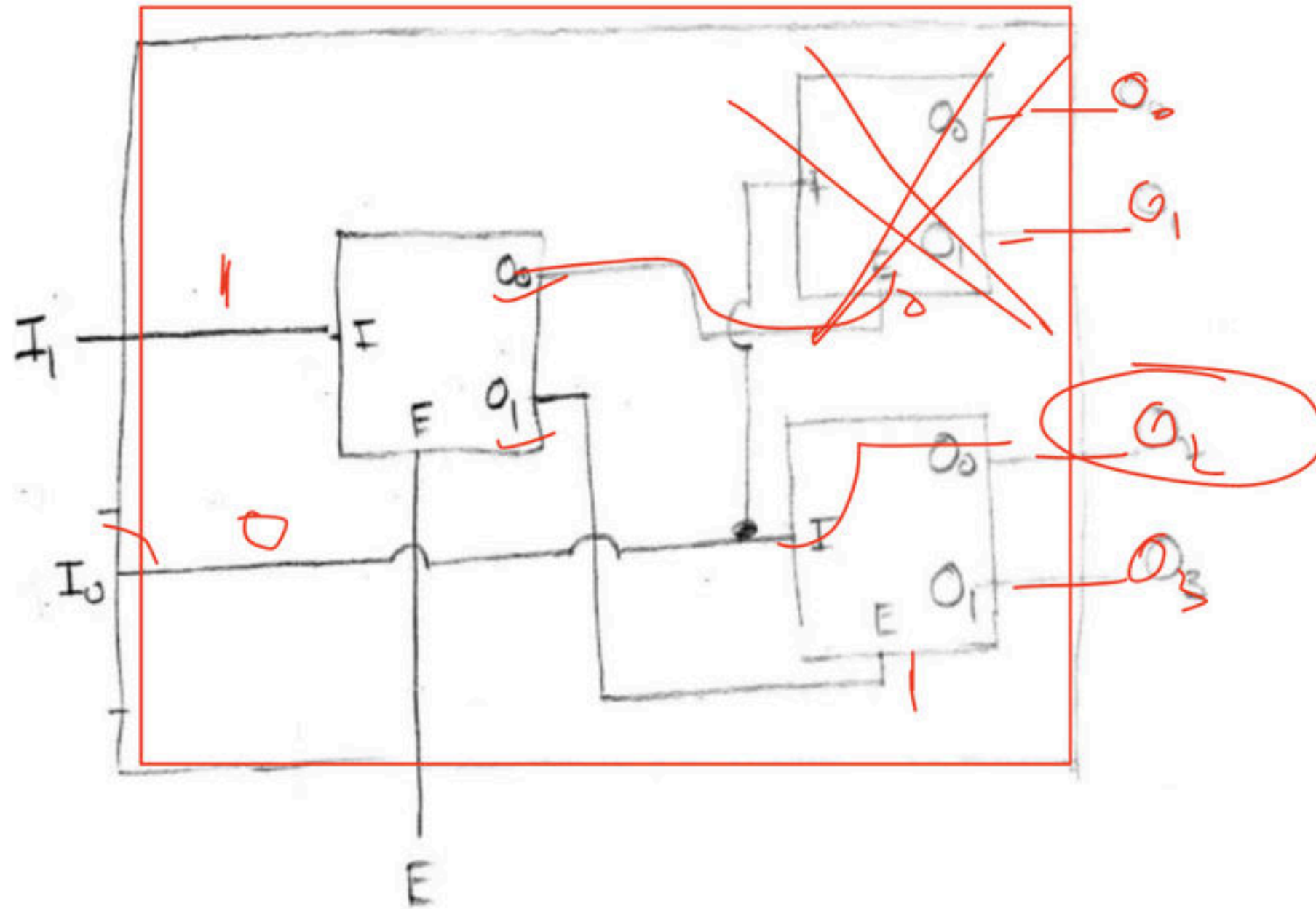
$$(1 \times 2) \rightarrow (2 \times 4)$$

Decoder Expansion

Decoder Expansion

I_1	I_0
1	0

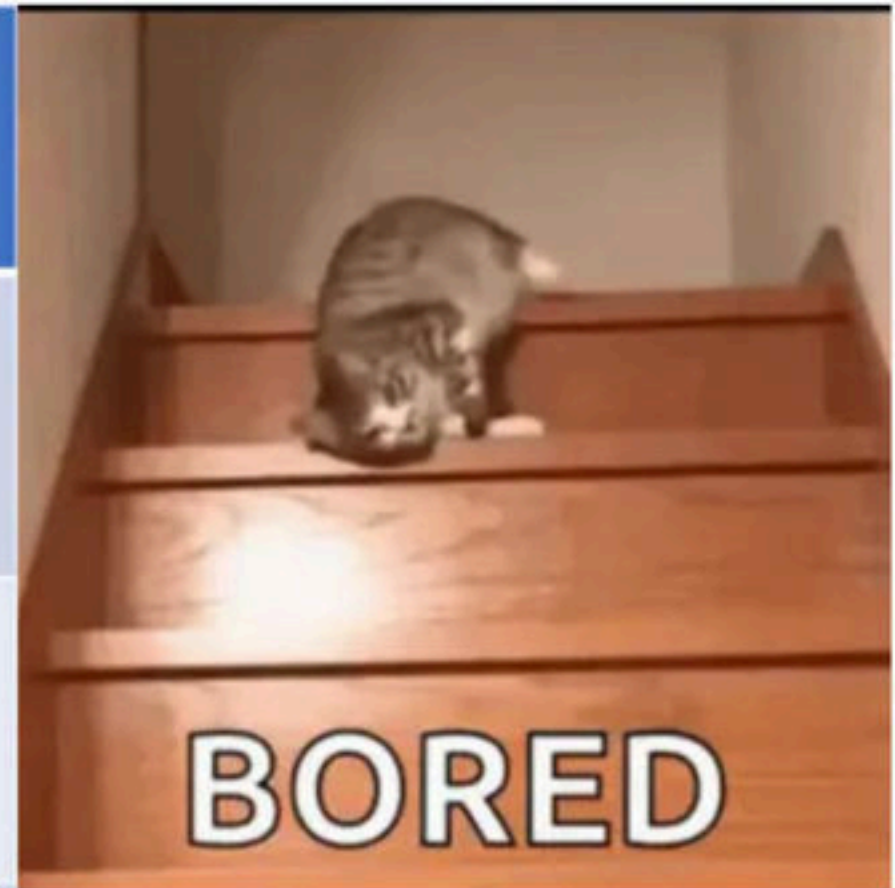
O_2



Q 6-to-64 from 2-to-4 ?

Q 7-to-128 from 3-to-8 ?

Target:	$m \times n$
Given:	$p \times q$
No of levels(K)	$\text{floor}(m/p)$
No of Mux at i^{th} level (x_i)	(n/q^{k-i+1})
Total Mux required	$\sum_{i=1}^{i=k} x_i$
Maximum capacity	$(p \times k) \times q^k$



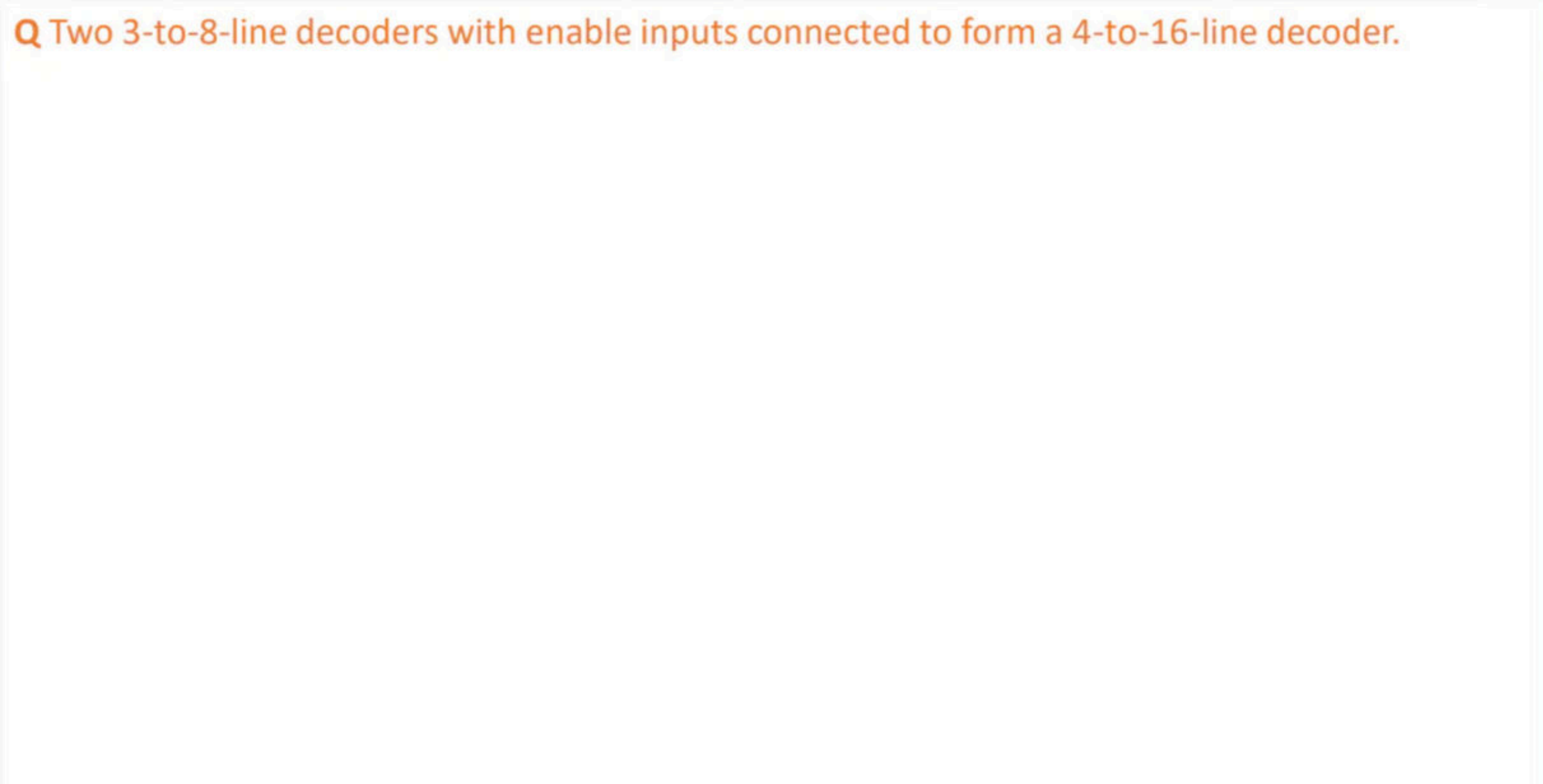
Q How many 3-to-8-line decoders with an enable input are needed to construct a 6-to-64-line decoder without using any other logic gates? **(GATE-2007) (2 Marks)**

(A) 7

(B) 8

(C) 9

(D) 10

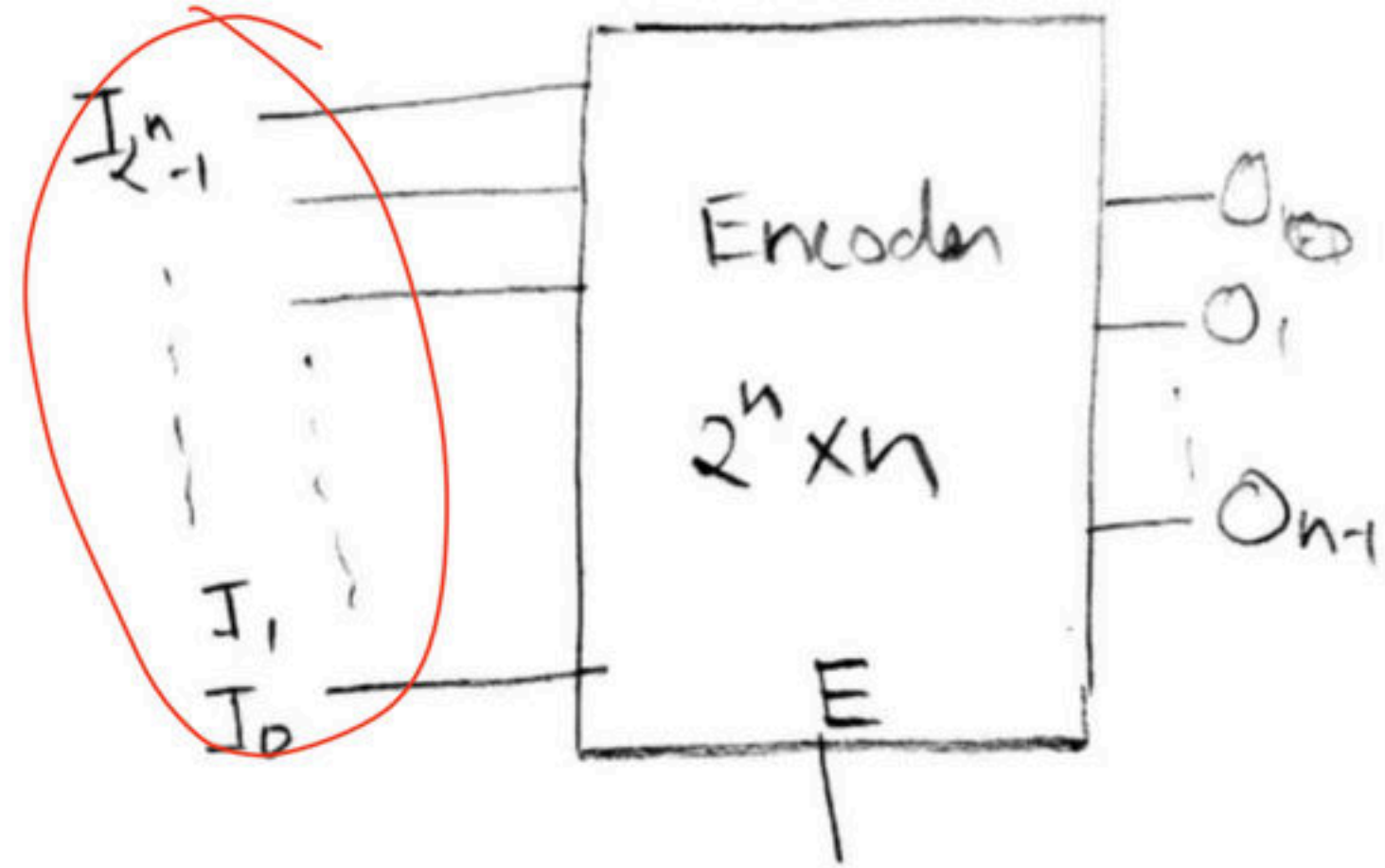


Q Two 3-to-8-line decoders with enable inputs connected to form a 4-to-16-line decoder.

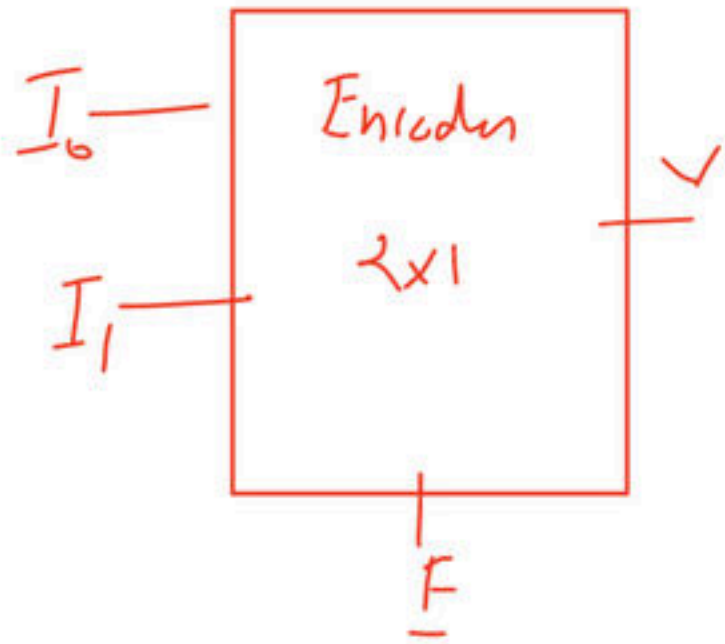
Break

Encoder

- An encoder is a combinational circuit that encode binary information from one of a 2^N input lines and encode it into N output lines, which represent N bit code for the input.
- For simple encoders, it is assumed that only one input line is active at a time.
- Encoder performs the inverse operation of a decoder.

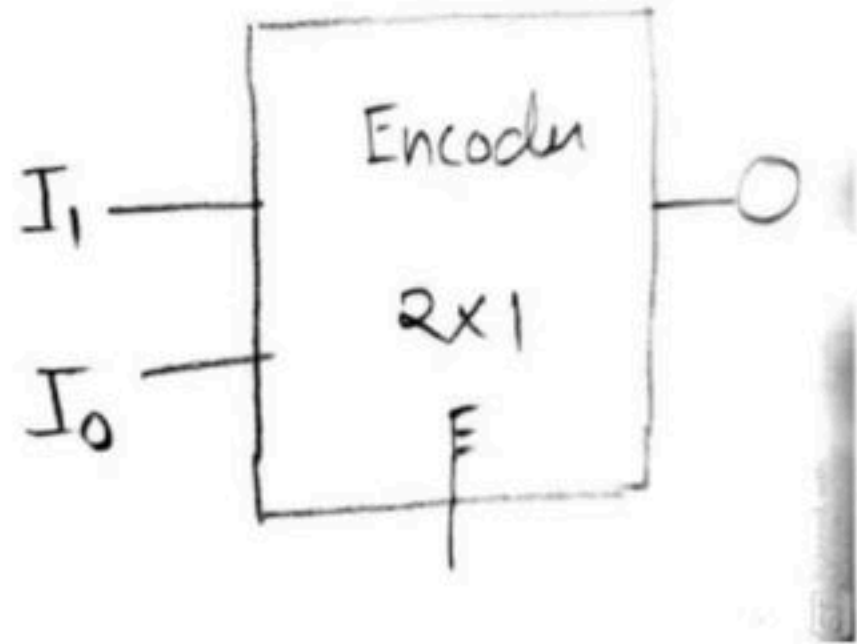


2-to-1 Encoder



i/p		o/p
I_1	I_0	Y
0	1	0
1	0	1

2-to-1 Encoder



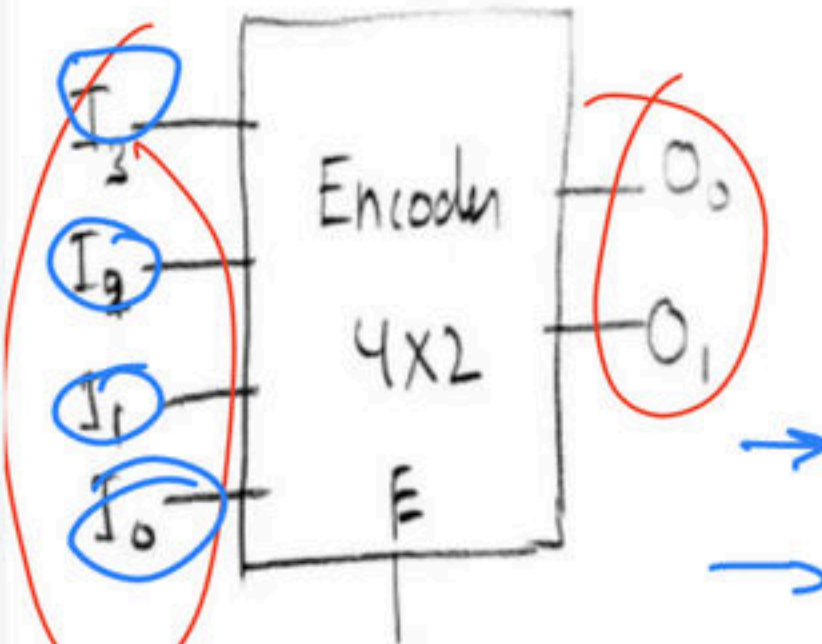
I_1	I_0	O_0
0	1	0
1	0	1

Blue circles highlight the input pairs (0, 1) and (1, 0). A blue arrow points from the first row to the second row, indicating the mapping from input combinations to the output.

$$O_0 = I_1$$

4-to-2 Encoder

4-to-2 Encoder



I_3	I_2	I_1	I_0	O_1	O_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

$$\bullet O_0 = (I_3 \oplus I_1) \bar{I}_2 \bar{I}_0$$

$$\bullet O_1 = (I_3 \oplus I_2) \bar{I}_1 \bar{I}_0$$

$$\bar{I}_3 I_2 \bar{I}_1 \bar{I}_0 + I_3 \bar{I}_2 \bar{I}_1 \bar{I}_0$$

$$(\bar{I}_3 I_2 + I_3 \bar{I}_2) \bar{I}_1 \bar{I}_0$$

- $O_0 = I_0' I_1' (I_2 \oplus I_3)$
- $O_1 = I_0' I_2' (I_1 \oplus I_3)$

Break

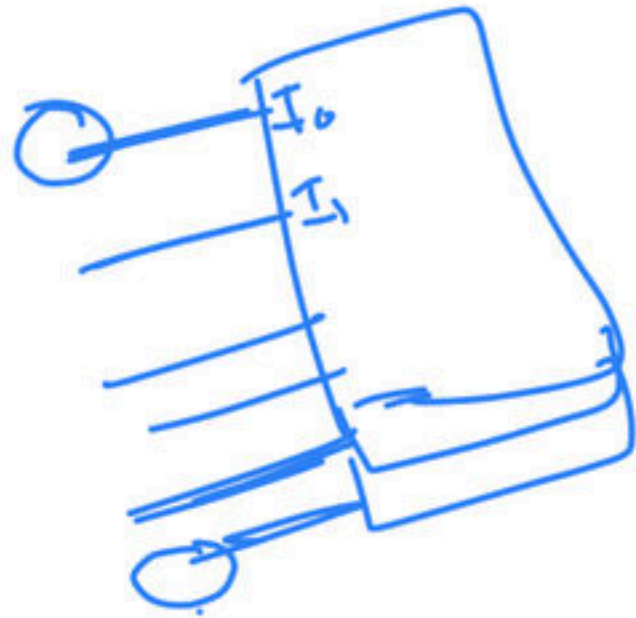
Priority Encoder

- In some practical cases more than one input can be high at a time, there we can not use simple encoder
- In a priority encoder more than one input can be high at a time. A priority encoder is an encoder circuit that includes the priority function.
- The operation of the priority encoder is such that if two or more inputs are equal to 1 at the same time, the input having the highest priority will take precedence.

- They are often used to control interrupt requests by acting on the highest priority interrupt input.
- The *priority encoders* output corresponds to the currently active input which has the highest priority.
- So when an input with a higher priority is present, all other inputs with a lower priority will be ignored.

I_0 I_1 I_2 I_3
a b c d

$$I_2 > I_3 > I_0 > I_1$$



	I_3	I_2	I_1	I_0	Q_1	Q_0
→	0	0	d	1	0	0
→	0	0	1	0	0	1
→	d	1	d	d	1	0
→	1	0	d	d	1	1

$$Q_1 = I_2 + I_3 \bar{I}_2$$

$$Q_0 = I_3 \bar{I}_2 + \bar{I}_3 \bar{I}_1 I_1 \bar{I}_0$$

$$I_3 > I_2 > I_1 > I_0$$

I_3	I_2	I_1	I_0	O_1	O_0
0	0	0	1		
0	0	1	d		
0	1	d	d		
1	d	d	d		

$$O_0 =$$

$$O_1 =$$

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$$I_3 > I_2 > I_1 > I_0$$

I_3	I_2	I_1	I_0	O_1	O_0
0	0	0	1	0	0
0	0	1	d	0	1
0	1	d	d	1	0
1	d	d	d	1	1

$$O_0 = I_3 + I_1 I_2'$$

$$O_1 = I_2 + I_3$$

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Q Match the terms in List - I with the options given in List - II: (NET-JULY-2018)

List - I	List - II
(1) Decoder	(i) 1 line to 2^n lines
(2) Multiplexer	(ii) n lines to 2^n lines
(3) De multiplexer	(iii) 2^n lines to 1 line
	(iv) 2^n lines to 2^n-1 lines

=

	1	2	3
a)	ii	i	iii
b)	ii	iii	i
c)	ii	i	iv
d)	iv	ii	i

Handwritten notes to the right of the table:
a) — 25 $\times 4$
b) — 62
c) — 6
d) — 7

Q In the following truth table, $V = 1$ if and only if the input is valid. **(GATE-2013) (2 Marks)**
What function does the truth table represent?

(A) Priority encoder

(B) Decoder

(C) Multiplexer

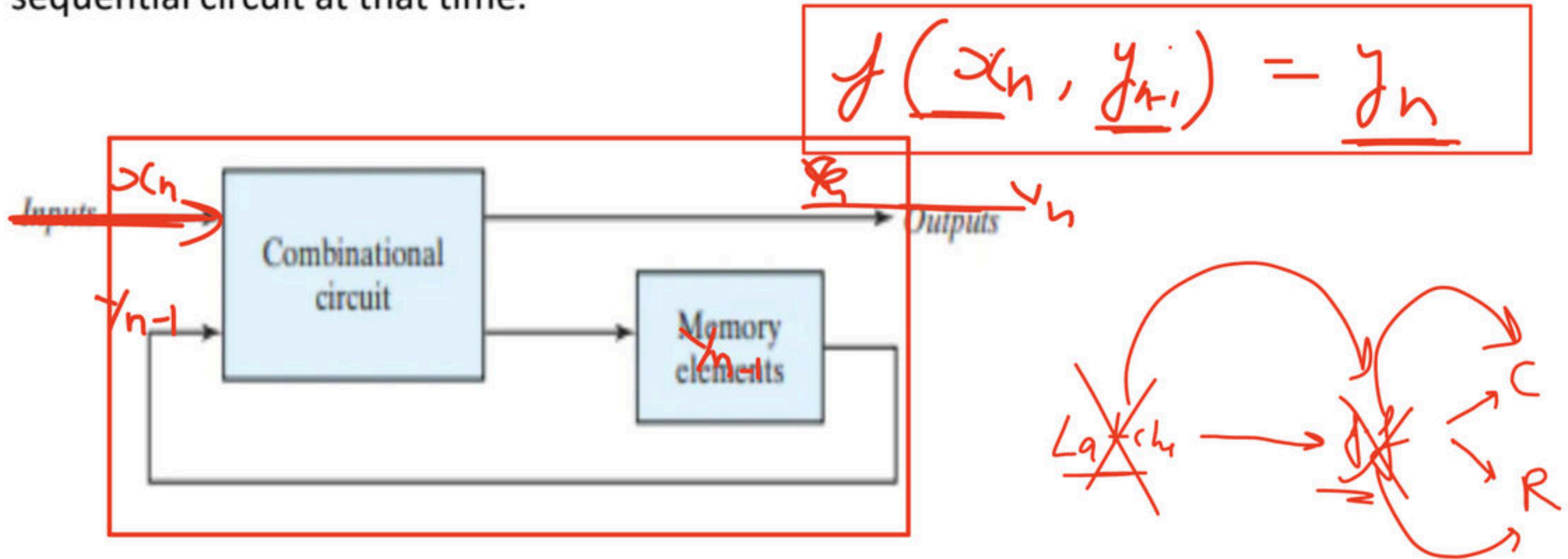
(D) Demultiplexer

72

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

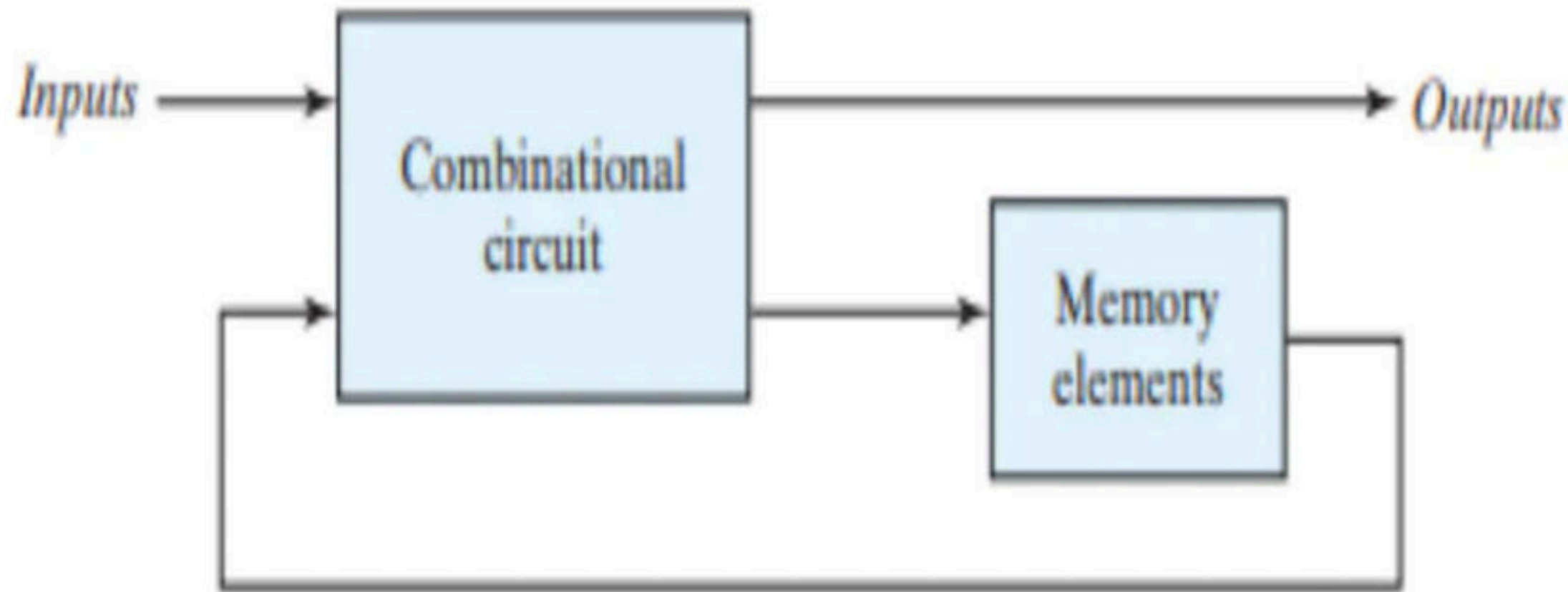
SEQUENTIAL CIRCUITS

- Sequential Circuits consists of a combinational circuit to which memory elements are connected to form a feedback path. The memory elements are devices capable of storing binary information.
- The binary information stored in these elements at any given time defines the state of the sequential circuit at that time.



- The sequential circuit receives binary information from external inputs (x_n) that, together with the present state (y_{n-1}) of the memory elements, determine the binary value of the outputs (y_n).
- A sequential circuit is specified by a time sequence of inputs, outputs, and internal states.

$$f(x_n, y_{n-1}) = y_n$$

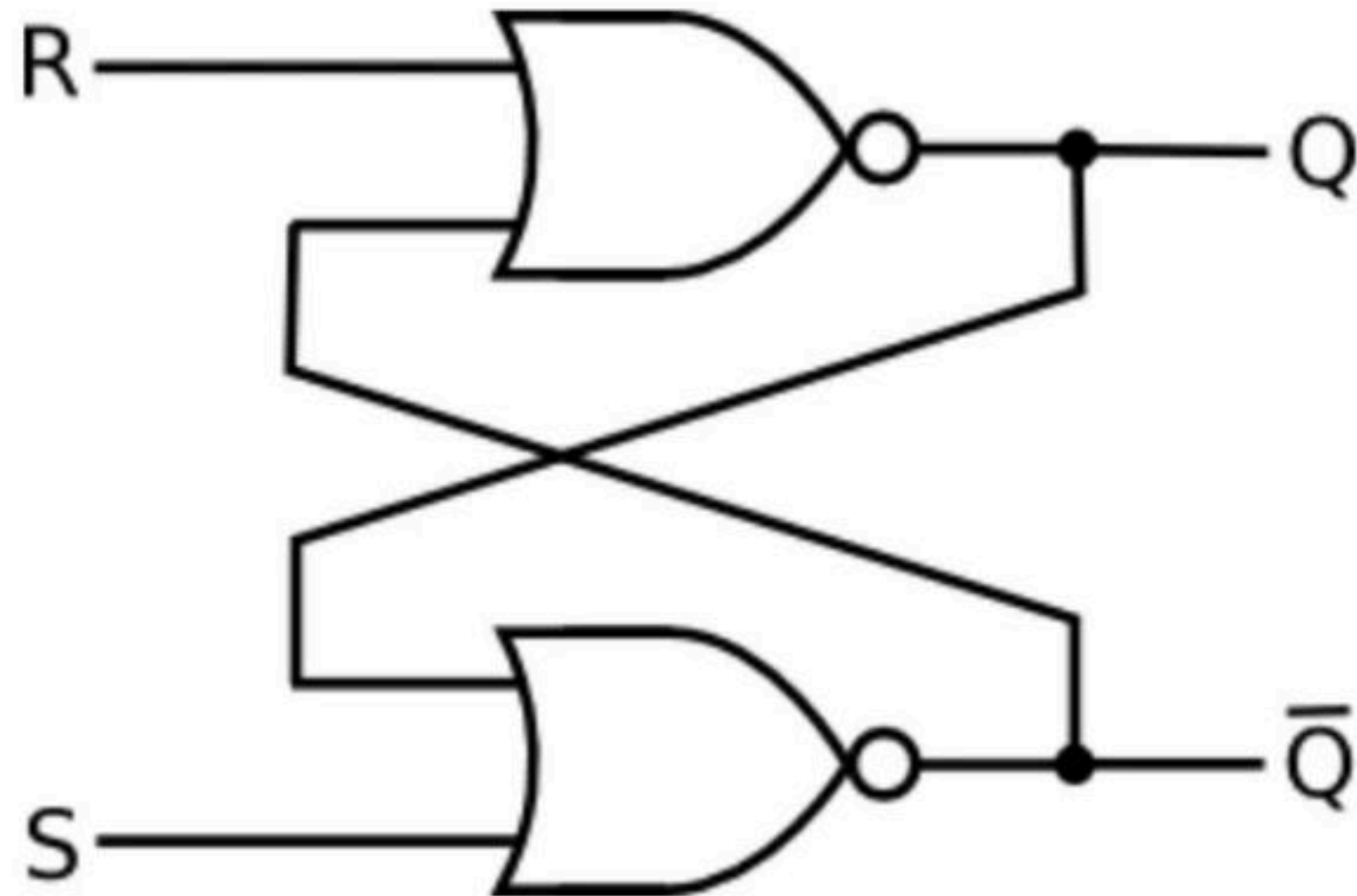


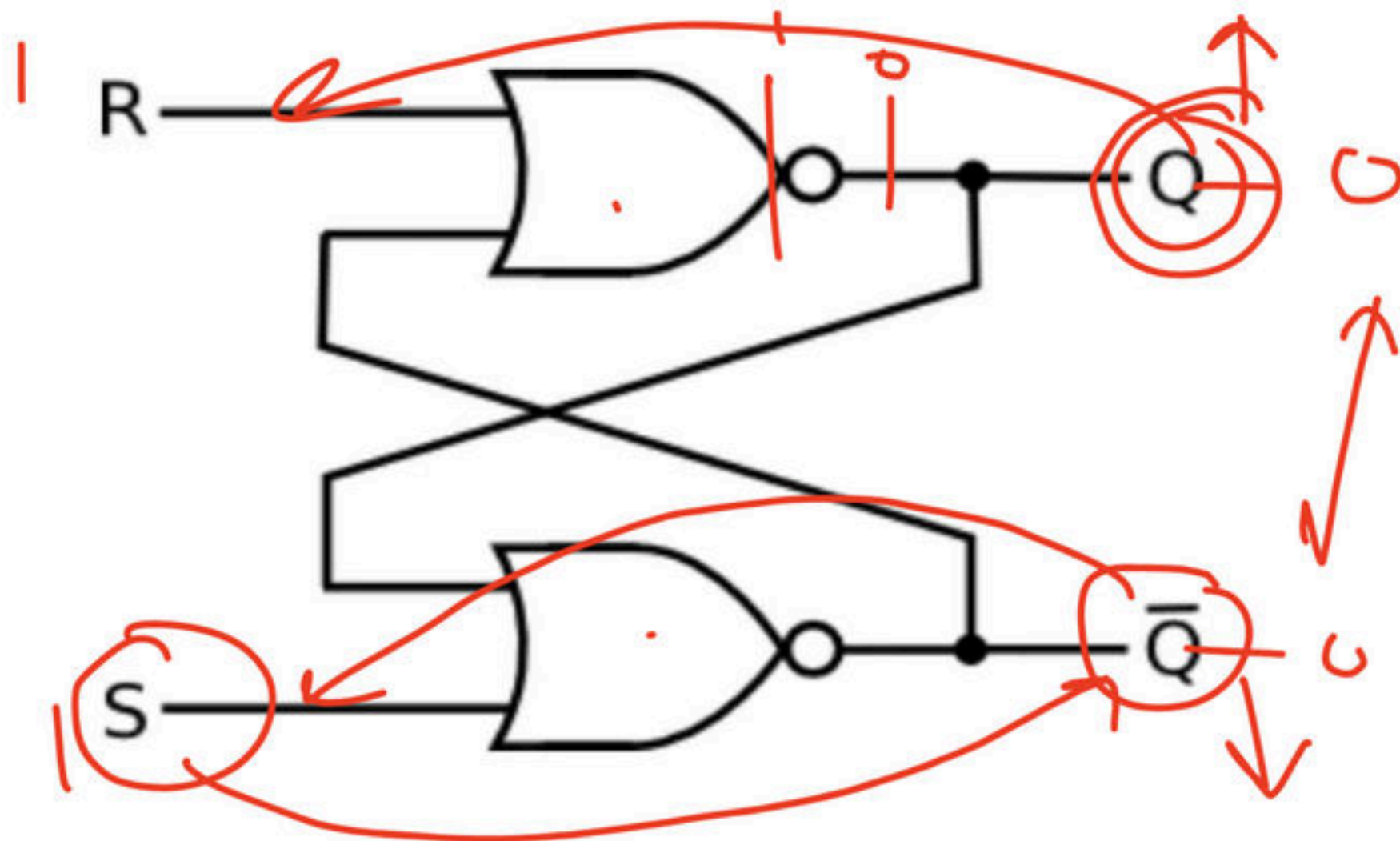
Latches

- Latch means to hold something or something which do not change.
- ✗
• latches are the basic building blocks of any flip flop and they are capable of holding 1 bit until necessary.
- Storage elements that operate with signal levels are referred to as latches.
- Latches are level sensitive devices.

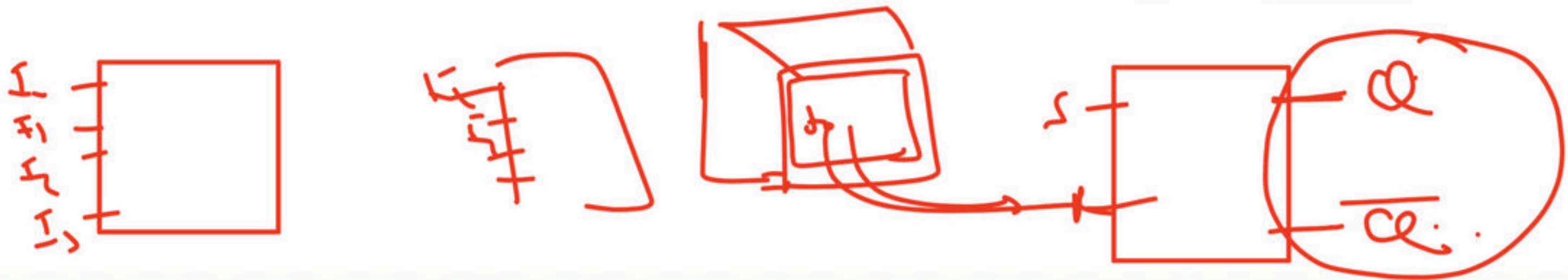
NOR Latch

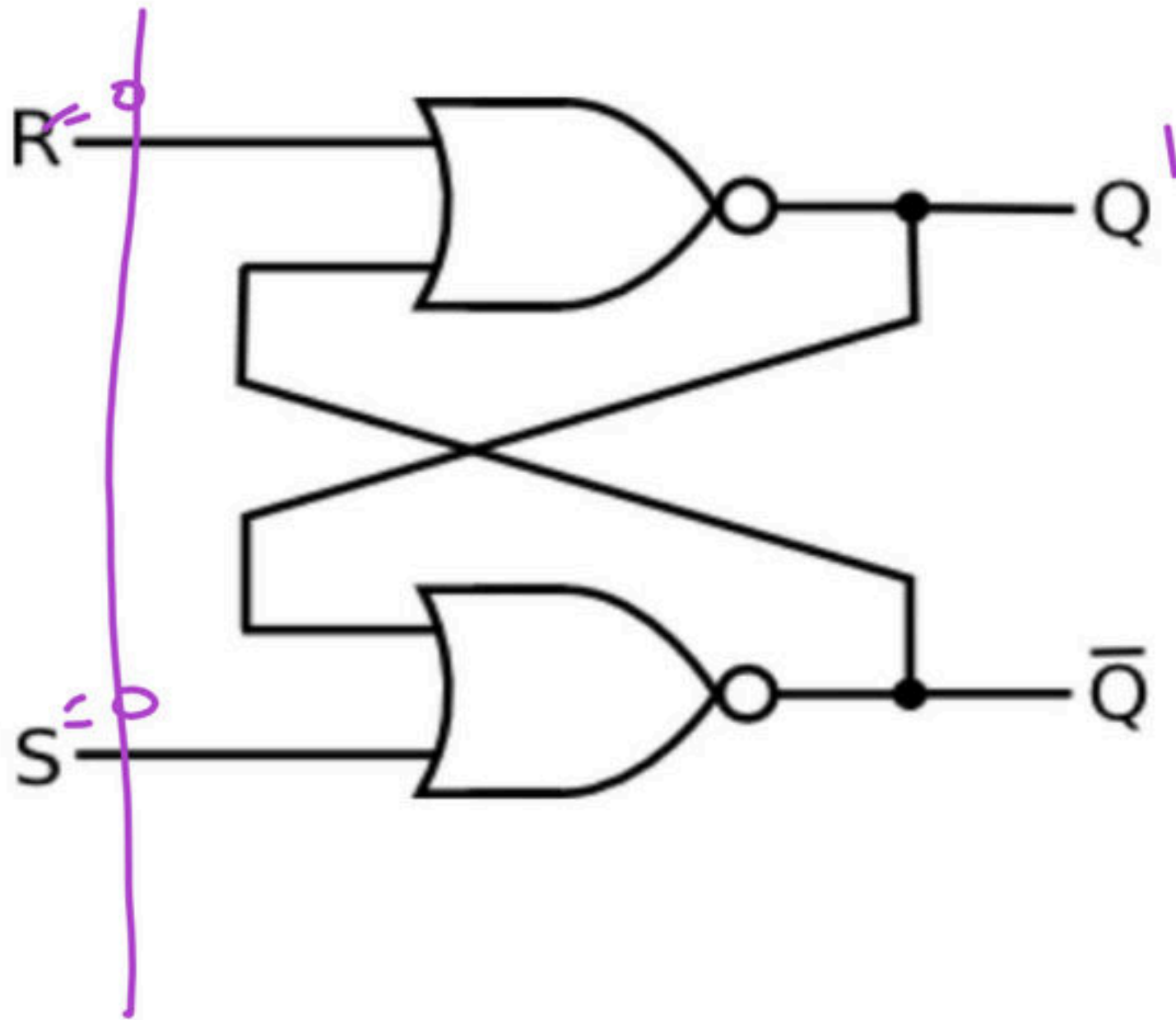
- The SR latch is a circuit with two cross-coupled NOR gates or two cross-coupled NAND gates, and two inputs labeled S for set and R for reset.
- Outputs Q_n and Q_n' are the complement of each other, in valid scenario.





S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	invalid
1	1	1	invalid

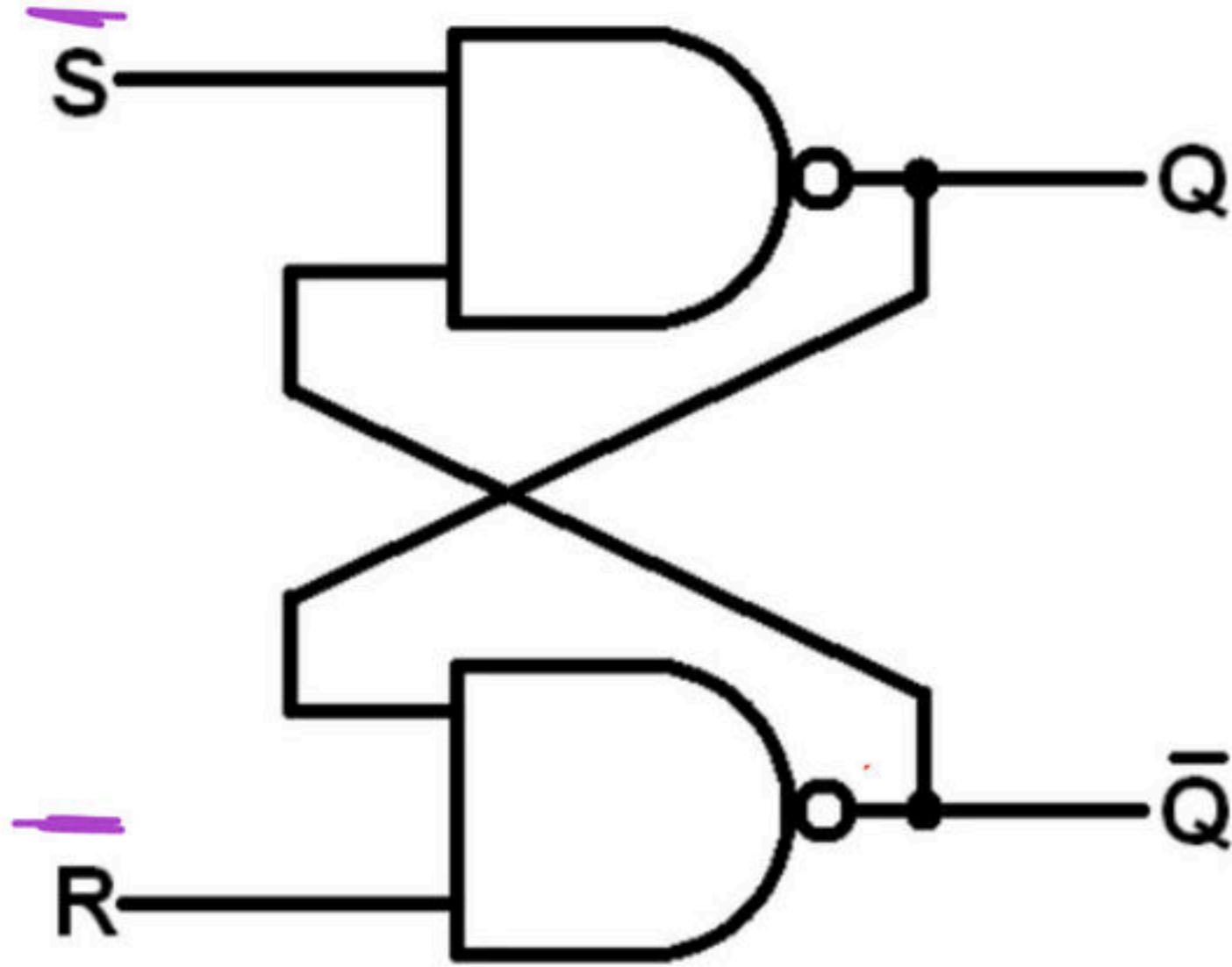




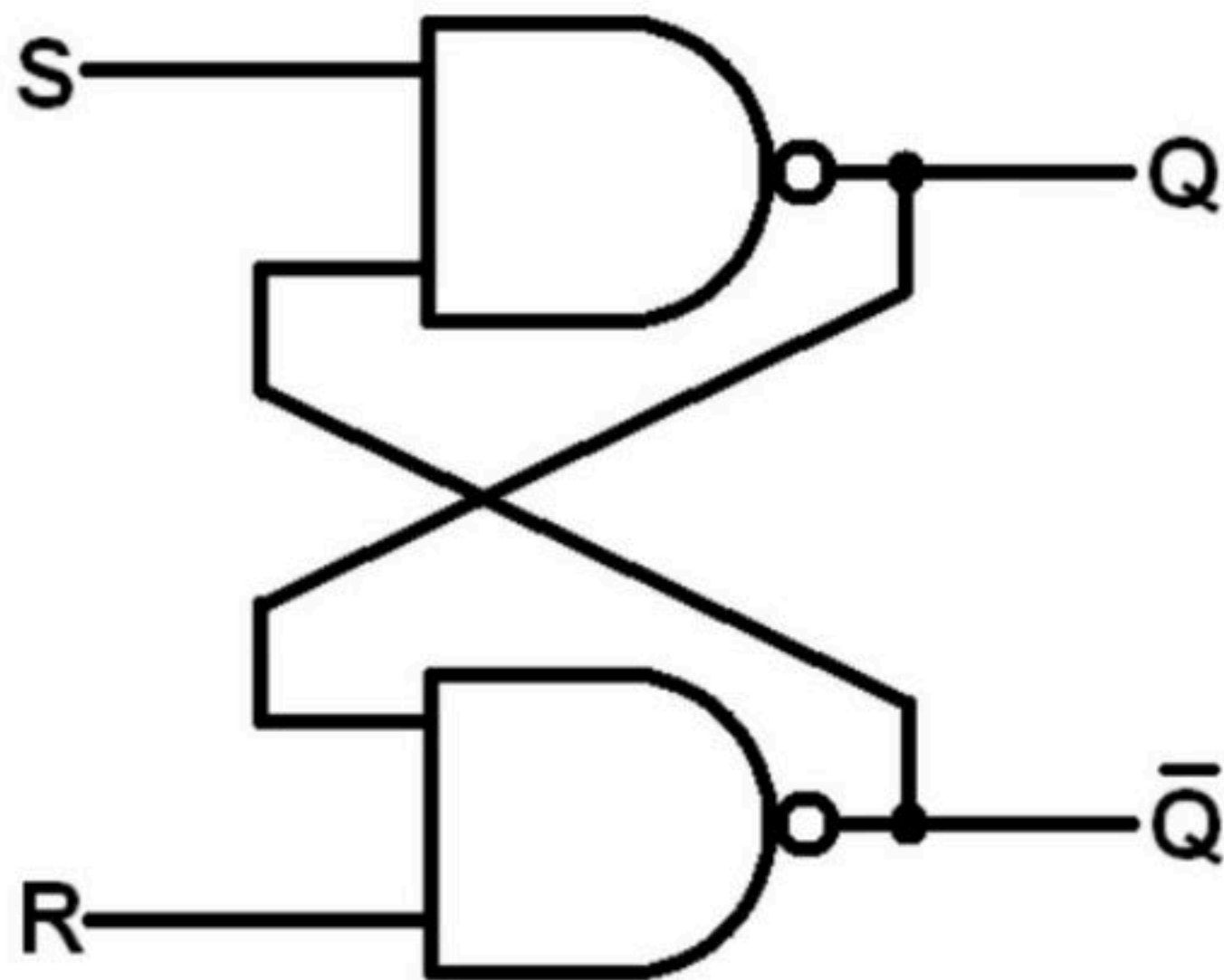
S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

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NAND Latch



S	R	Q_n	Q_{n+1}
0	0	0	invalid
0	0	1	invalid
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



S	R	Q_n	Q_{n+1}
0	0	0	X
0	0	1	X
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

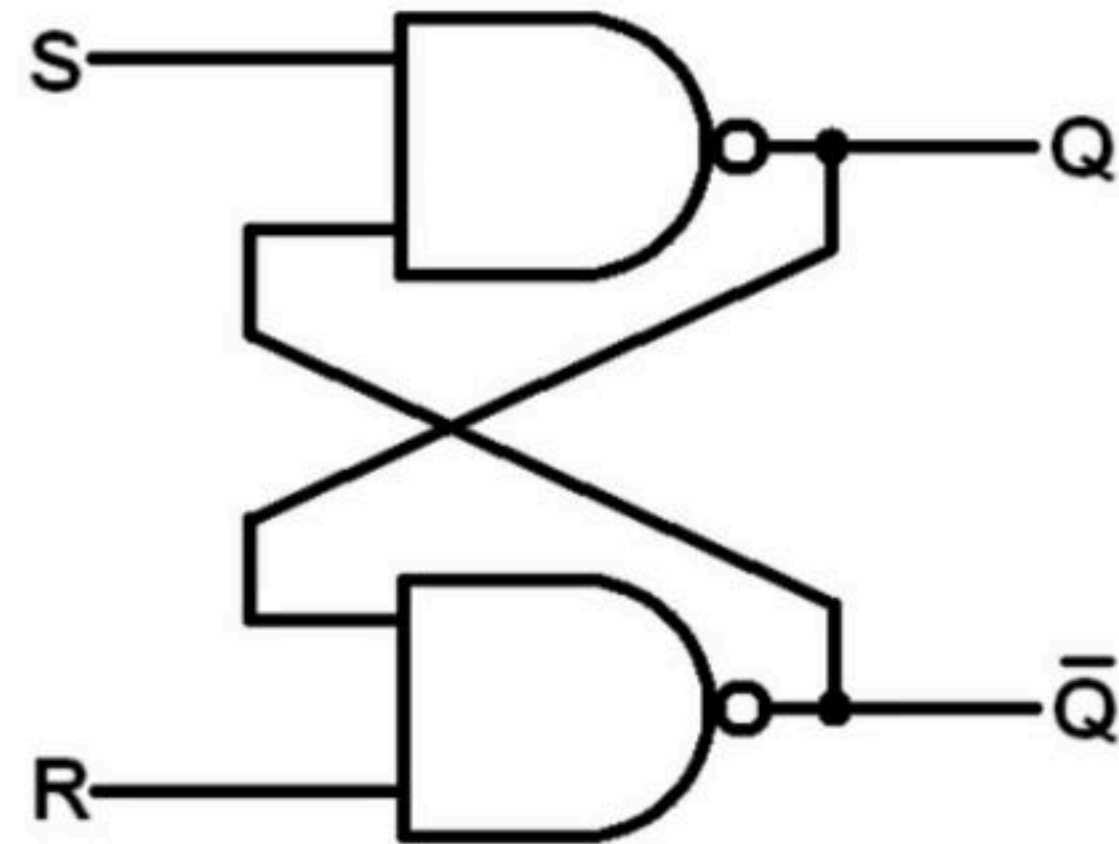
Q In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in **(GATE-2004) (2 Marks)**

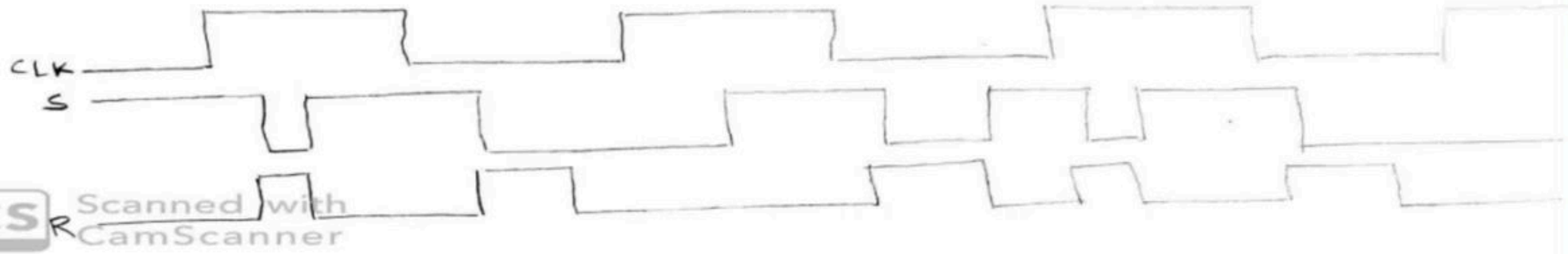
(A) $Q = 0, Q' = 1$

(B) $Q = 1, Q' = 0$

(C) $Q = 1, Q' = 1$

(D) Indeterminate states



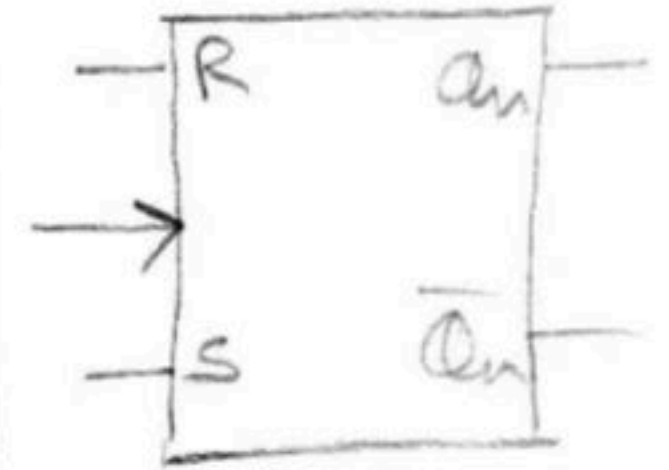
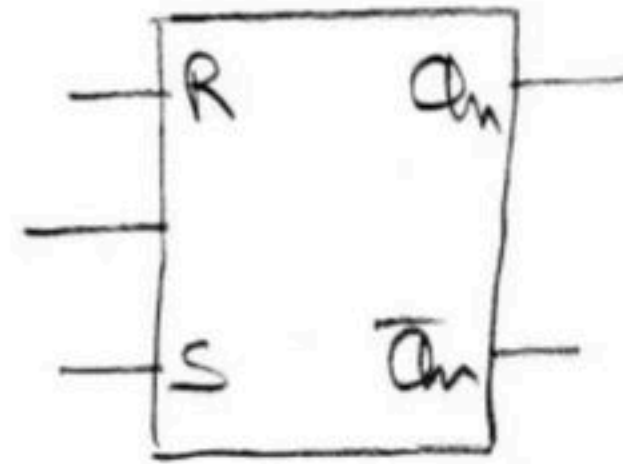


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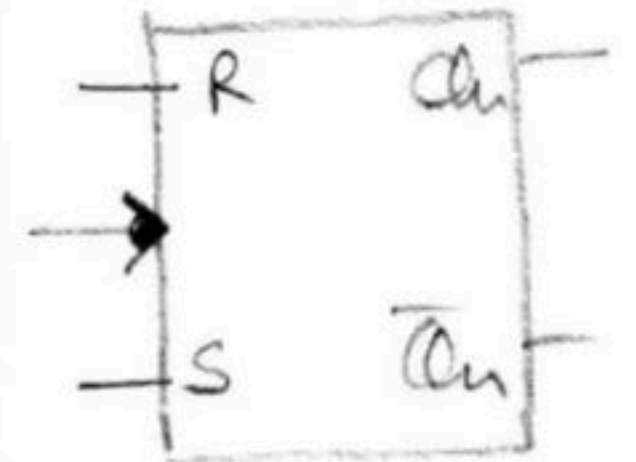
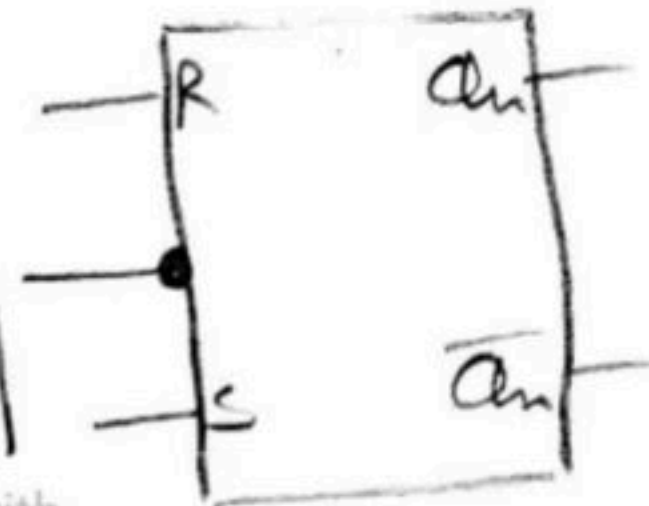
Order of Priority

- ve edge
- +ve edge
- ve level
- +ve level

+ve

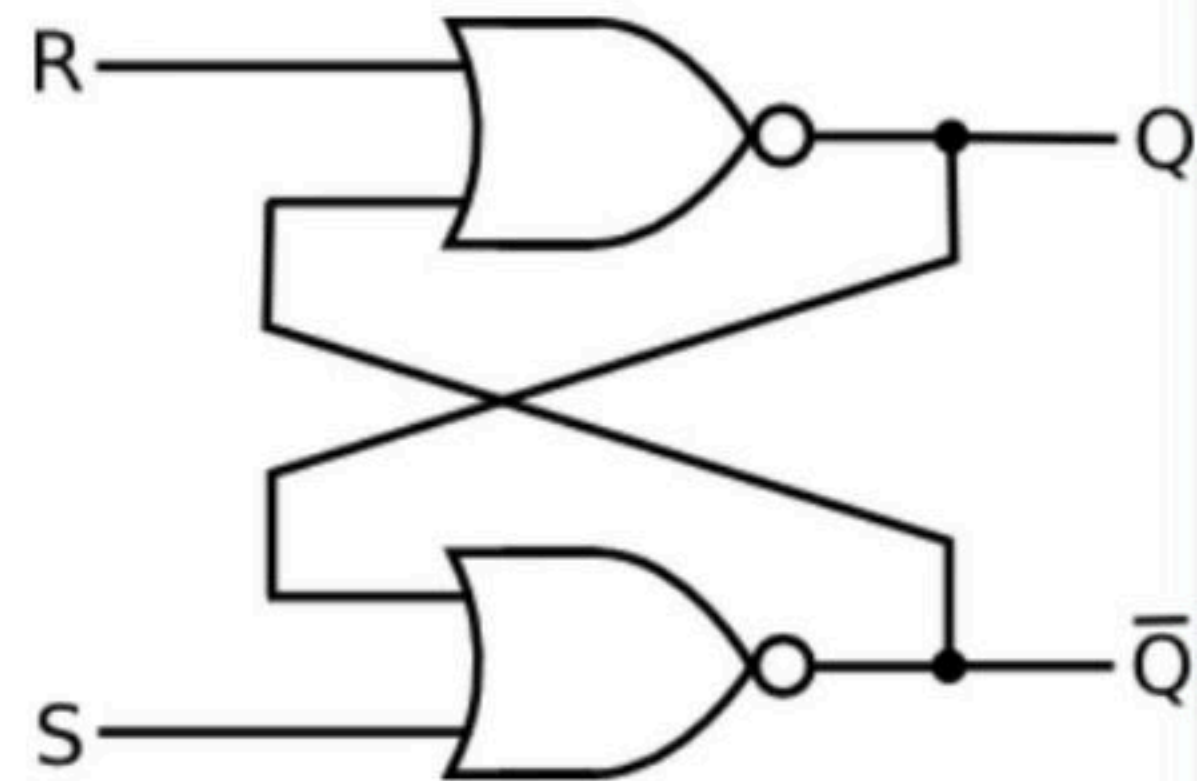


-ve



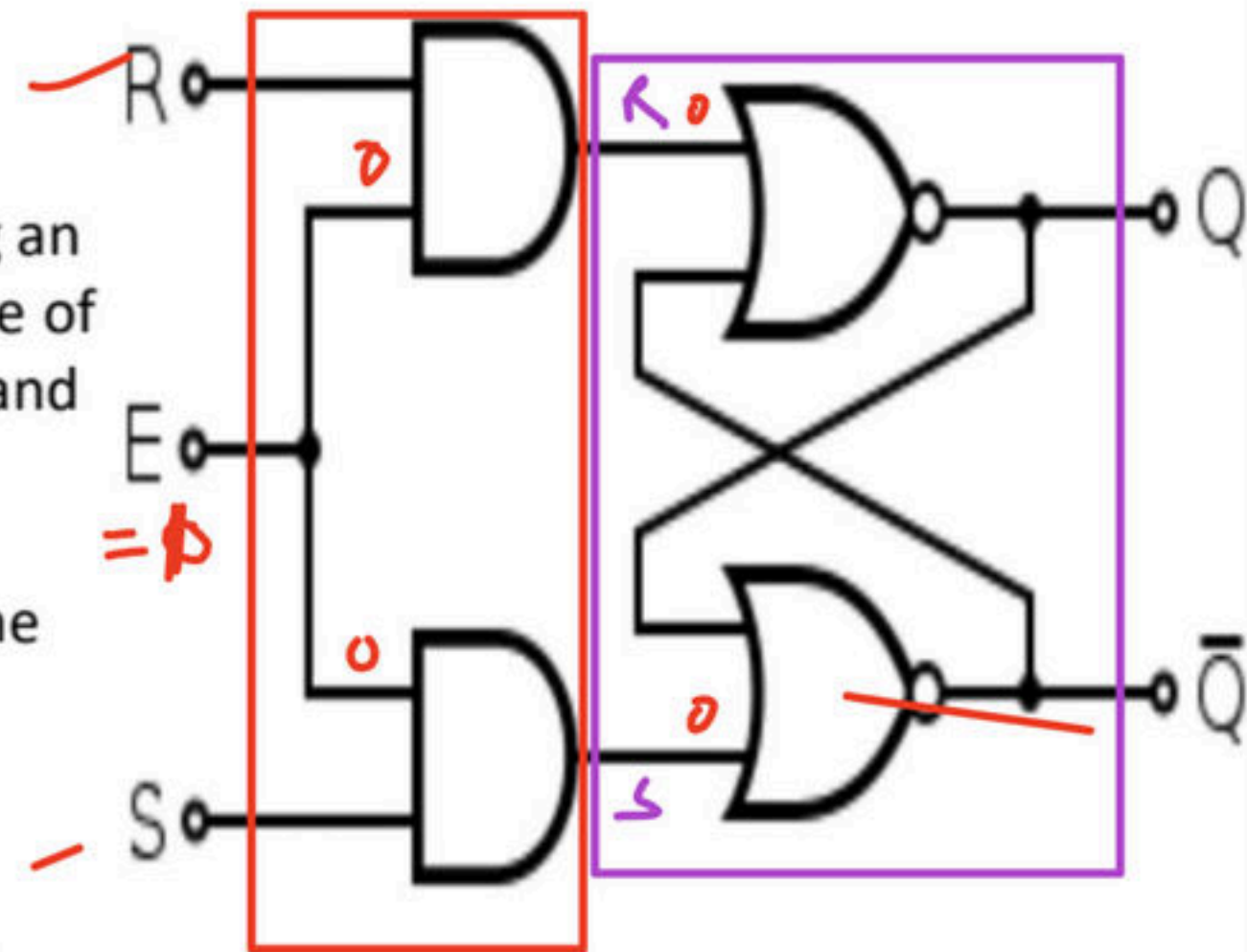
- Edge triggering is always preferred over level triggering because in edge triggering system is in transparent mode for very less amount of time, so system will be stable.

1. The key to the proper operation of a flip-flop is to trigger it ***only during a signal transition(edge)***.
2. This can be accomplished by eliminating the feedback path that is inherent in the operation of the sequential circuit using latches.
3. A clock pulse goes through two transitions: from 0 to 1(+ve edge) and the return from 1 to 0 (-ve edge).
4. The way that a latch can be modified to form a flip-flop. To produce a flip-flop that triggers only during a signal transition(edge) (from 0 to 1 or from 1 to 0) of the synchronizing signal (clock) and is disabled during the rest of the clock pulse.



SR flip flop

1. The operation of the basic SR latch can be modified by providing an additional input signal that determines (controls), when the state of the latch can be changed by determining whether S and R (or S and R) can affect the circuit.
2. It consists of the basic SR latch and two additional AND gates. The control input CP / E_n acts as an enable signal for the other two inputs.
3. The outputs of the AND gates stay at the 0 as long as the enable signal remains at 0 as one input of AND gate gets 0 resulting in 0 as output. When the enable input goes to 1, information from the S or R input is allowed to affect the latch.



Transparent mode
 $E = 1$

Latch mode
 $E = 0$

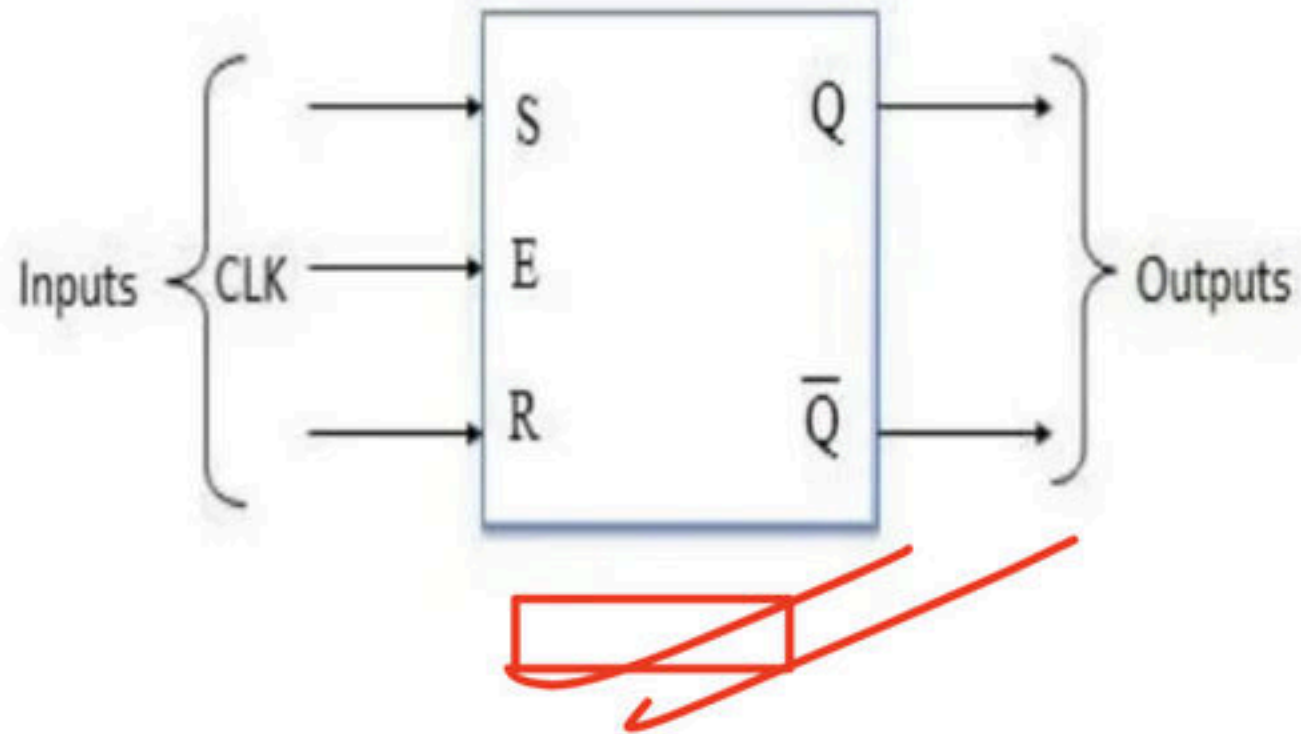
dit

Flip flop

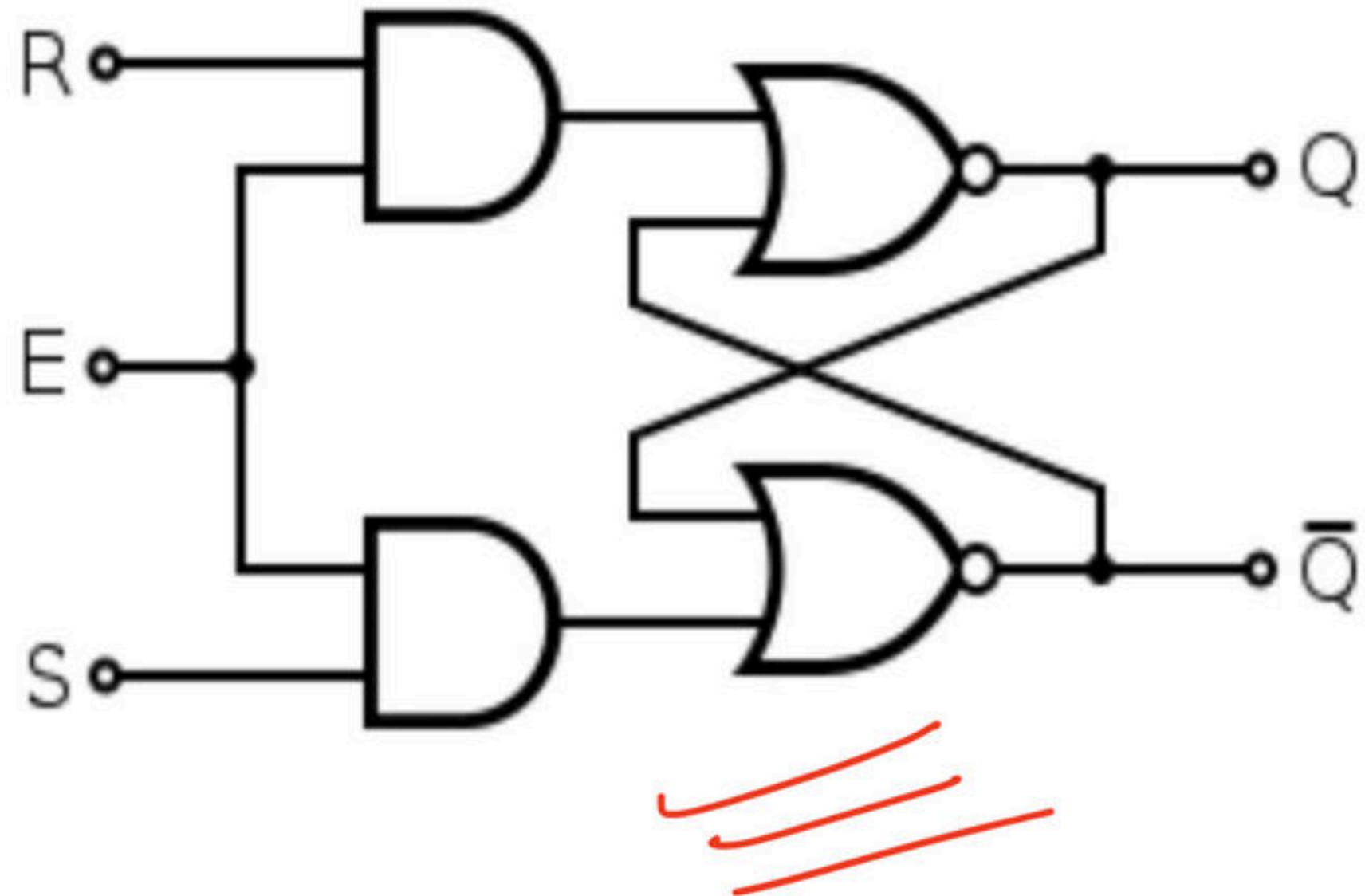
1. Latches are the basic circuits from which all flip-flops are constructed.
2. The storage elements (memory) used in clocked sequential circuits are called flipflops.
3. A flip-flop is a binary storage device capable of storing one bit of information. In a stable state, the output of a flip-flop is either 0 or 1 (it is called bi-stable multi-vibrator).
4. A flip-flop is said to be stable if it has complementary behavior.
5. Latches are level sensitive devices; flip-flops are edge-sensitive devices.
6. Storage elements that operate with signal levels are referred to as latches; those controlled by a clock transition are flip-flops.

- Flip-flops can be either level-triggered (asynchronous, transparent or opaque) or edge-triggered (**synchronous**, or **clocked**).
- The term flip-flop has historically referred generically to both level-triggered and edge-triggered circuits that store a single bit of data using gates.
- Recently, some authors reserve the term *flip-flop* exclusively for discussing clocked circuits;
 - the simple ones are commonly called *transparent latches*. Using this terminology, a level-sensitive flip-flop is called a transparent latch
 - Whereas an edge-triggered flip-flop is simply called a flip-flop.
- Using either terminology, the term "flip-flop" refers to a device that stores a single bit of data. The terms "edge-triggered", and "level-triggered" may be used to avoid ambiguity.

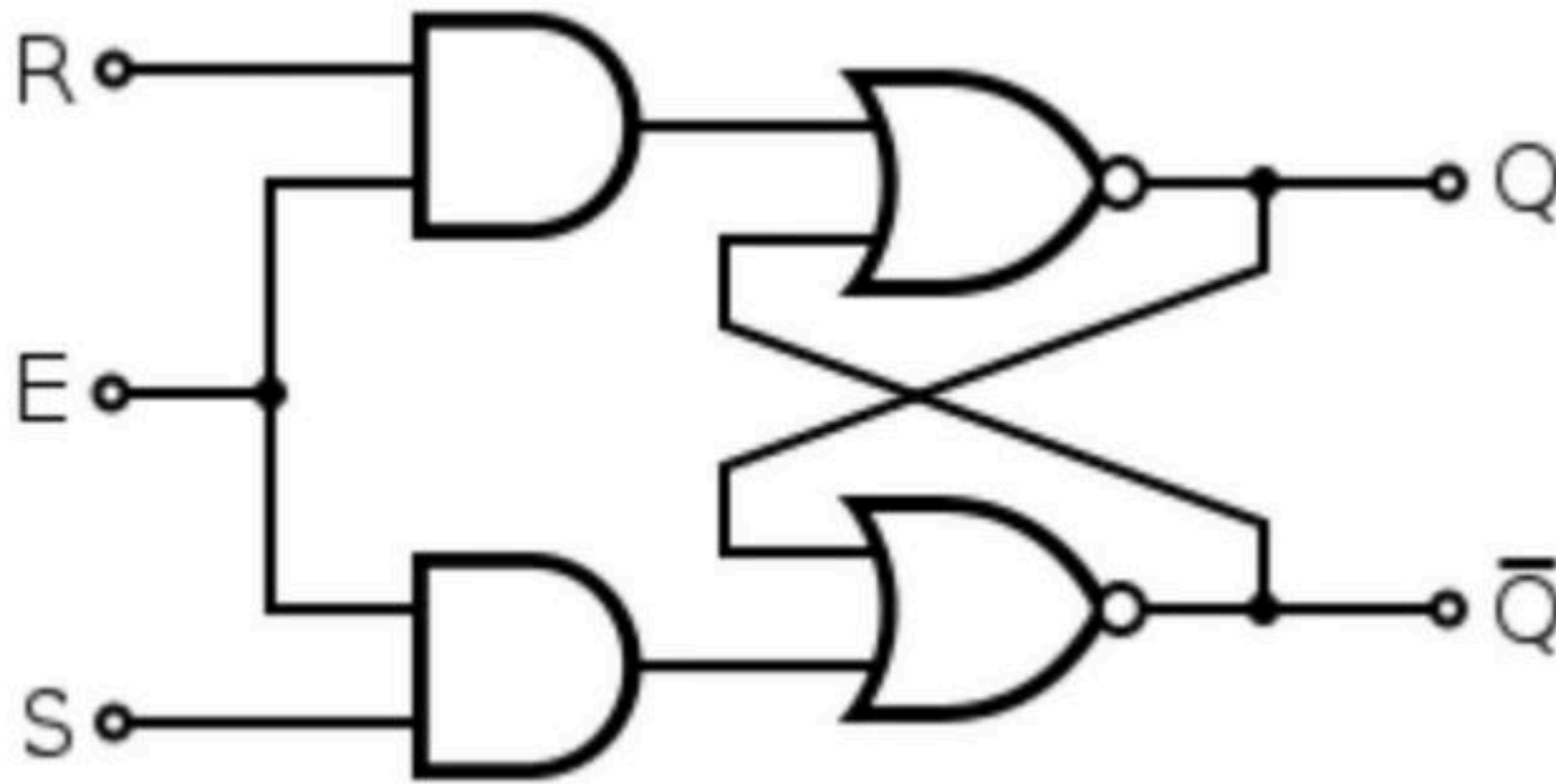
Block Diagram



Implementation



Implementation



Truth Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	d
1	1	1	d

1
4
5
6
7

Truth Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

K-Map

Q_{n+1}

	sr	$s'r'$	$s'r$	sr	sr'
q		00	01	11	10
q'	00			X	1
q	01	1		X	1

Characteristics Equation

$$Q_{n+1} = S + \bar{R}Q_n$$

Truth Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Characteristics Equation

$$Q_{n+1} = S + R' Q_n$$


Function Table

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	indeterminate

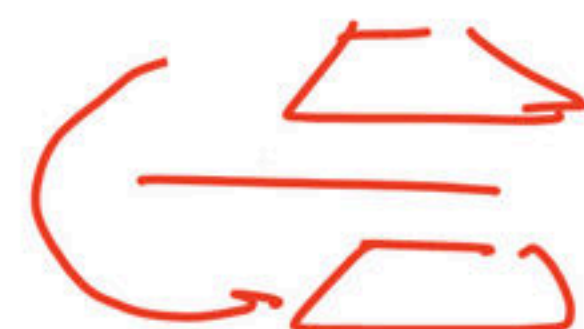
Truth Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Excitation Table



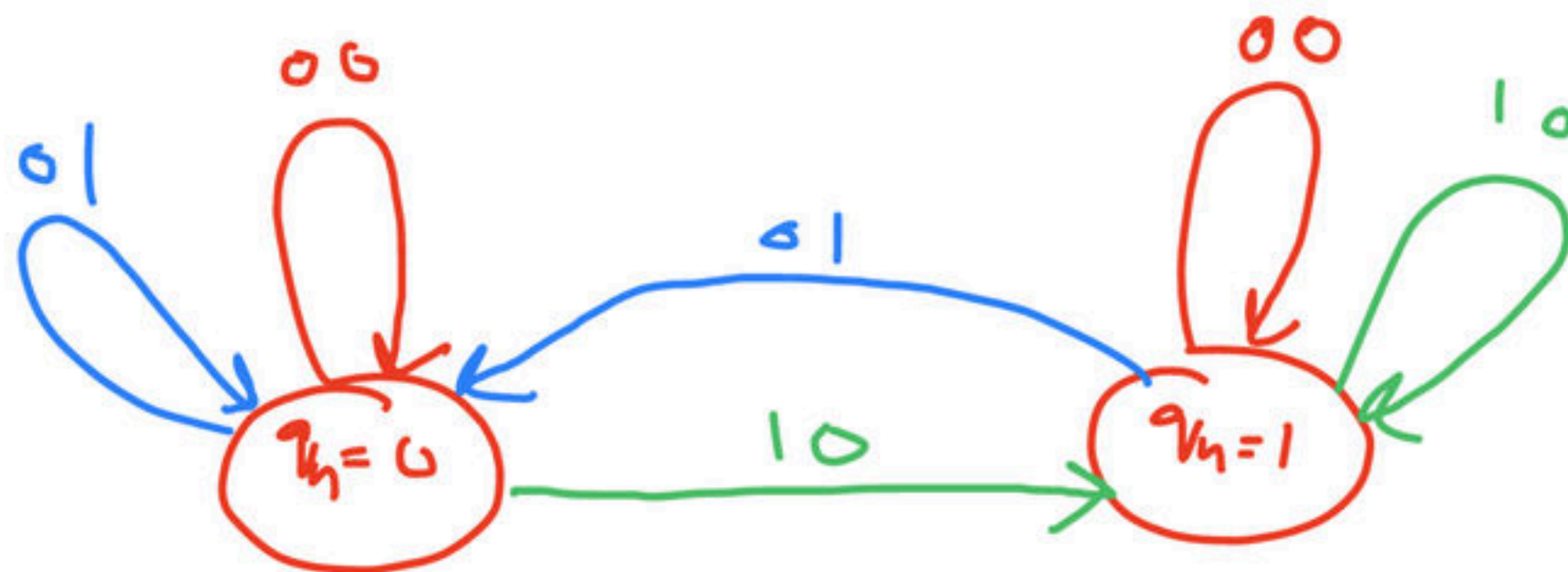
Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0



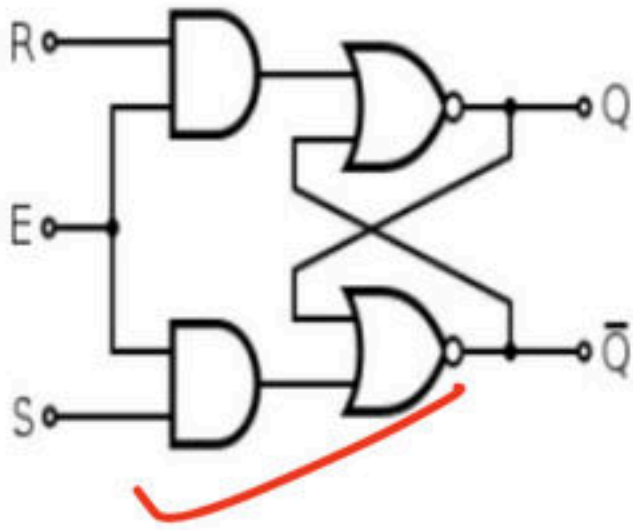
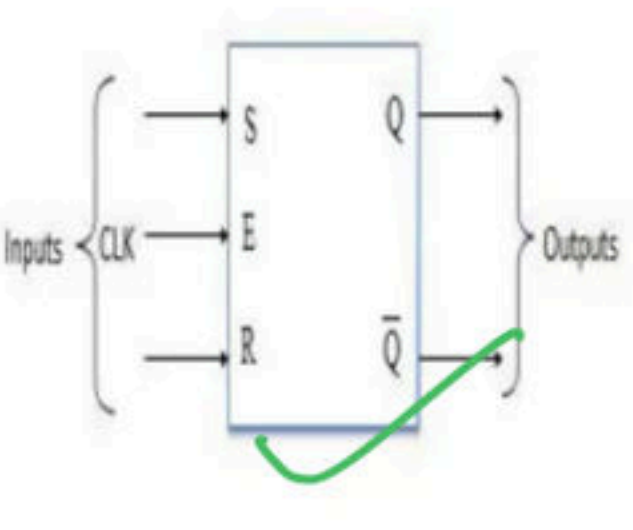
Truth Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

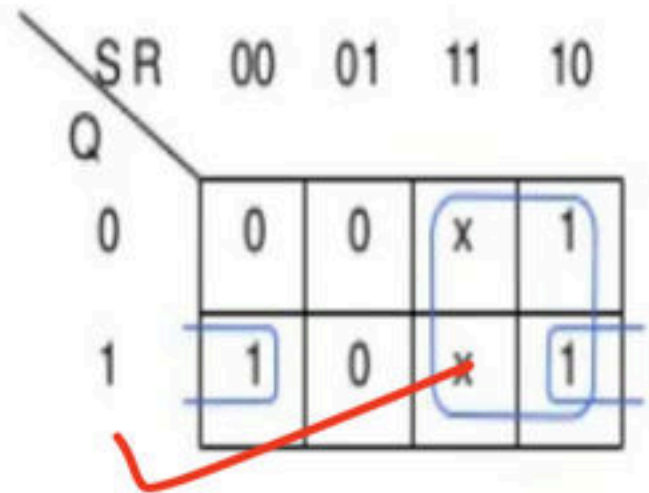
State Diagram



Block Diagram	Implementation	Truth Table	K-Map
---------------	----------------	-------------	-------



S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

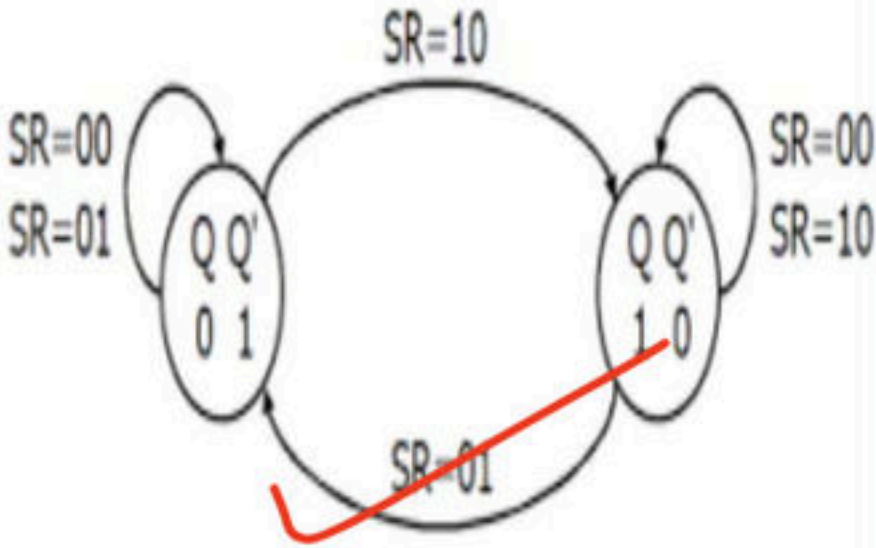


Characteristics Equation	Function Table	Excitation Table	State Diagram
--------------------------	----------------	------------------	---------------

$Q_{n+1} = S + R' Q_n$

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	X

Q_n	Q_{n+1}	S	R
0	0	0	d
0	1	1	0
1	0	0	1
1	1	d	0





Question

from God of Mischief



Omg Where's W? 😲



Xavier Commentar

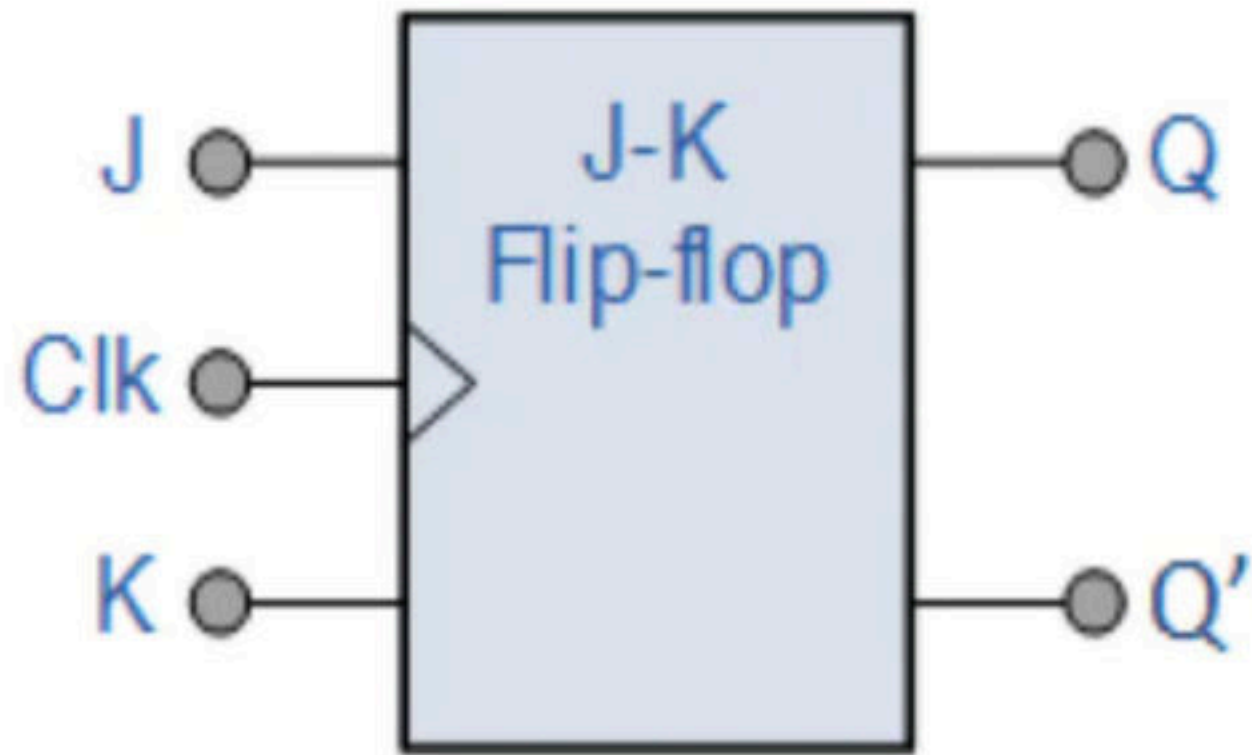
This is why aliens don't visit earth

2-3 pm

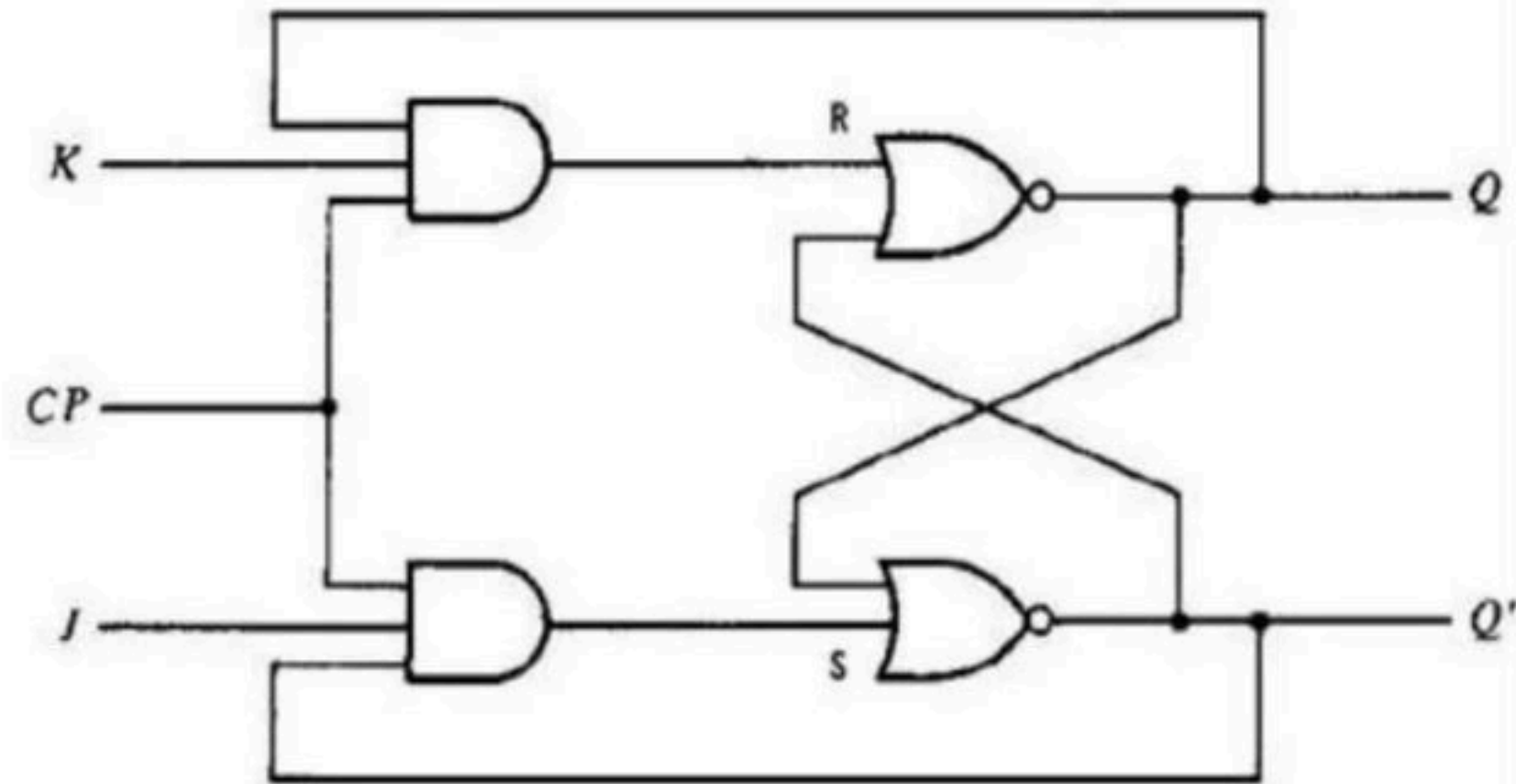
JK flip flop

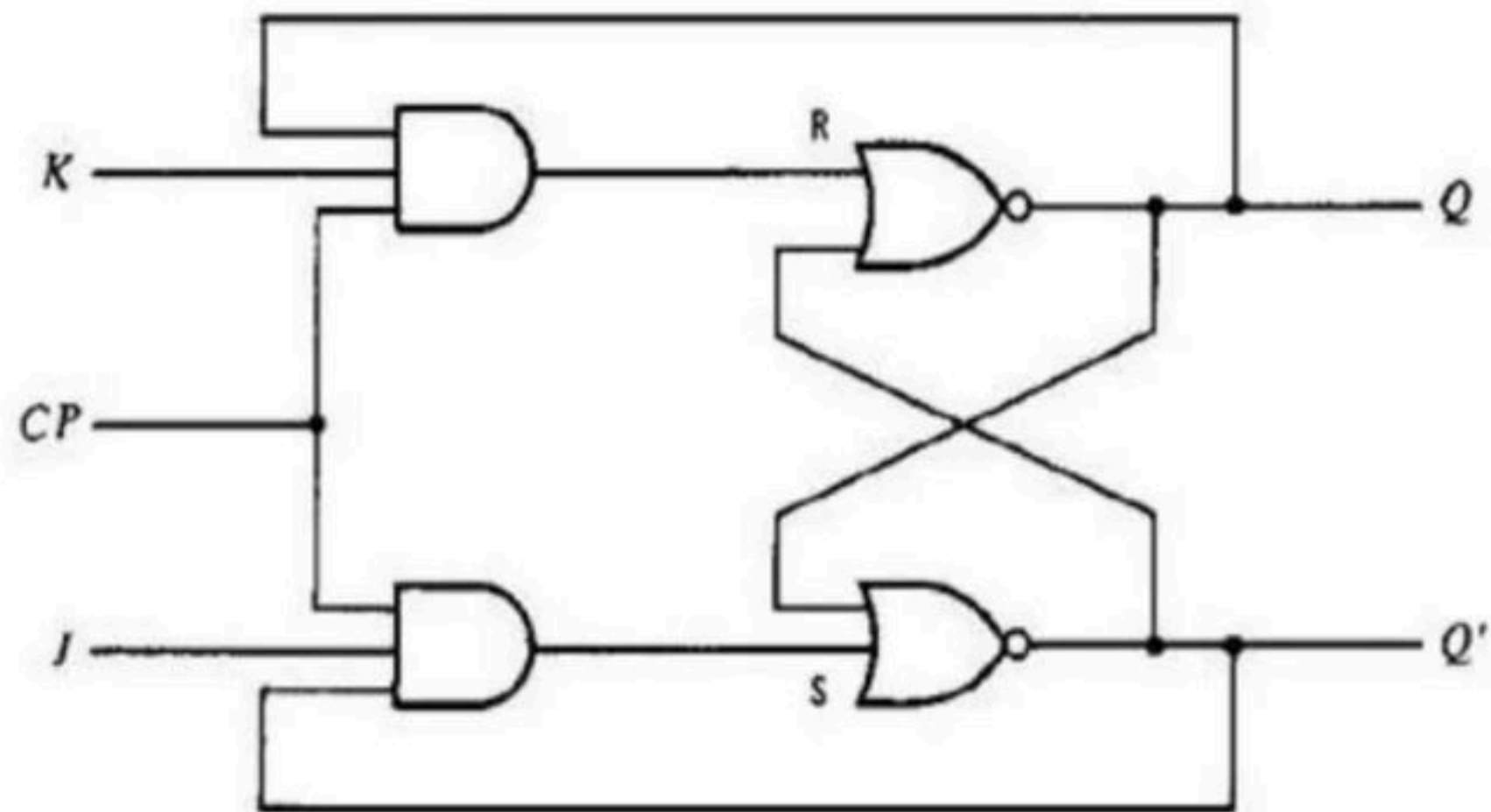
- SR flip flop when both S and R = 1, results in invalid output. To resolve the problem we use JK Flip Flop. We take a feedback from the outputs.

Block Diagram



Implementation





Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

K-Map

	JK	$J'K'$	$J'K$	JK	JK'
q		00	01	11	10
q'	00				
q	01				

Characteristics Equation

$$Q_{n+1} =$$

Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Characteristics Equation

$$Q_{n+1} = J Q'_n + K' Q_n$$

Function Table

J	K	Q_{n+1}
0	0	
0	1	
1	0	
1	1	

Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation Table

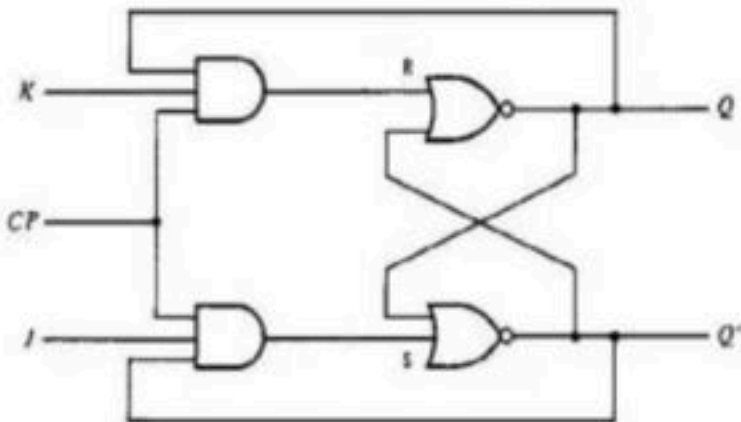
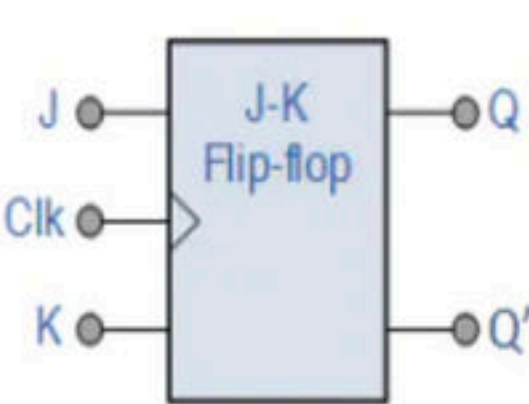
Q_n	Q_{n+1}	J	K
0	0		
0	1		
1	0		
1	1		

Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

State Diagram

Block Diagram	Implementation	Truth Table	K-Map
---------------	----------------	-------------	-------



J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

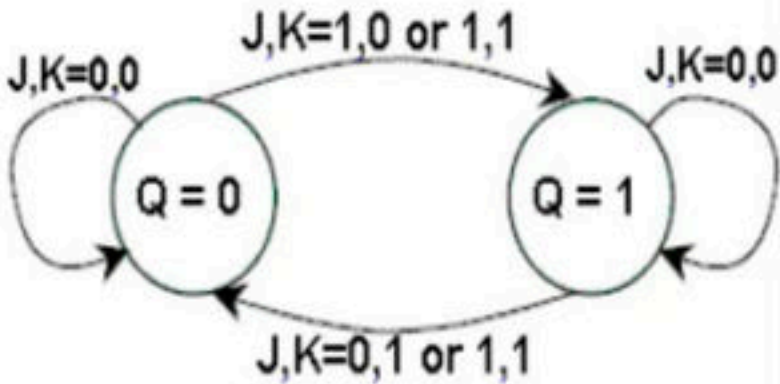
JK					
		$J'K'$	$J'K$	JK	JK'
Q_{n+1}	Q'_n			1	1
	Q_n	1			1

Characteristics Equation	Function Table	Excitation Table	State Diagram
--------------------------	----------------	------------------	---------------

$$Q_{n+1} = J Q'_n + K' Q_n$$

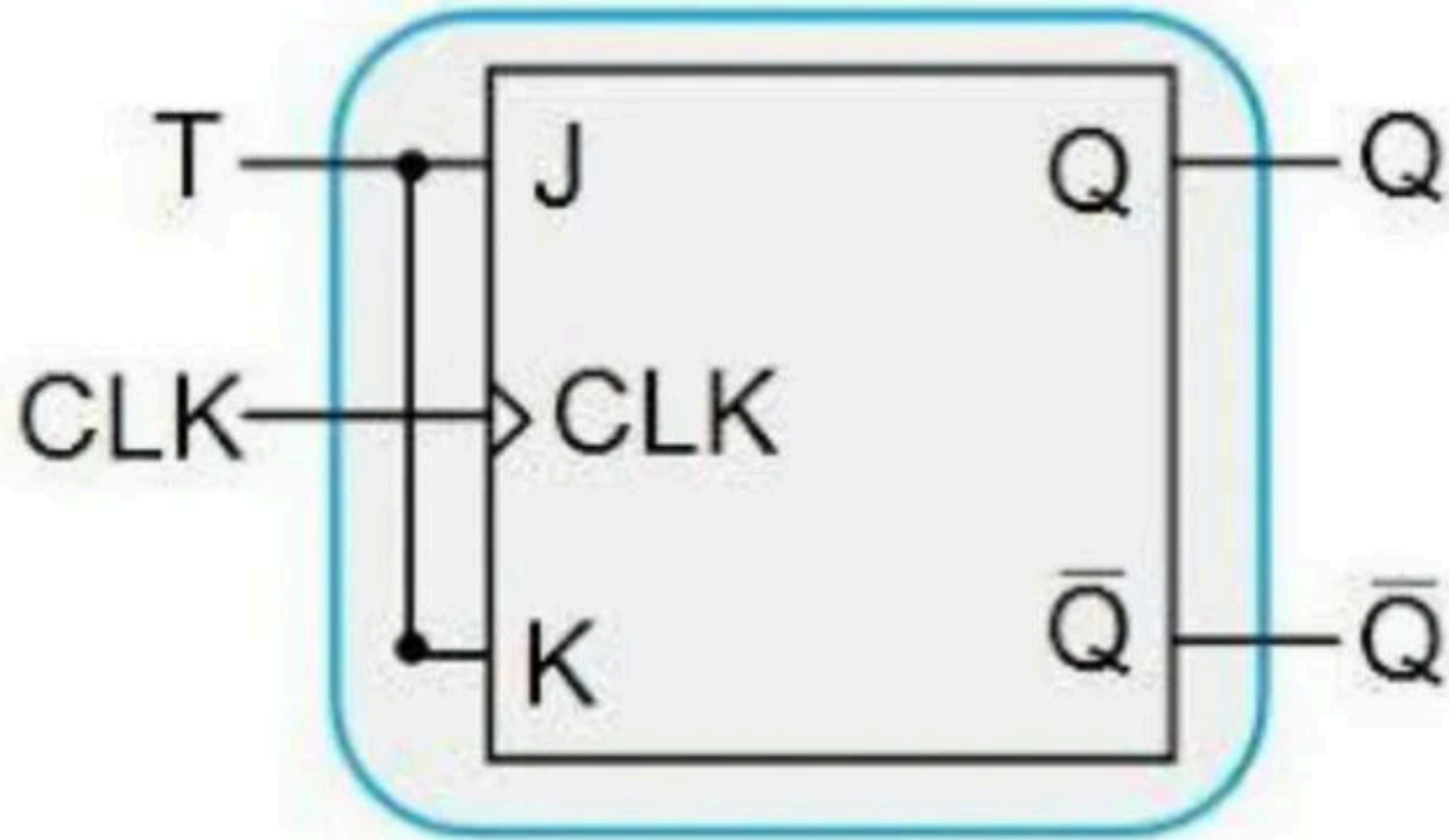
J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Q'_n

Q_n	Q_{n+1}	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0



T(Toggle) flip flop

- The T (toggle) flip-flop is a complementing flip-flop and can be obtained from a JK flip-flop when inputs J and K are tied together.



J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Truth Table

T	Q_n	Q_{n+1}
0	0	
0	1	
1	0	
1	1	

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Truth Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

K-Map

	T	T'	T
q		0	1
q'	0		
q	1		

Characteristics Equation

$$Q_{n+1} =$$

Truth Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Characteristics Equation

$$Q_{n+1} = T \oplus Q_n$$

Function Table

T	Q_{n+1}
0	
1	

Truth Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table

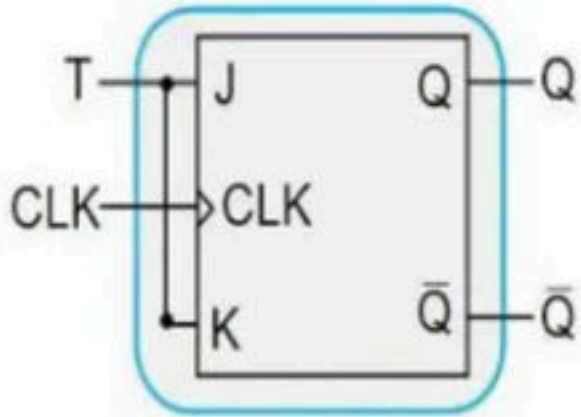
Q_n	Q_{n+1}	T
0	0	
0	1	
1	0	
1	1	

Truth Table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

State Diagram

Block Diagram	Truth Table	K-Map
---------------	-------------	-------



T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

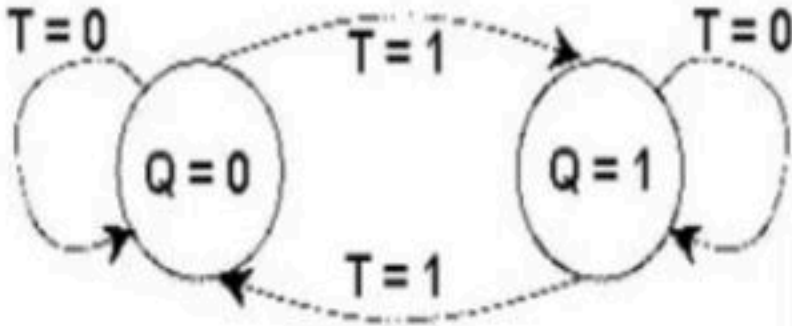
	T	T'	T
q		0	1
q'	0		1
q	1	1	

Characteristics Equation	Function Table	Excitation Table	State Diagram
--------------------------	----------------	------------------	---------------

$$Q_{n+1} = T \oplus Q_n$$

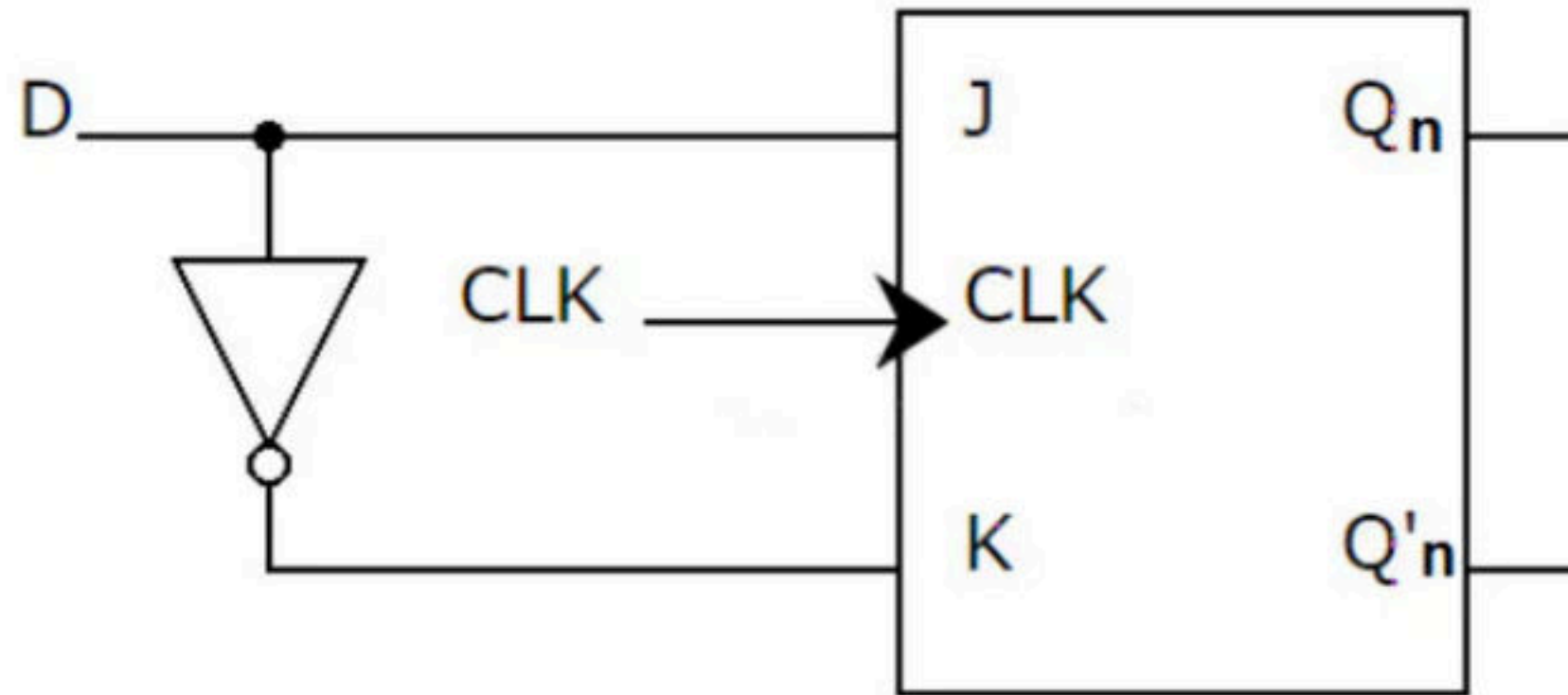
T	Q_{n+1}
0	Q_n
1	Q_n'

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



D flip flop

- The D (Data/Delay) flip-flop, tracks the input at D and produces the same value as output.



Truth Table

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

D	Q_n	Q_{n+1}
0	0	
0	1	
1	0	
1	1	

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Truth Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

K-Map

	D	D'	D
q		0	1
q'	0		
q	1		

Characteristics Equation

$$Q_{n+1} =$$

Truth Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Characteristics Equation

$$Q_{n+1} = D$$

Function Table

D	Q_{n+1}
0	
1	

Truth Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table

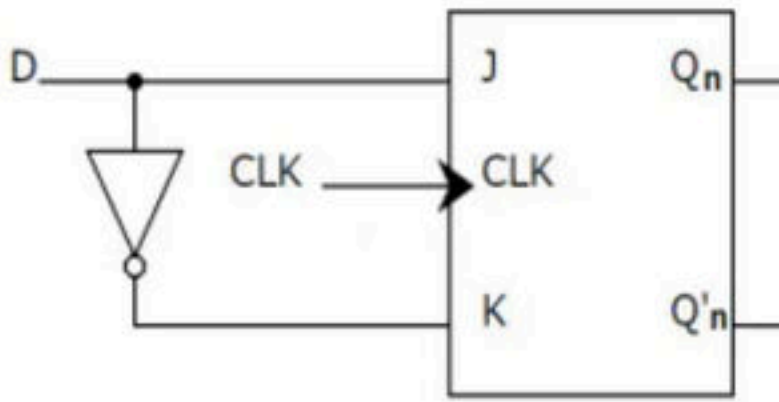
Q_n	Q_{n+1}	D
0	0	
0	1	
1	0	
1	1	

Truth Table

D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

State Diagram

Block Diagram	Truth Table	K-Map
---------------	-------------	-------



D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1

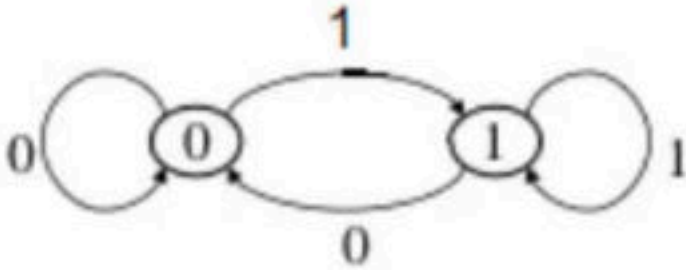
	D	D'	D
q		0	1
q'	0		1
q	1		1

Characteristics Equation	Function Table	Excitation Table	State Diagram
--------------------------	----------------	------------------	---------------

$$Q_{n+1} = D$$

D	Q_{n+1}
0	0
1	1

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1



Flip Flops Conversion

1. We will require the **Characteristics Table** of **target flip flop**.
2. We will require the **Excitation Table** of **given flip flop**.
3. Determine the excitation values for characteristics table.
4. Obtain the expressions for input of given flip flop in terms of target.

Convert D Flip Flop to T Flip Flop

Characteristics table of T- Flip Flop

T	Q_n	Q_{n+1}
0	0	
0	1	
1	0	
1	1	

Excitation Table of D flip flop

Q_n	Q_{n+1}	D
0	0	
0	1	
1	0	
1	1	

Characteristics table of T- Flip Flop

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

T	Q_n	Q_{n+1}	D
0	0	0	
0	1	1	
1	0	1	
1	1	0	

Excitation Table of D flip flop

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Convert D Flip Flop to T Flip Flop

Excitation values for characteristics table

T	Q_n	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

Obtaining simplified expression

T	Q_n	D
0	0	
0	1	
1	0	
1	1	

Obtaining simplified expression

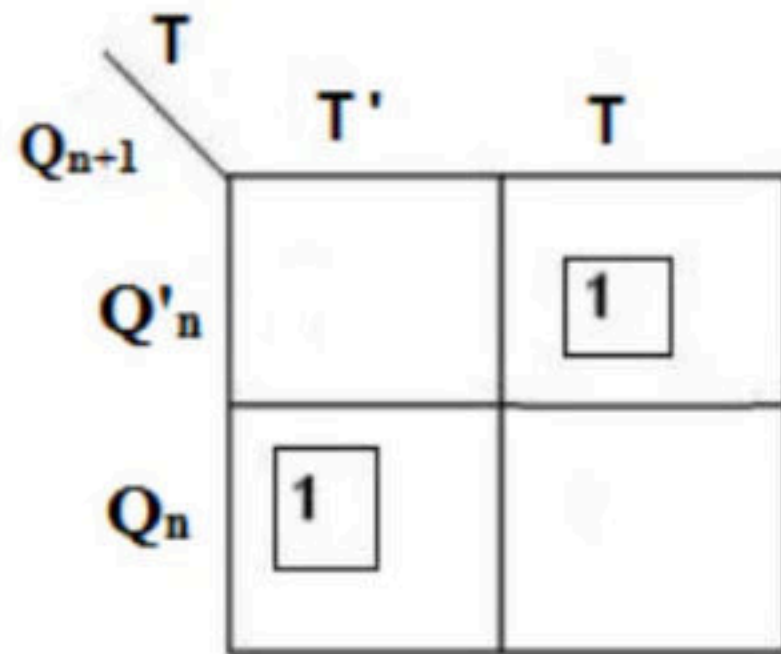
T	Q_n	D
0	0	0
0	1	1
1	0	1
1	1	0

Using K-Map to Simplify

	T	T'	T
q		0	1
q'	0		
q	1		

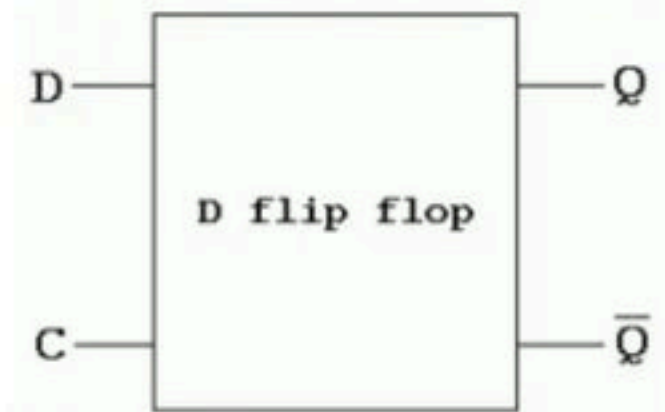
Convert D Flip Flop to T Flip Flop

Using K-Map to Simplify



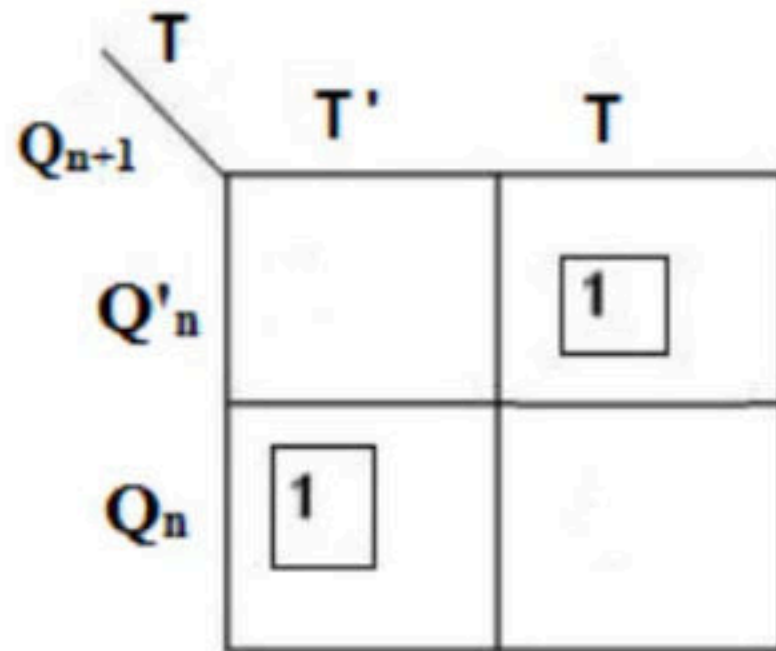
$$D = Q_n \oplus T$$

Block Diagram

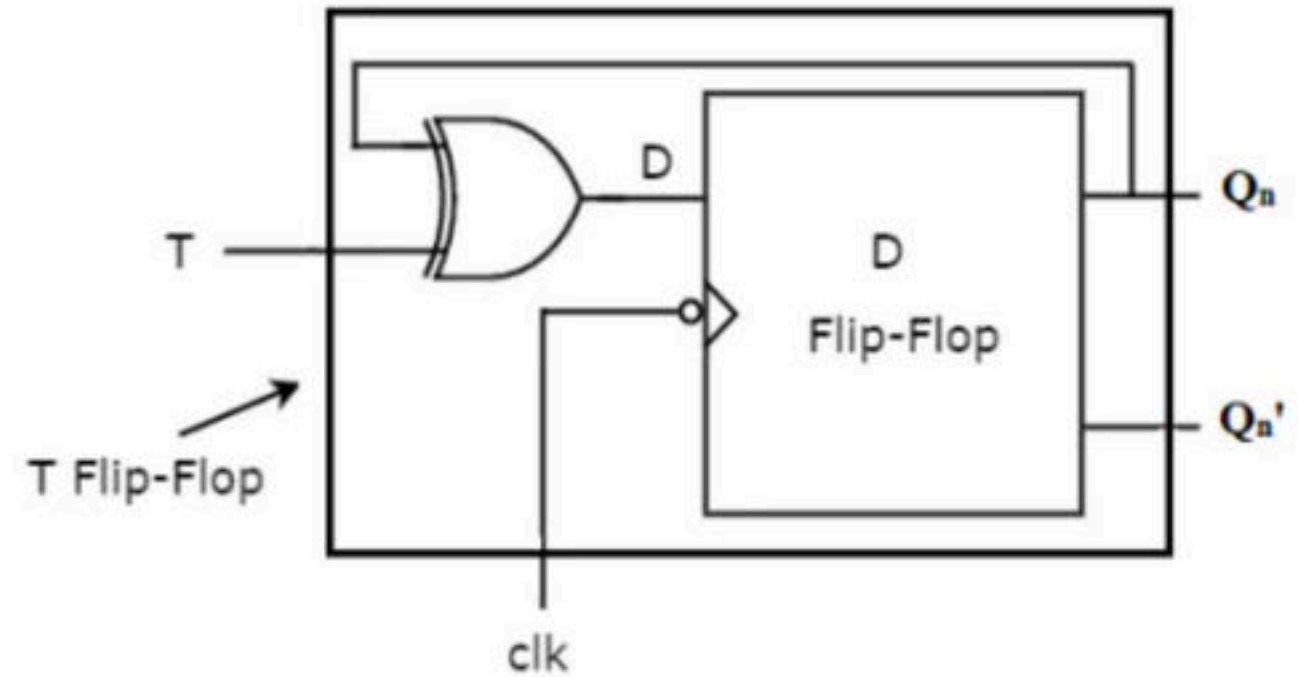


Convert D Flip Flop to T Flip Flop

Using K-Map to Simplify



Block Diagram



$$D = Q_n \oplus T$$

Flip Flops Conversion

1. We will require the **Characteristics Table** of **target flip flop**.
2. We will require the **Excitation Table** of **given flip flop**.
3. Determine the excitation values for characteristics table.
4. Obtain the expressions for input of given flip flop in terms of target.

Convert T flip flop to JK flip flop

Characteristics table of JK- Flip Flop

J	K	Q_n	Q_{n+1}
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Excitation Table of T flip flop

Q_n	Q_{n+1}	T
0	0	
0	1	
1	0	
1	1	

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J	K	Q_n	Q_{n+1}	T
0	0	0	0	
0	0	1	1	
0	1	0	0	
0	1	1	0	
1	0	0	1	
1	0	1	1	
1	1	0	1	
1	1	1	0	

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

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Convert D Flip Flop to T Flip Flop

Excitation values for characteristics table

J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Obtaining simplified expression

J	K	Q_n	T
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Convert D Flip Flop to T Flip Flop

Using K-Map to Simplify

J	K	Q_n	T
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

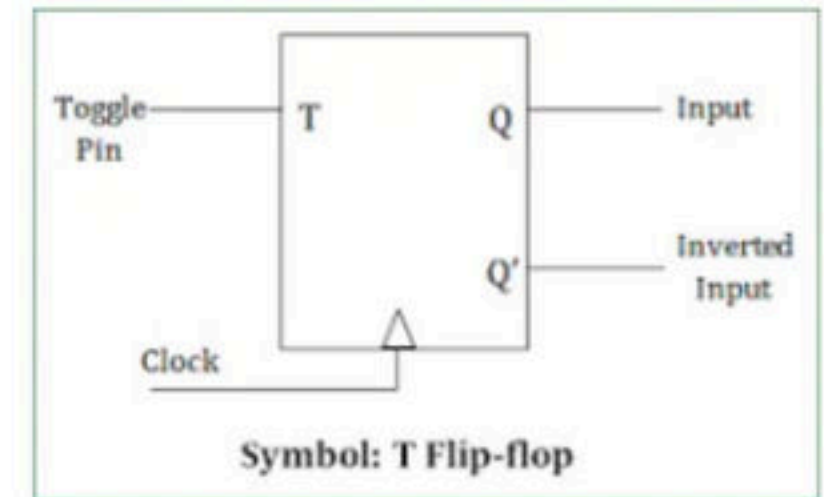
	JK	J'K'	J'K	JK	JK'
q		00	01	11	10
q'	00				
q	01				

T =

Convert D Flip Flop to T Flip Flop

T =

Block Diagram



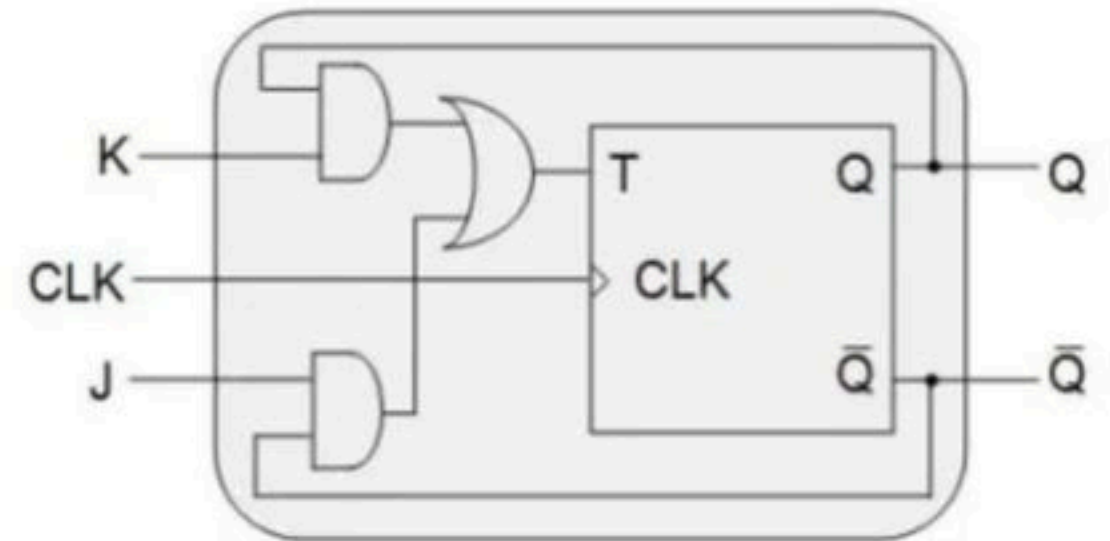
Convert D Flip Flop to T Flip Flop

Using K-Map to Simplify

T				
	J'K'	J'K	JK	JK'
Q _n '			1	1
Q _n		1	1	

$$T = J Q_n' + K Q_n$$

Block Diagram



Flip Flops Conversion

1. We will require the **Characteristics Table** of **target flip flop**.
2. We will require the **Excitation Table** of **given flip flop**.
3. Determine the excitation values for characteristics table.
4. Obtain the expressions for input of given flip flop in terms of target.

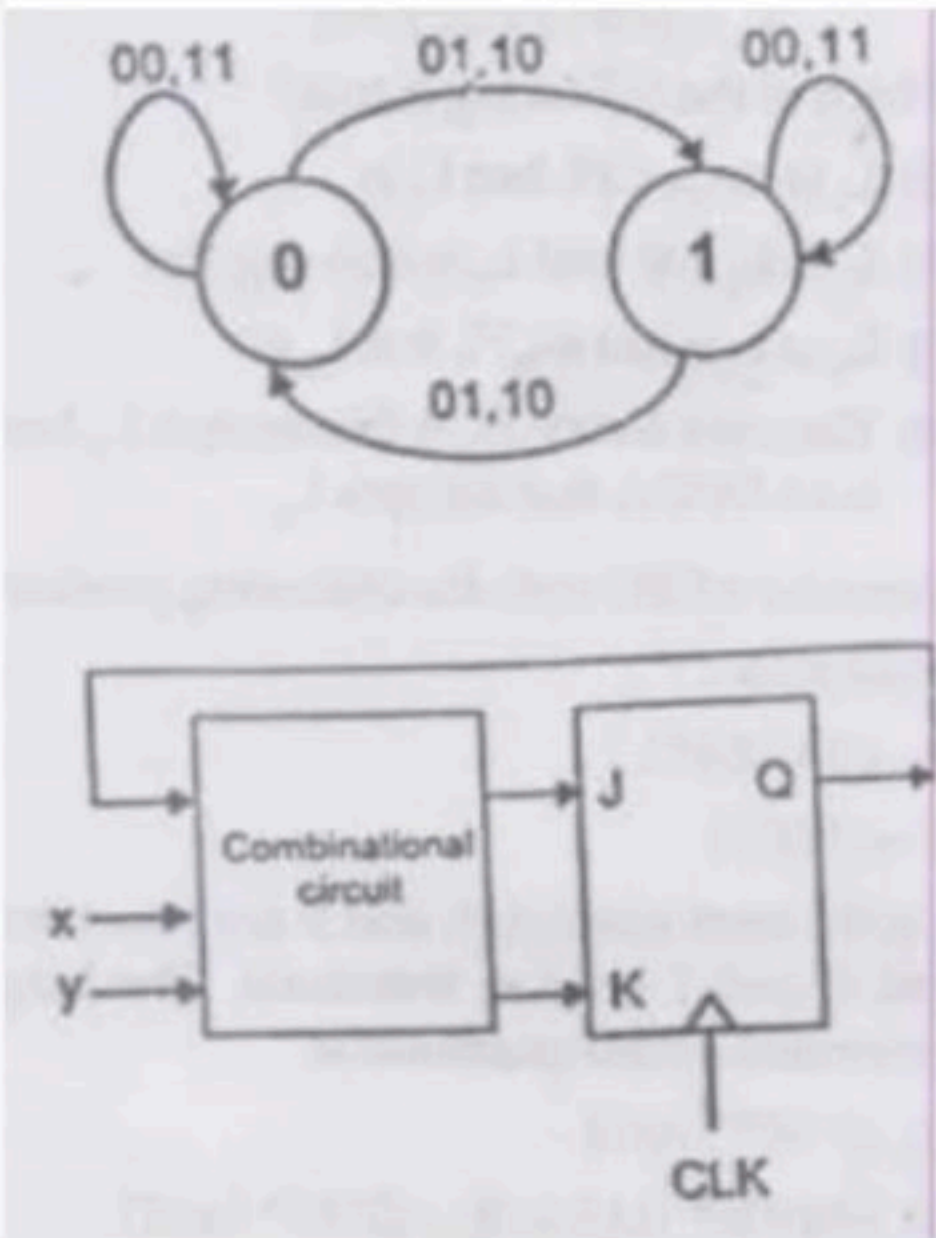
Q Consider the following state diagram and its realization by a JK flip flop, the combinational circuit generates J and K in terms of x, y and Q. The Boolean expressions for J and K are : **(GATE-2008) (2 Marks)**

(A) $(x \oplus y)'$ and $x' \oplus y'$

(C) $x \oplus y$ and $(x \oplus y)'$

(B) $(x \oplus y)'$ and $x \oplus y$

(D) $x \oplus y$ and $x \oplus y$



X	Y	Q_n	Q_{n+1}
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

X	Y	Q_n	Q_{n+1}	J	K
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Q_n	Q_{n+1}	J	K
0	0		
0	1		
1	0		
1	1		

X	Y	Q_n	J	K
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

	JK	J'K'	J'K	JK	JK'
q		00	01	11	10
q'	00				
q	01				

	JK	J'K'	J'K	JK	JK'
q		00	01	11	10
q'	00				
q	01				

Basics of counters

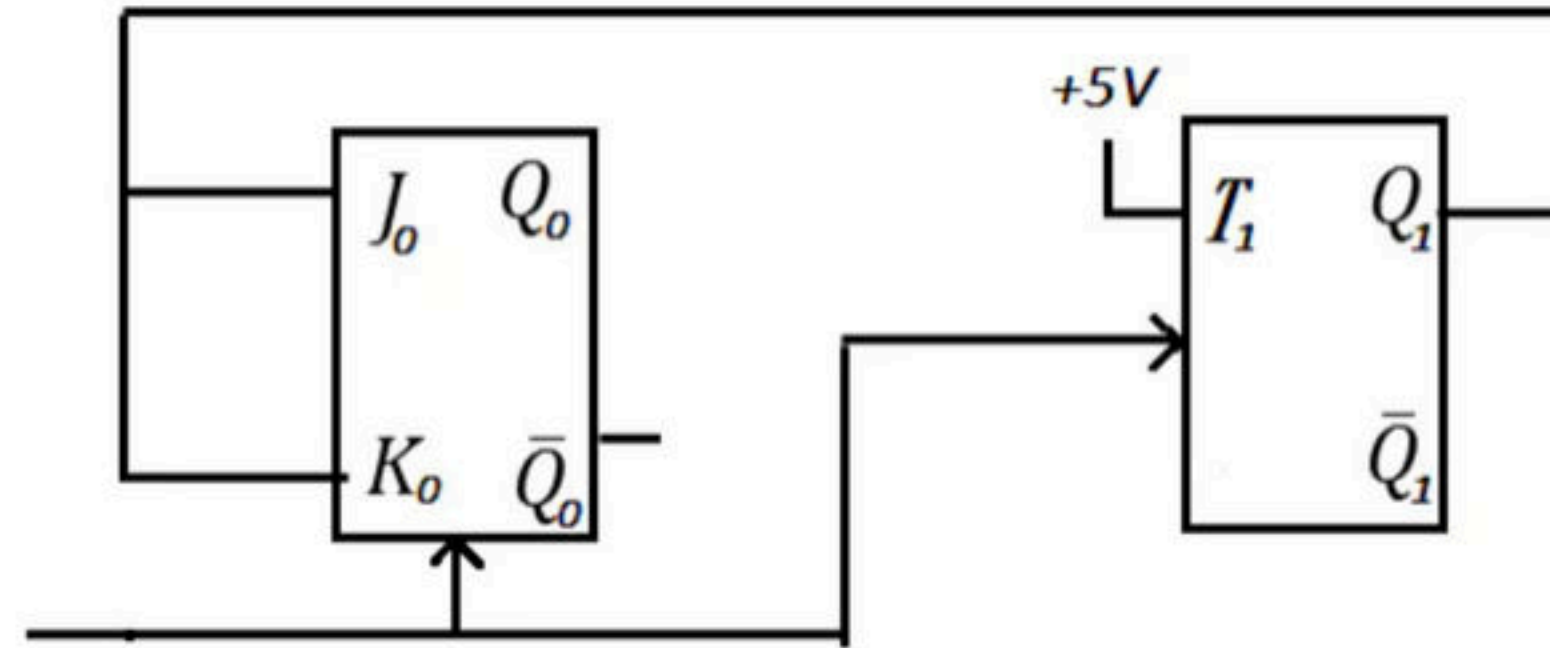
1. A counter is a sequential circuit that goes through a predetermined sequence of binary states upon the application of input pulses.
2. The gates in the counter are connected in such a way as to produce the prescribed sequence of states.
3. A counter that follows the binary number sequence is called a binary counter.
4. An n -bit binary counter consists of n flip-flops and can count in binary from 0 through $2^n - 1$.
5. Counters are available in two categories: **synchronous counters** and **Ripple counters (asynchronous)**.

Q Find out the counting sequence for the counter below?

The characteristics equation for JK Flip Flop is $Q_{n+1} =$
 $Q_{0N} =$

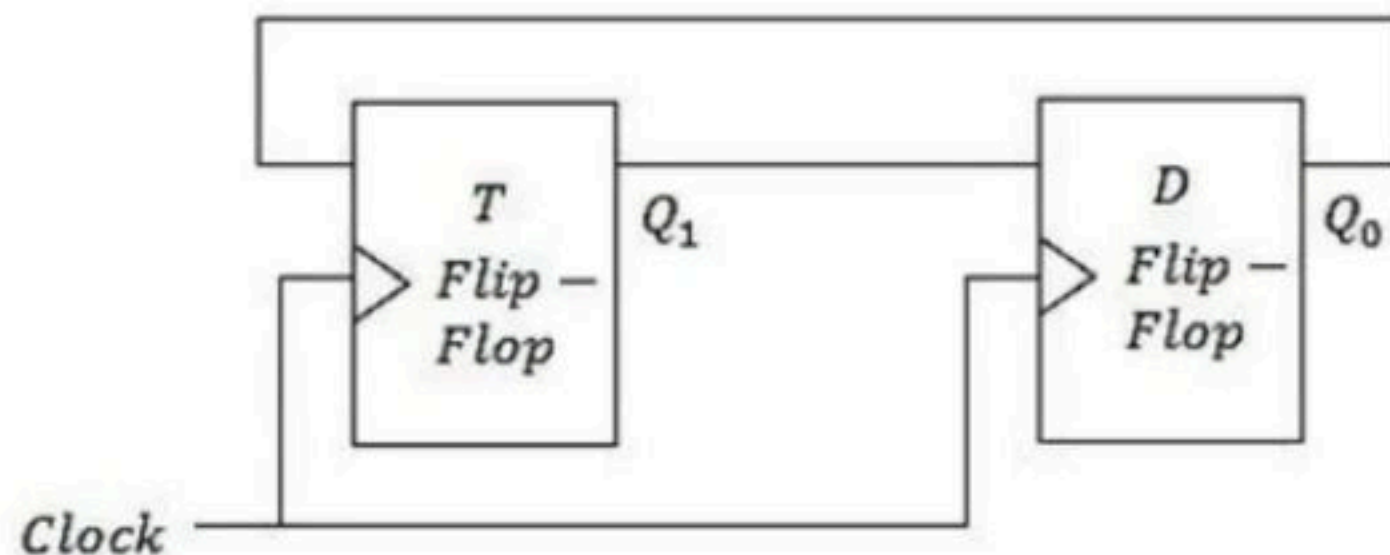
The characteristics equation for T Flip Flop is $Q_{n+1} =$
 $Q_{1N} =$

Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		



Q Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop. Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs **(GATE-2017) (2 Marks)**

- a) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
- b) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively
- c) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 11 respectively
- d) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively



Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

Q The number of flip-flops required to design a modulo - 272 counter is: **(NET-JUNE-2015)**

(a) 8

(b) 9

(c) 27

(d) 11

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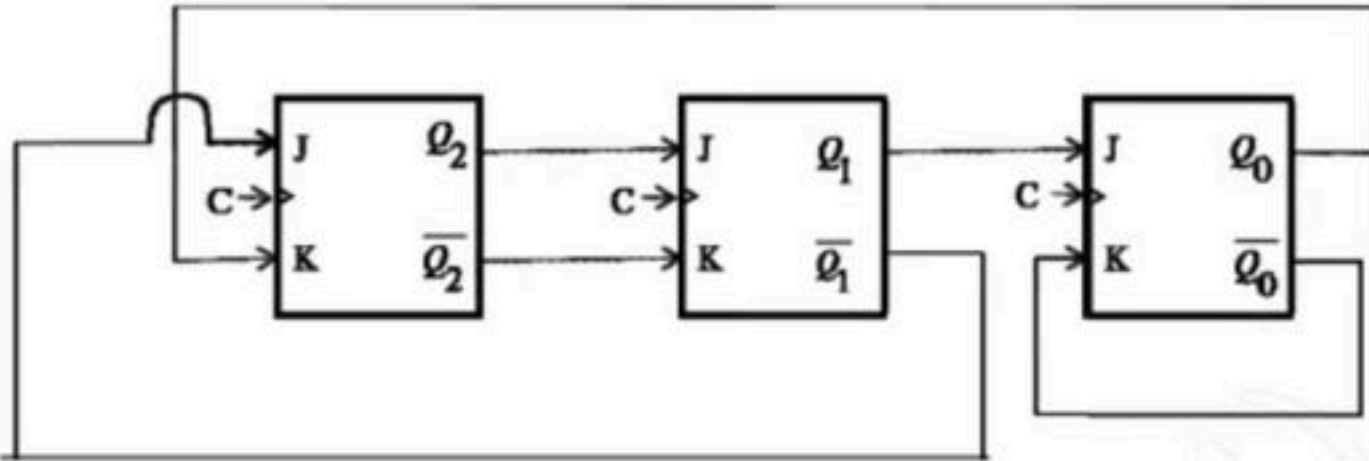
Q The above sequential circuit is built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycle is **(GATE-2014) (2 Marks)**

(A) 001, 010, 011

(B) 111, 110, 101

(C) 100, 110, 111

(D) 100, 011, 001



Present State			Next State		
Q_{2p}	Q_{1p}	Q_{0p}	Q_{2N}	Q_{1N}	Q_{0N}
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(A) 000

(B) 001

(C) 010

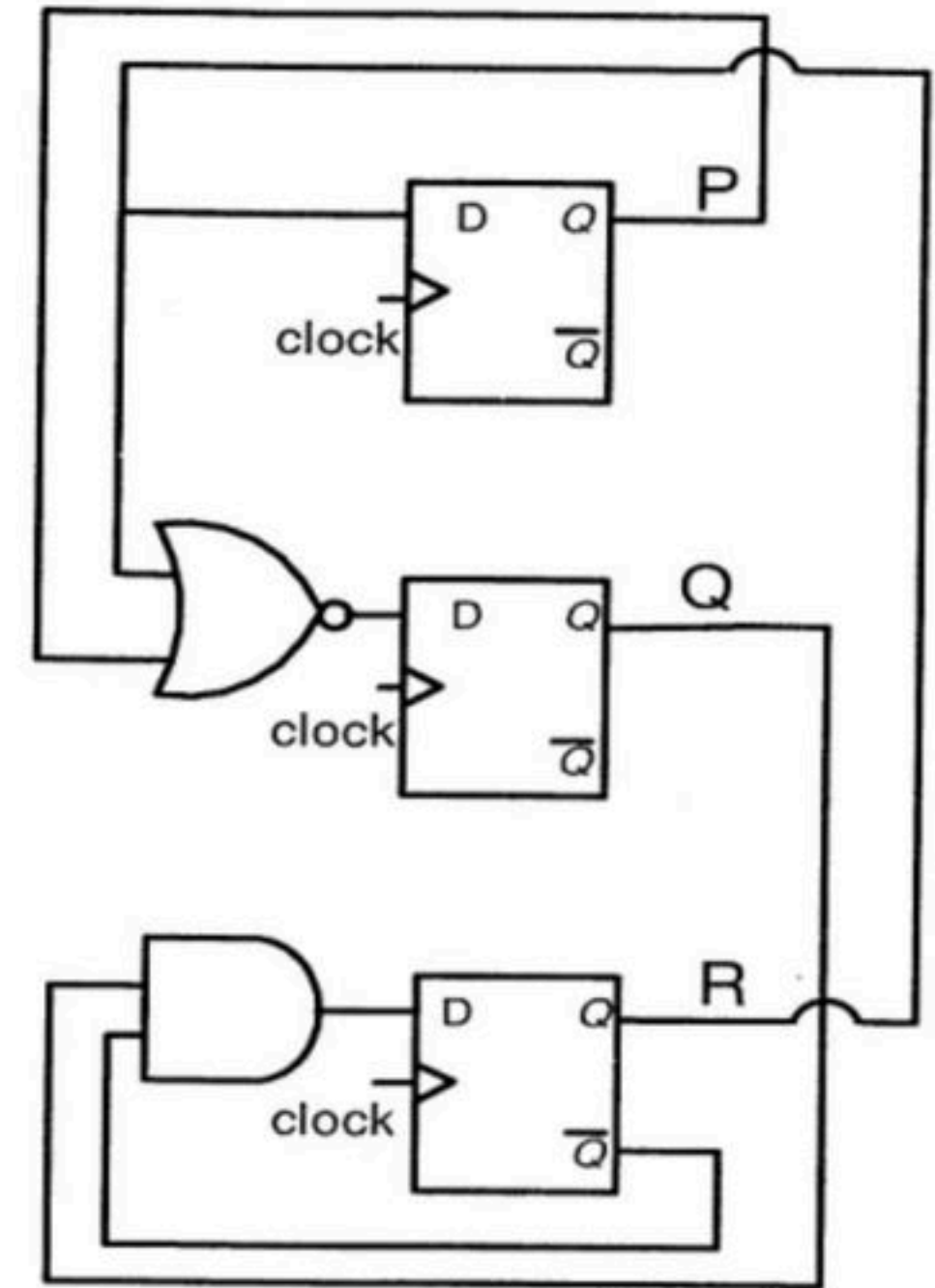
(D) 011

The diagram shows a 3-bit shift register implemented with three D flip-flops. The first flip-flop has its D input connected to the output of the second flip-flop (Q) and its clock input connected to the common clock. Its output is labeled P. The second flip-flop has its D input connected to the output of the third flip-flop ($\overline{Q}$) and its clock input connected to the common clock. Its output is labeled Q. The third flip-flop has its D input connected to the output of the first flip-flop (P) and its clock input connected to the common clock. Its output is labeled R. The outputs P, Q, and R are taken from the Q outputs of the three flip-flops.

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Q Consider the data given in previous question. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter? **(GATE-2011) (2 Marks)**

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6



Q The minimum number of D flip-flops needed to design a mod-258 counter is (GATE-2011) (1 Marks)

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Q In order to build a MOD-18 counter, the minimum number of flip flops needed is equal to:
(NET-DEC-2007)

(A) 18

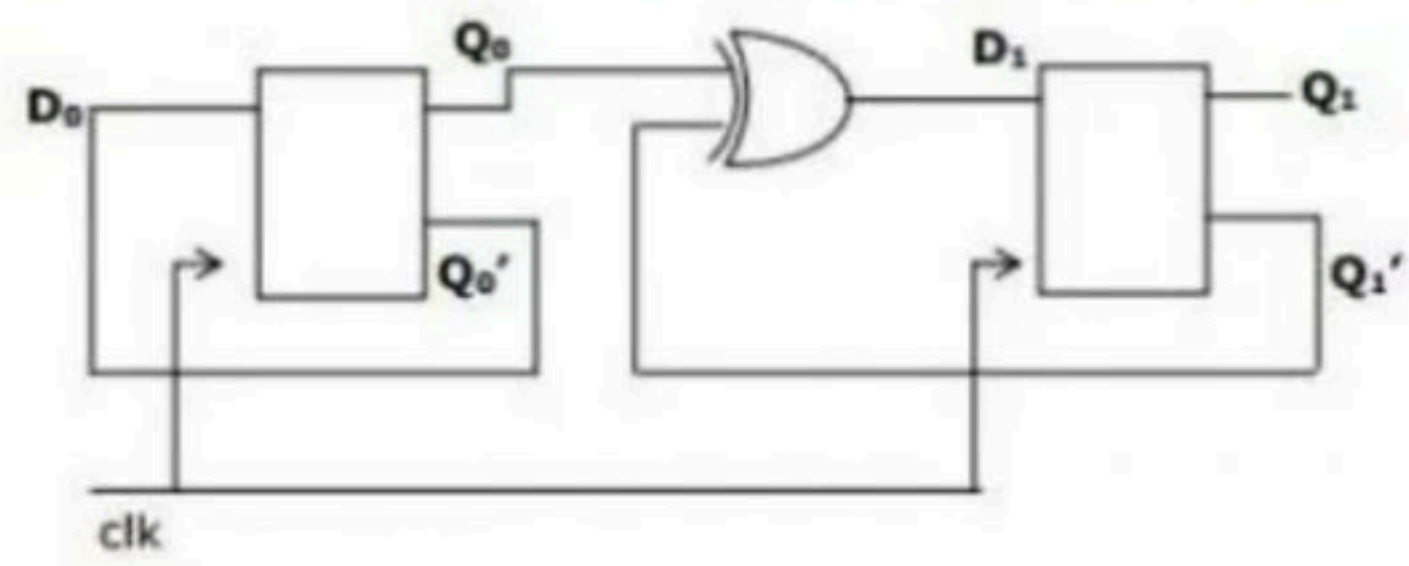
(B) 9

(C) 5

(D) 4

Q Consider the following circuit, The flip-flops are positive edge triggered D FFs. Each state is designated as a two-bit string Q_0Q_1 . Let the initial state be 00. The state transition sequence is: **(GATE - 2005) (2 Marks)**

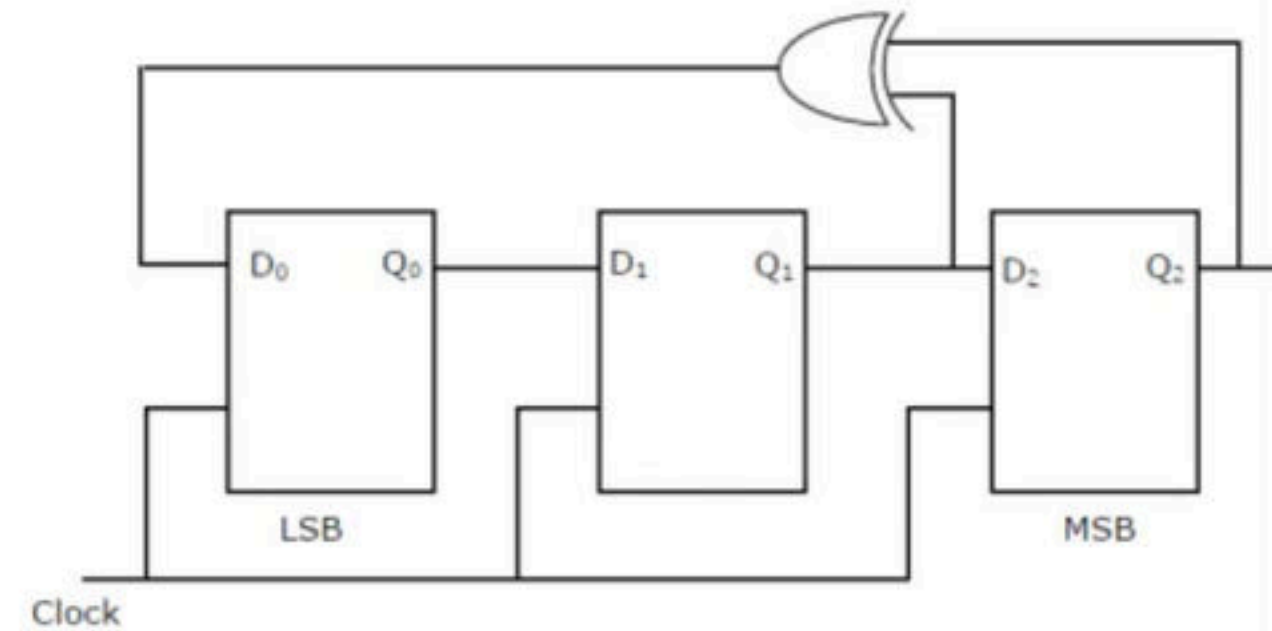
- a)** 00, 11, 01
- b)** 00, 11
- c)** 00, 10, 01, 11
- d)** 00, 11, 01, 10



Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

Q Consider the circuit given below with initial state $Q_0 = 1, Q_1 = Q_2 = 0$. The state of the circuit is given by the value $4Q_2 + 2Q_1 + Q_0$, Which one of the following is the correct state sequence of the circuit? **(GATE-2001) (2 Marks)**

(A) 1,3,4,6,7,5,2 **(B)** 1,2,5,3,7,6,4 **(C)** 1,2,7,3,5,6,4 **(D)** 1,6,5,7,2,3,4



Q Design a synchronous counter for sequence: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$, using T flip flop.

Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

Q Example: Design a synchronous counter for sequence: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$, using D flip flop.

Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

Q Design synchronous counter for sequence: $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 0$, using T flip-flop.

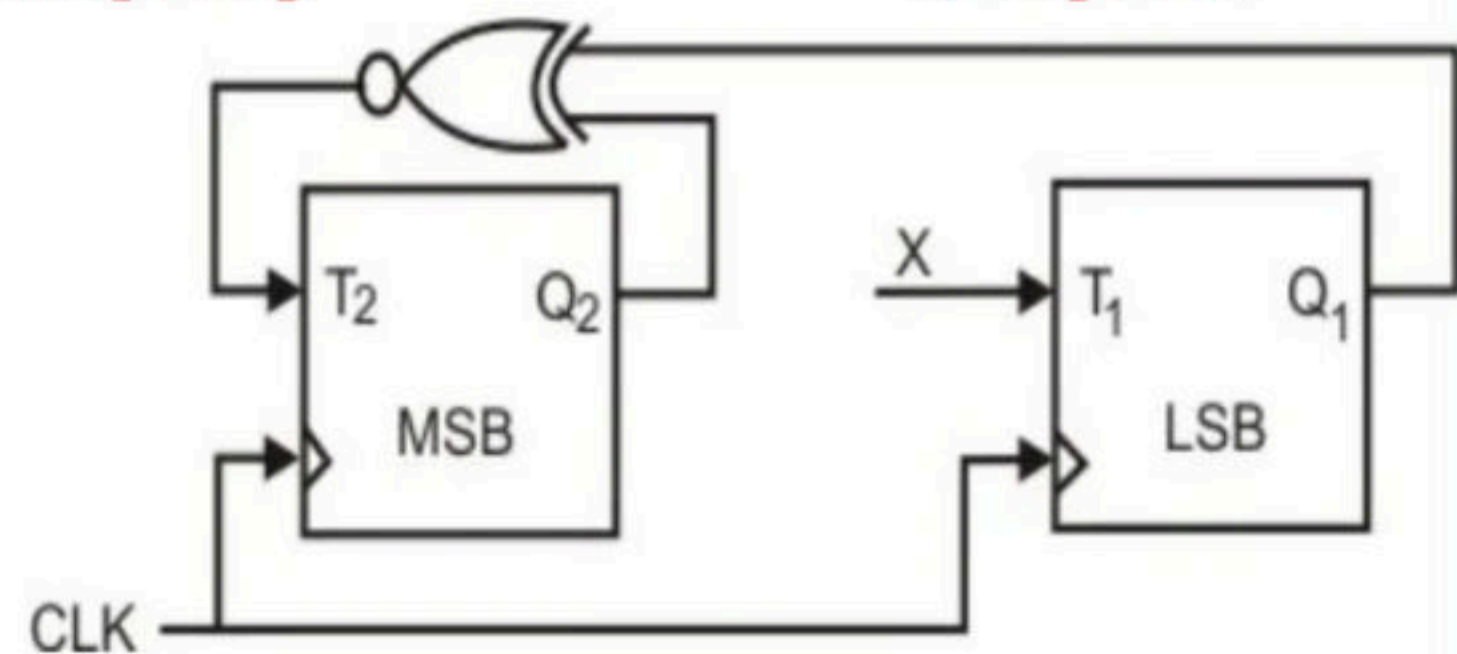
Present State			Next State		
Q_{2p}	Q_{1p}	Q_{0p}	Q_{2N}	Q_{1N}	Q_{0N}
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Q Consider the partial implementation of a 2-bit counter using T flip-flops following the sequence 0-2-3-1-0, as shown below **(GATE-2004) (2 Marks)**

To complete the circuit, the input X should be

(A) Q_2' **(B)** $Q_2 + Q_1$ **(C)** $(Q_1 \oplus Q_2)'$ **(D)** $Q_1 \oplus Q_2$

Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		



Q The next state table of a 2-bit saturating up-counter is given below. **(GATE-2017) (2 Marks)**
 The counter is built as synchronous sequential circuit using T flip-flops. The value for T1 and T0 are

(A) $T_1 = Q_0Q_1$ $T_0 = Q'_0Q'_1$
 (B) $T_1 = Q'_1Q_0$ $T_0 = Q'_1 + Q'_0$
 (C) $T_1 = Q_1 + Q_0$ $T_0 = Q'_1 + Q'_0$
 (D) $T_1 = Q'_1Q_0$ $T_0 = Q_1 + Q_0$

Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

Q A binary 3-bit down counter uses J-K flip-flops, F_i with inputs J_i , K_i and outputs Q_i , $i = 0, 1, 2$ respectively. The minimized expression for the input from following, is **(NET-JAN-2017)**

I. $J_0 = K_0 = 0$

II. $J_0 = K_0 = 1$

III. $J_1 = K_1 = Q_0$

IV. $J_1 = K_1 = Q_0'$

$$V. J_2 = K_2 = Q_1 Q_0$$

VI. $J_2 = K_2 = Q_1' Q_0'$

(a) I, III, V

(b) I, IV, VI

(c) II, III, V

(d) II, IV, VI

Present State			Next State		
Q_{2p}	Q_{1p}	Q_{0p}	Q_{2N}	Q_{1N}	Q_{0N}
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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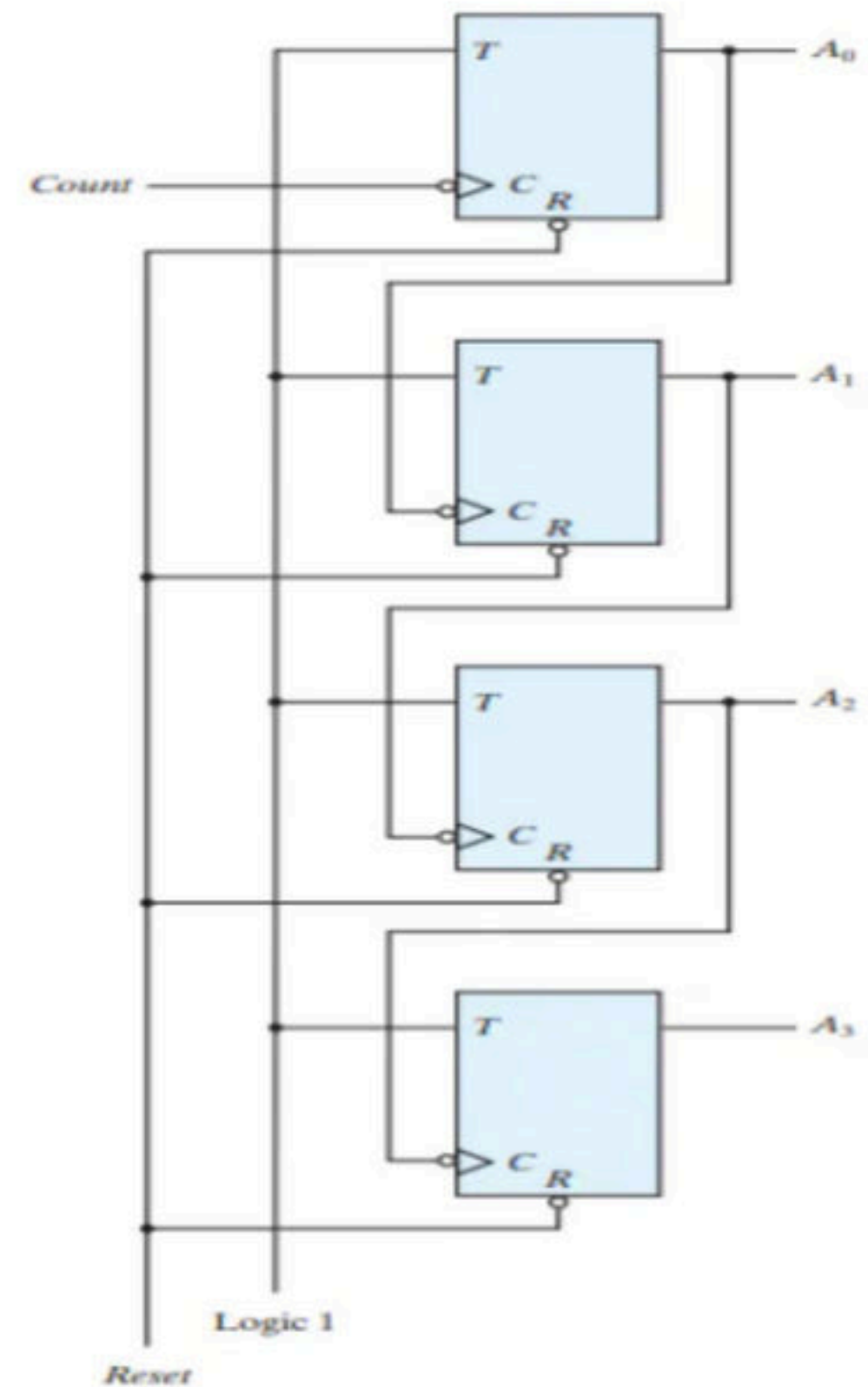
Asynchronous counters

SYNCHRONOUS COUNTER	ASYNCHRONOUS COUNTER
In synchronous counter, all flip flops are triggered with same clock simultaneously.	In asynchronous counter, different flip flops are triggered with different clock, not simultaneously.
Synchronous Counter is faster than asynchronous counter in operation. $T_{\text{delay}} = T_{\text{FF}} + T_{\text{CC}}$	Asynchronous Counter is slower than synchronous counter in operation. $T_{\text{delay}} = n \times T_{\text{FF}} + T_{\text{CC}}$
Synchronous Counter is also called Parallel Counter.	Asynchronous Counter is also called Serial Counter.
Synchronous Counter designing as well implementation are complex due to increasing the number of states.	Asynchronous Counter designing as well as implementation is very easy.
Synchronous Counter will operate in any desired count sequence.	Asynchronous Counter will operate only in fixed count sequence (UP/DOWN).
Synchronous Counter examples are: Ring counter, Johnson counter.	Asynchronous Counter examples are: Ripple UP counter, Ripple DOWN counter.
A synchronous sequential circuit is a system whose behavior can be defined from the knowledge of its signals at discrete instants of time.	The behavior of an asynchronous sequential circuit depends upon the input signals at any instant of time and the order in which the inputs change.

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RIPPLE COUNTERS

1. A binary ripple counter consists of a series connection of complementing flip-flops, with the output of each flip-flop connected to the input of the next higher order flip-flop.
2. The flip-flop holding the least significant bit receives the incoming count pulses.



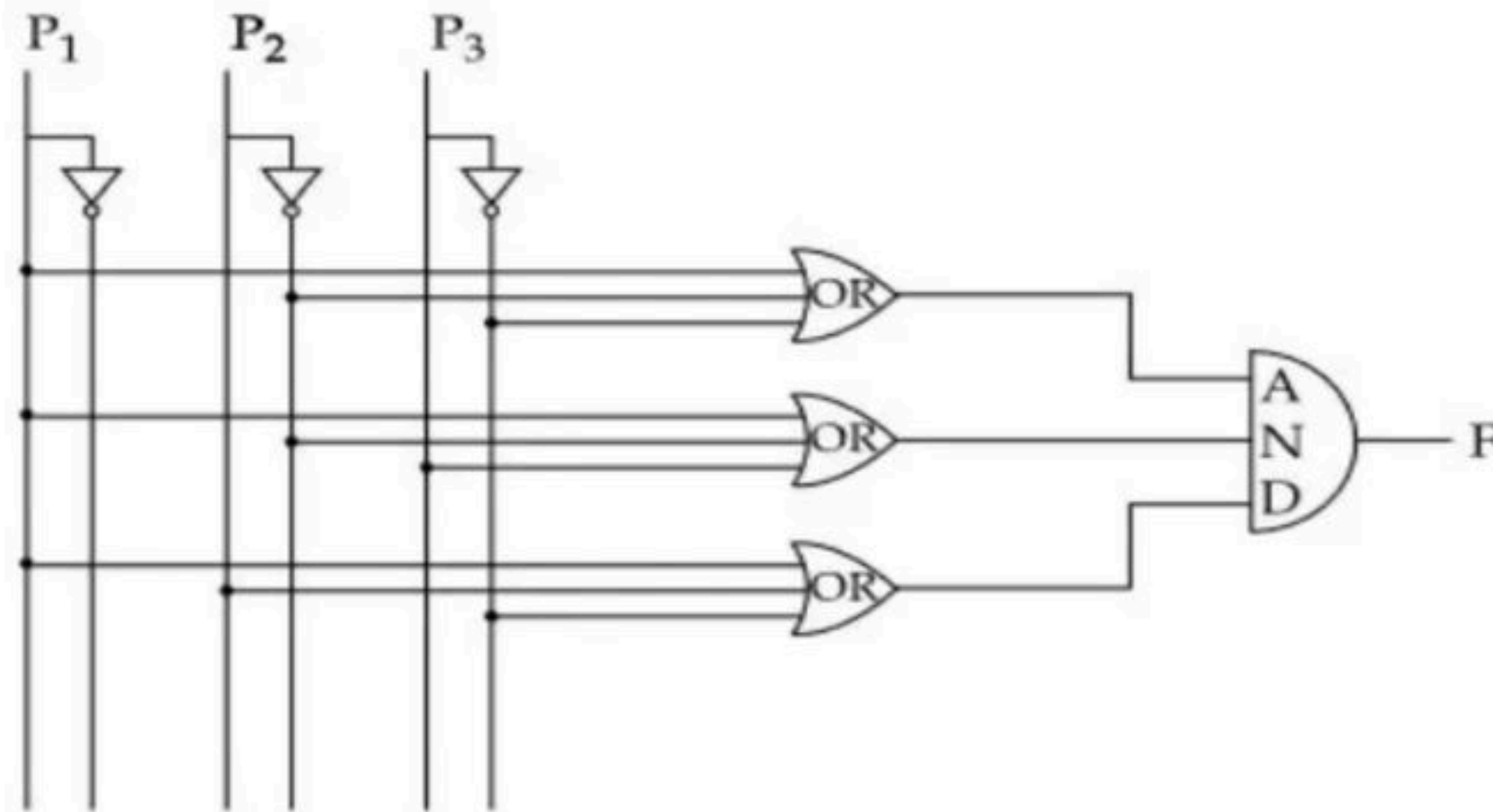
Q What does the following logic diagram represent? **(NET-JULY-2018)**

(a) Synchronous Counter

(c) Combinational Circuit

(b) Ripple Counter

(d) Mod 2 Counter



Q In a positive-edge-triggered JK flip-flop, if J and K both are high then the output will be _____ on the rising edge of the clock. **(NET-JULY-2016)**

(a) No change

(b) Set

(c) Reset

(d) Toggle

Q Which of the following is a sequential circuit? **(NET-JULY-2016)**

(a) Multiplexer

(b) Decoder

(c) Counter

(d) Full adder

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Q A ripple counter is a (n) : **(NET-AUG-2016)**

(a) Synchronous Counter

(c) Parallel counter

(b) Asynchronous counter

(d) None of the above

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Q A binary ripple counter is required to count up to 16383. How many flip-flops are required?
(NET-SEPT-2013)

(A) 16382

(B) 8191

(C) 512

(D) 14

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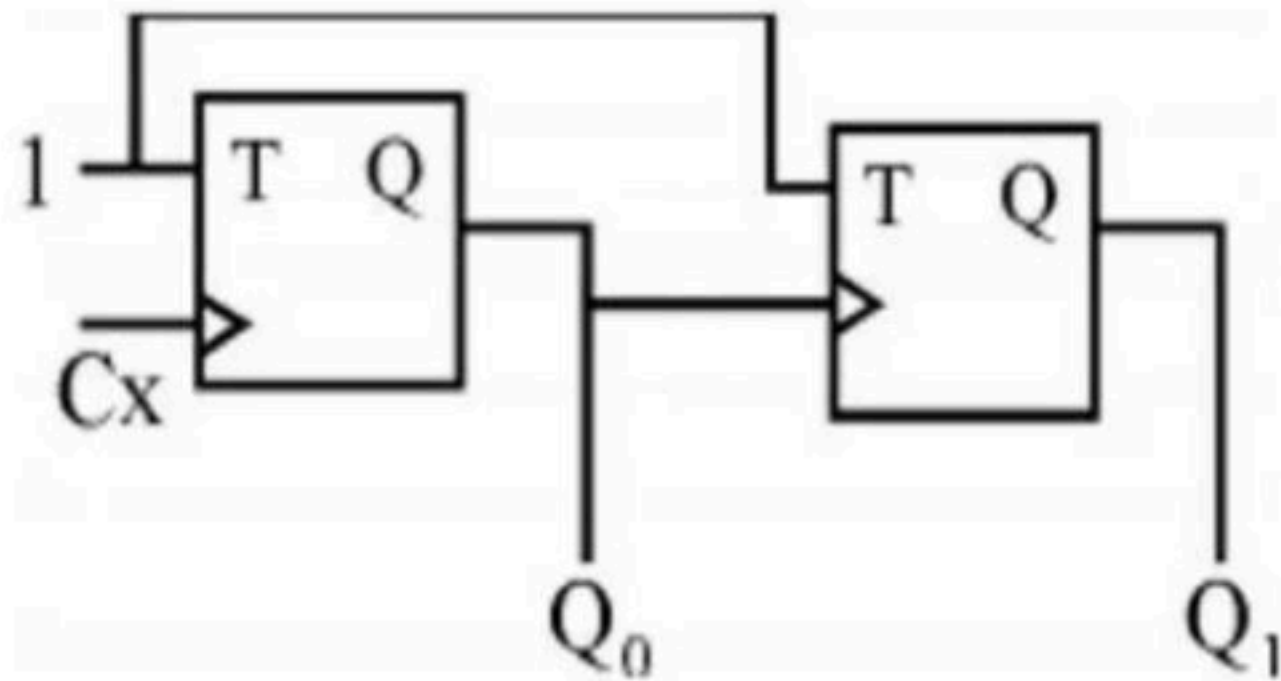
Q In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0 ? **(GATE-2010) (2 Marks)**

(A) 11, 10, 01, 00

(B) 10, 11, 01, 00

(C) 10, 00, 01, 11

(D) 11, 10, 00, 01



Present State		Next State	
Q_{1p}	Q_{0p}	Q_{1N}	Q_{0N}
0	0		
0	1		
1	0		
1	1		

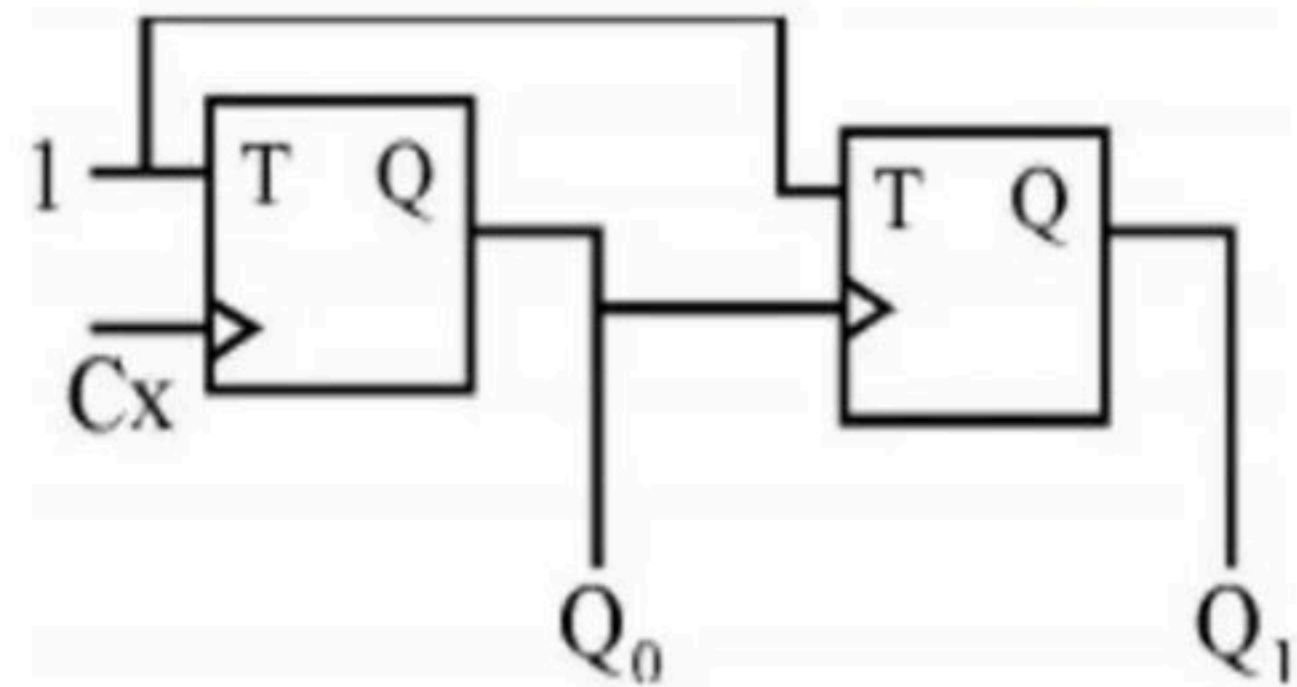
Q What are the final values of Q_1 and Q_0 after 4 clock cycles, if initial values are 00 in the sequential circuit shown below: **(NET-DEC-2013)**

(A) 11

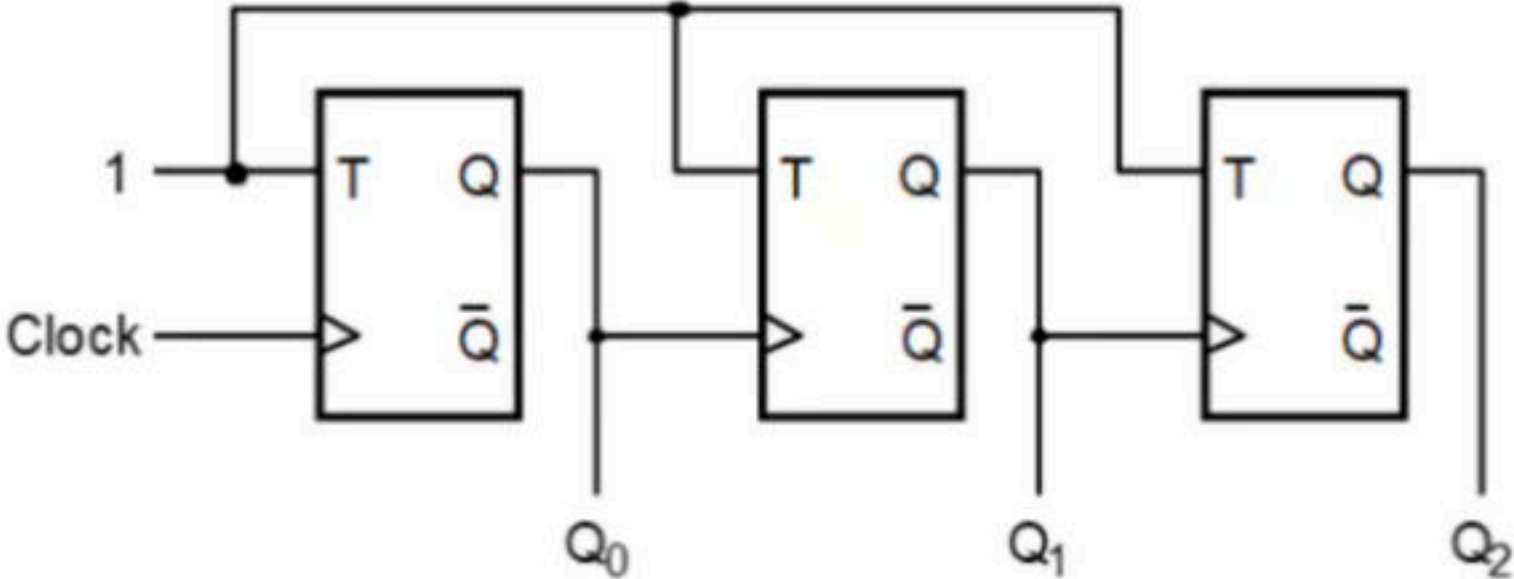
(B) 10

(C) 01

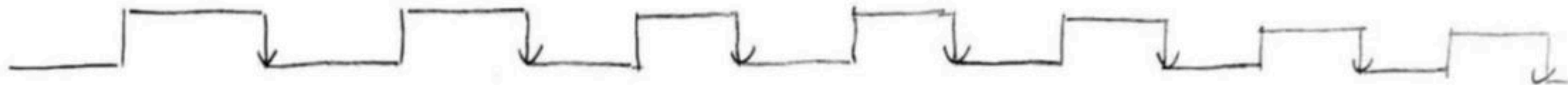
(D) 00



Q What is the counting behavior of the giving counter



Present State			Next State		
Q_{2p}	Q_{1p}	Q_{0p}	Q_{2N}	Q_{1N}	Q_{0N}
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			



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Nature of Clock	Nature of Feedback	Nature of counting
+ve	Q_n'	UP Counting
+ve	Q_n	Down Counting
-ve	Q_n	UP Counting
-ve	Q_n'	Down Counting

- **Self-Starting Counter**: - A counter is said to be self-starting if it provides the counting sequence irrespective of initial state.
- **Free running counter**: - a counter is said to be free running If it maintains all possible states in the counting sequence.
- $0 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$
- $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 1$
- $0 \rightarrow 3 \rightarrow 0, 1 \rightarrow 2 \rightarrow 3$
- $0 \rightarrow 1 \rightarrow 2 \rightarrow 0, 3 \rightarrow 3$
- $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 2, 7 \rightarrow 6$
- $1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 2 \rightarrow 1, 7 \rightarrow 6$
- **Conclusion**: - if a counter is free running, it is also self-starting, but vice-versa not required to be true.

$0 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ (SS, FR)

$0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 1$ (SS)

$0 \rightarrow 3 \rightarrow 0, 1 \rightarrow 2 \rightarrow 3$ (SS)

$0 \rightarrow 1 \rightarrow 2 \rightarrow 0, 3 \rightarrow 3$

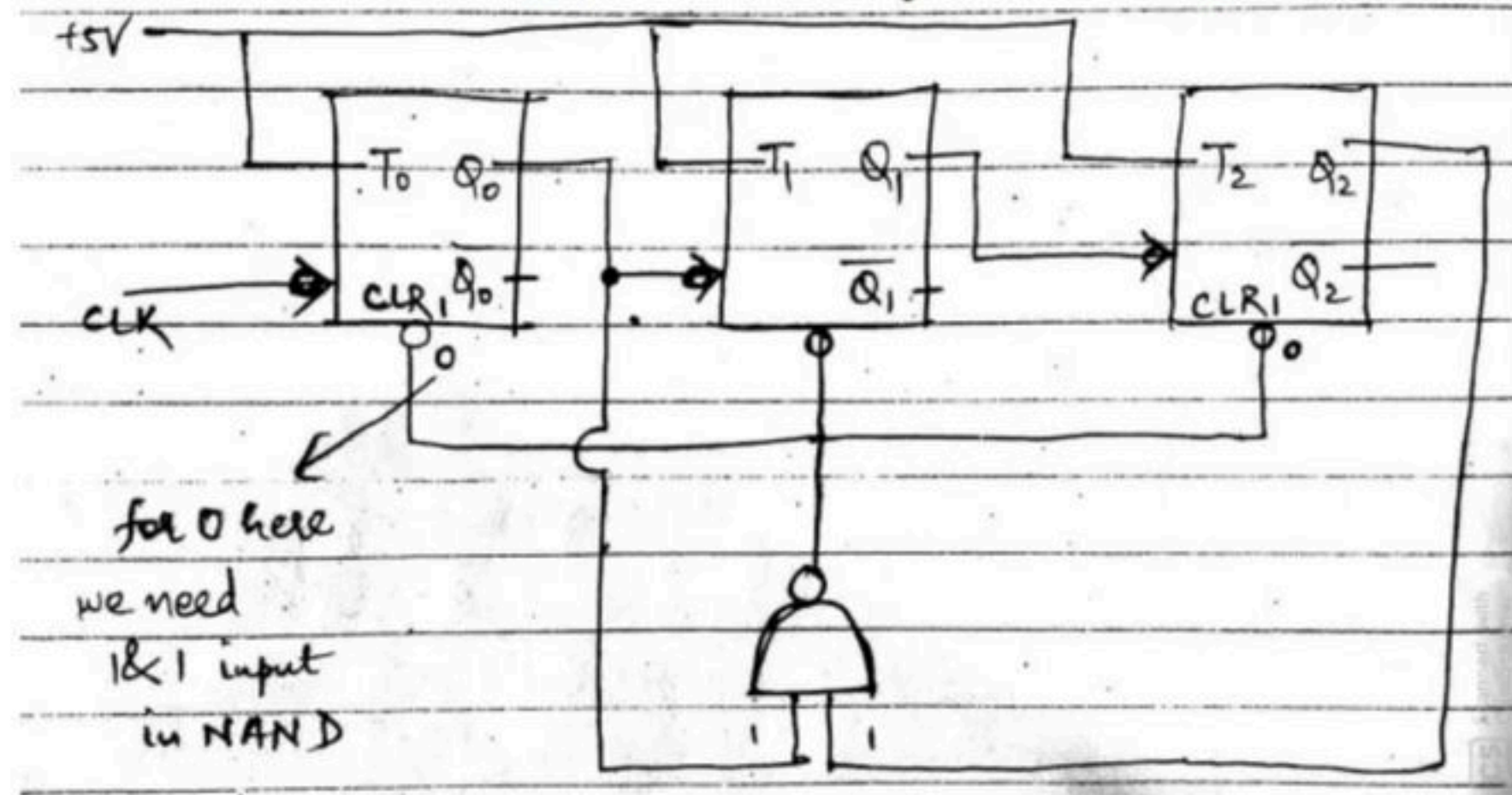
$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 2, 7 \rightarrow 6$ (SS)

$1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 2 \rightarrow 1, 7 \rightarrow 6$ (SS)

Restricted mod counter

Q Consider a restricted mod counter and find what sequence it is counting?

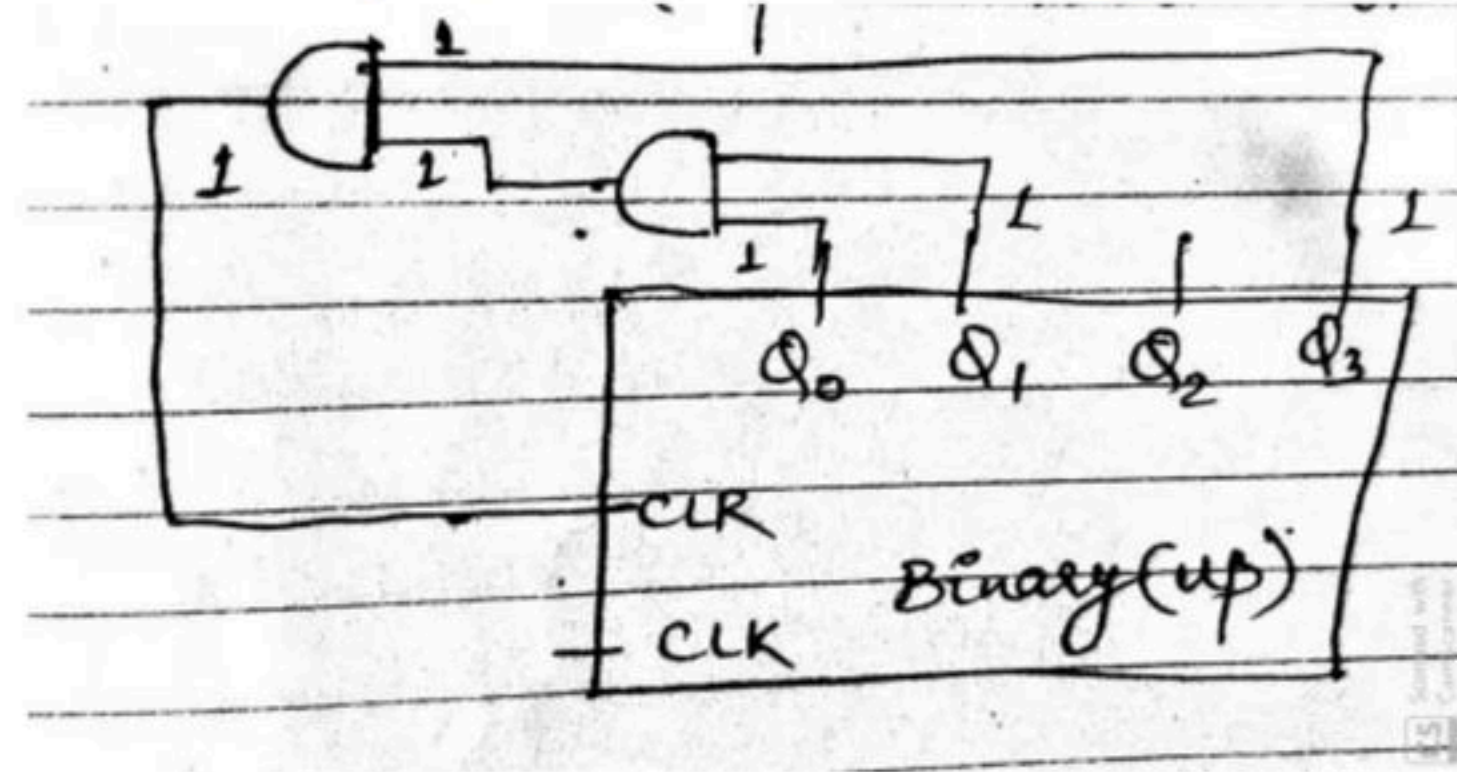
Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1



Restricted mod counter

Q Consider a restricted mod counter and find what sequence it is counting?

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1



Registers

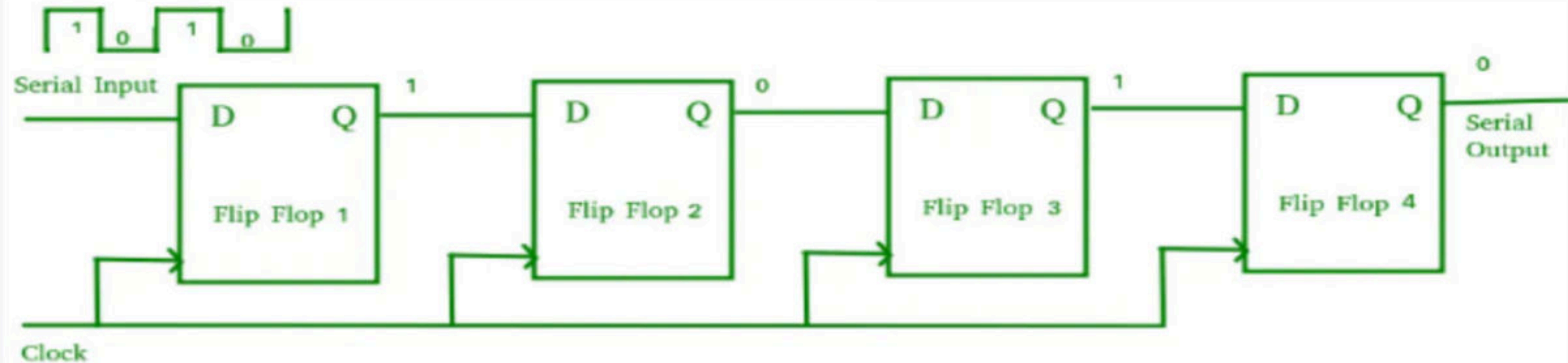
1. Registers are basically storing devices which are also designed using basic element called flip-flop.
2. D-flip-flop are most popular choice for register because they don't perform any functionality and output is simply based on current input so, they act as a buffer.
3. Apart from storing registers sometimes also be used in performing basic mathematical operation like multiply by 2 by left shift and divide by 2 by right shift.

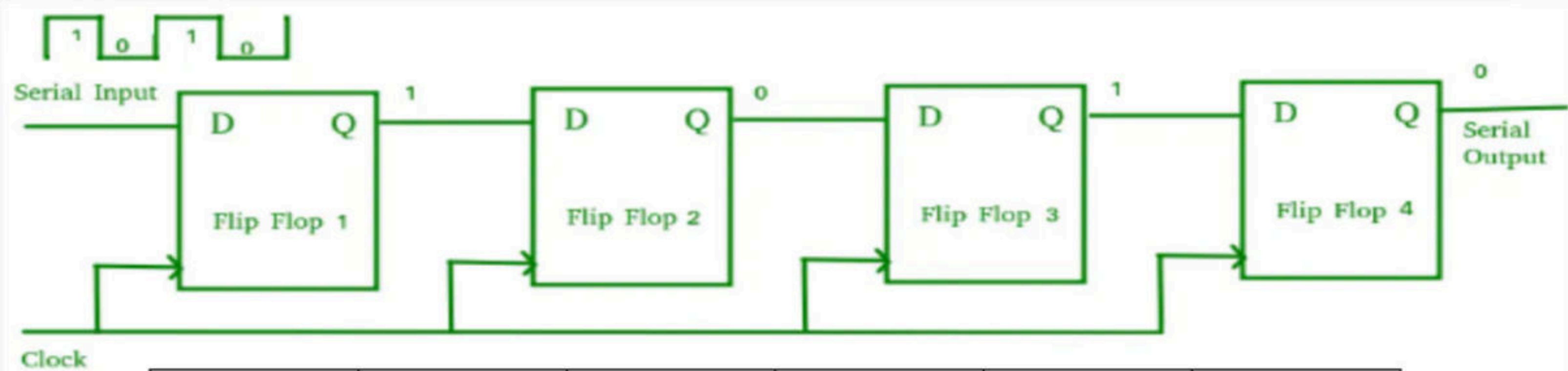
Shift Register Types & Operations

- There are four different modes in which registers operate.
 1. Serial In-Serial Out (SISO)
 2. Serial In-Parallel Out (SIPO)
 3. Parallel In-Serial Out (PISO)
 4. Parallel In- Parallel Out (PIPO)

Serial In-Serial Out (SISO)

- The shift register, which allows serial input (one bit after the other through a single data line) and produces a serial output is known as Serial-In Serial-Out shift register.
- Since there is only one output, the data leaves the shift register one bit at a time in a serial pattern, thus the name Serial-In Serial-Out Shift Register.





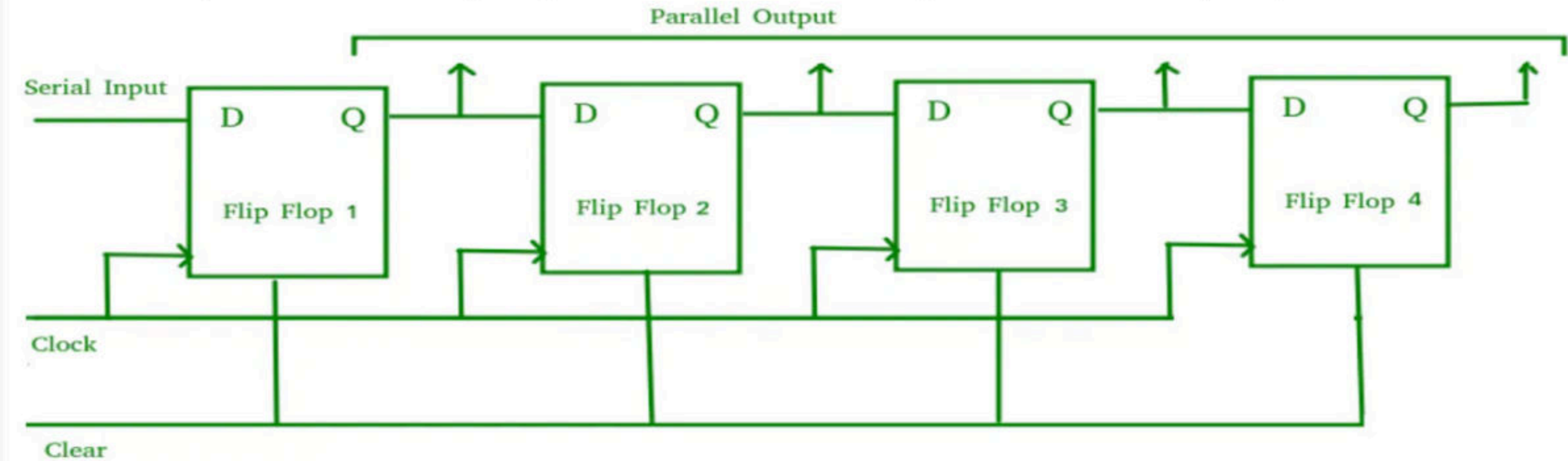
Sequence	Input	Q_2	Q_1	Q_0	Output
		Q	Q	0	0
1	1	1	Q	Q	Q
2	1	1	1	Q	Q
3	0	0	1	1	1
4	Q	Q	0	1	1
5	Q	Q	Q	0	0

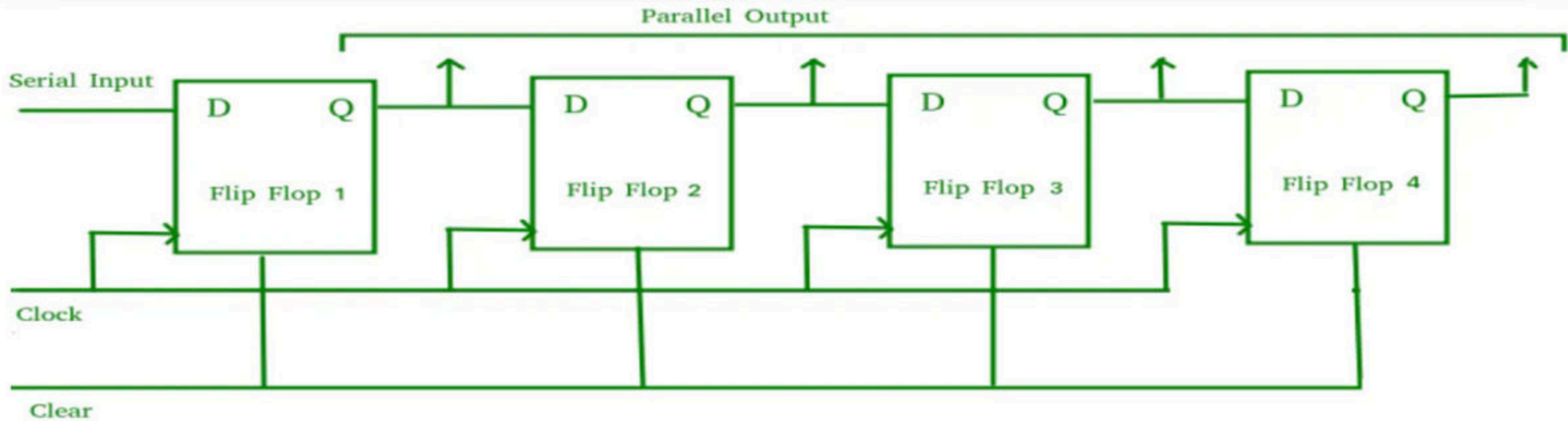
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- The main use of a SISO is to act as a delay element.
- In SISO registers to provide n bit data serially in it requires n clock pulse and to provide serial output it requires $n - 1$ clock pulses.

Serial-In Parallel-Out shift Register (SIPO)

1. The shift register, which allows serial input (one bit after the other through a single data line) and produces a parallel output is known as Serial-In Parallel-Out shift register.
2. The circuit consists of four D flip-flops which are connected.
3. The output of the first flip flop is connected to the input of the next flip flop and so on.





- They are used in communication lines where demultiplexing of a data line into several parallel lines is required because the main use of the SIPO register is to convert serial data into parallel data.

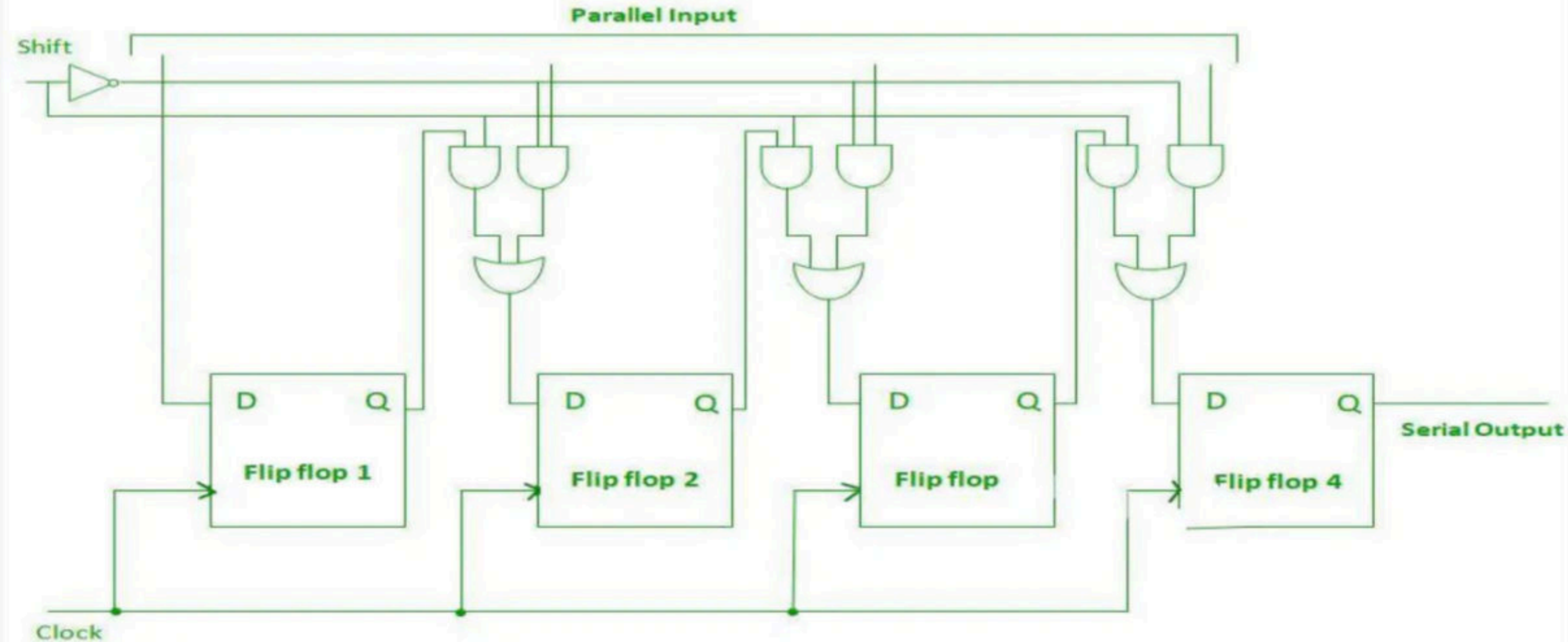
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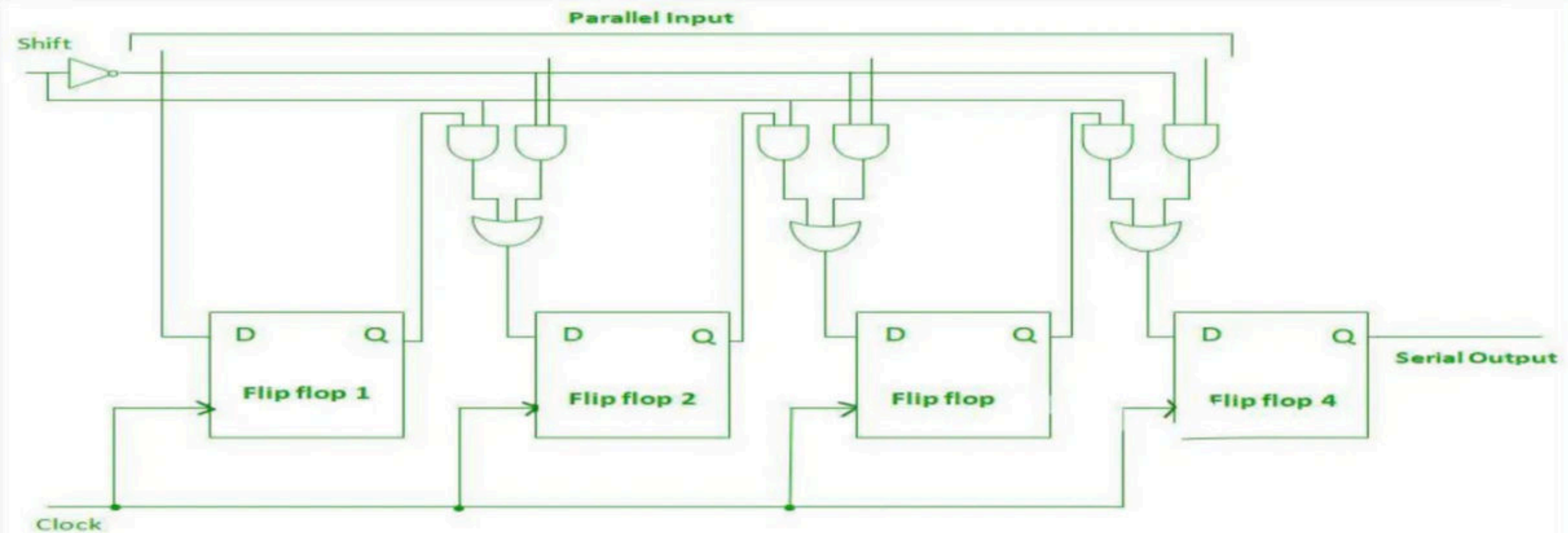
Sequence	I/p	Q ₂	Q ₁	Q ₀	O/p ₁	O/p ₂	O/p ₃
		Q	Q	Q	Q	Q	Q
1	1	1	Q	Q	1	Q	Q
2	1	1	1	Q	1	1	Q
3	0	0	1	1	0	1	1

- In SIPO registers to provide n bit data serially in it requires n clock pulse and to provide parallel output it requires 0 clock pulses.

Parallel-In Serial-Out Shift Register (PISO)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and produces a serial output is known as Parallel-In Serial-Out shift register.





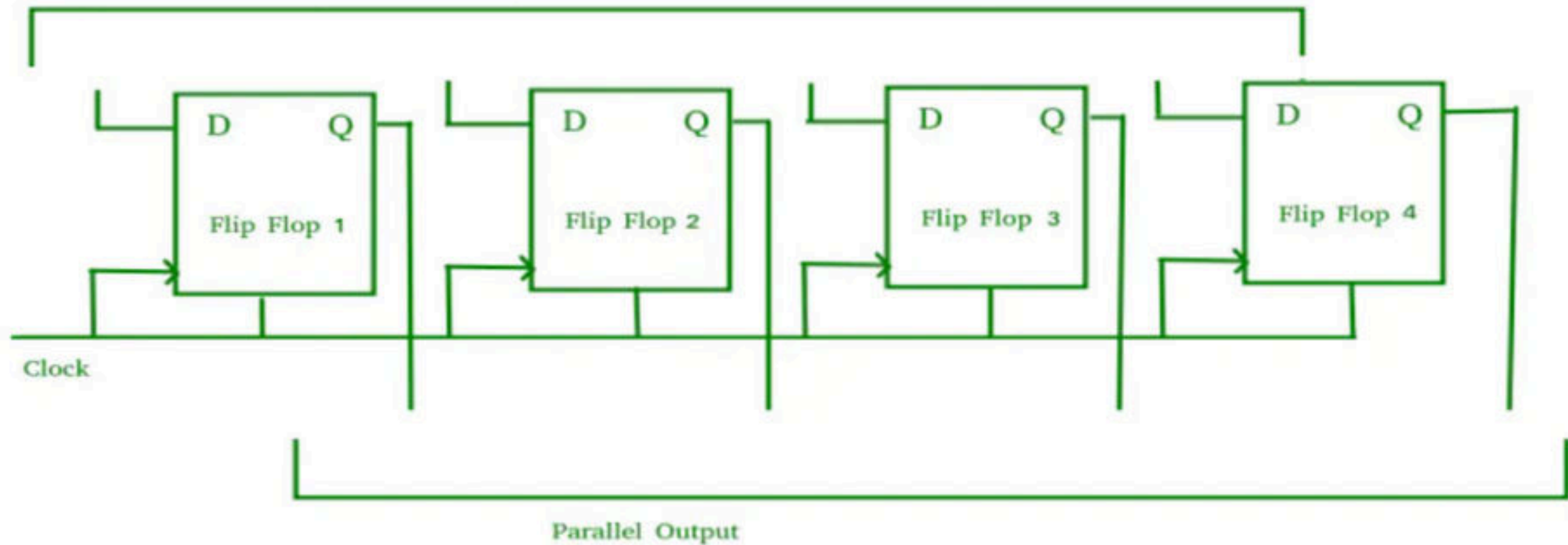
- The circuit consists of four D flip-flops which are connected.
- The clock input is directly connected to all the flip flops but the input data is connected individually to each flip flop through a multiplexer at the input of every flip flop.
- The output of the previous flip flop and parallel data input are connected to the input of the MUX and the output of MUX is connected to the next flip flop.
- A Parallel in Serial out (PISO) shift register is used to convert parallel data to serial data.

- In PISO register to provide parallel input it requires 1 clock pulse and to provide serial output (n-1) clock.

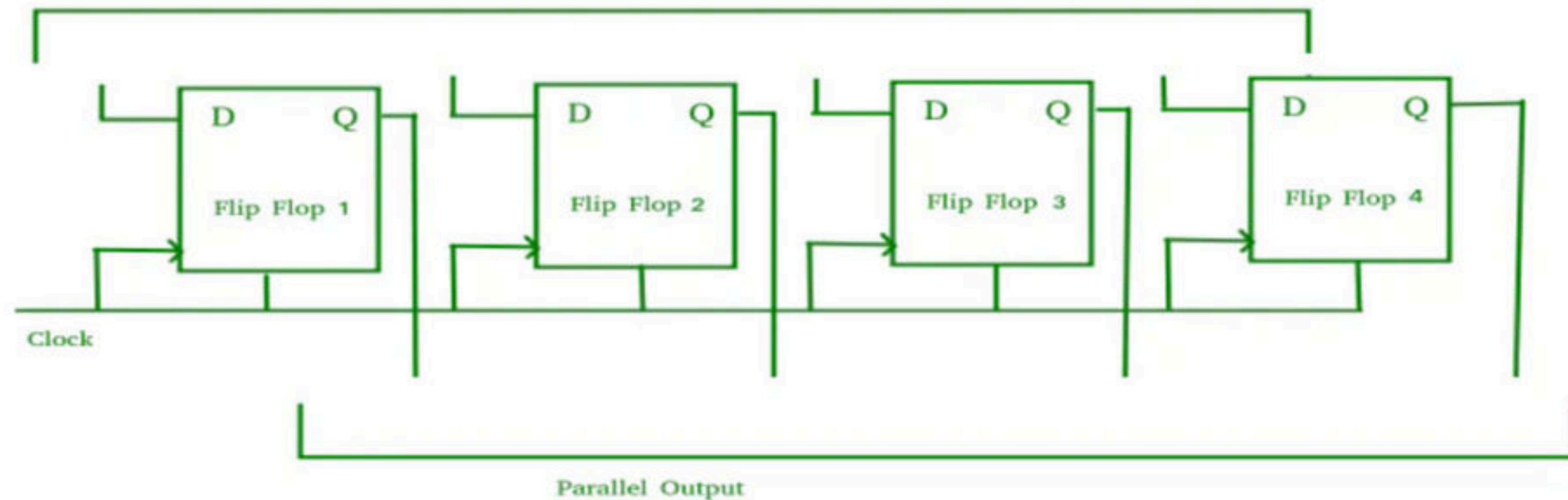
Sequence	Input	Q_2	Q_1	Q_0	Output
1	1	1	Q	Q	
2	1	1	1	Q	
3	0	0	1	1	
4	Q	Q	0	1	
5	Q	Q	Q	0	

Parallel-In Parallel-Out Shift Register (PIPO)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and also produces a parallel output is known as Parallel-In parallel-Out shift register.



- The circuit consists of four D flip-flops which are connected.
- In this type of register, there are no interconnections between the individual flip-flops since no serial shifting of the data is required.
- Data is given as input separately for each flip flop and in the same way, output also collected individually from each flip flop.
- A Parallel in Parallel out (PIPO) shift register is used as a temporary storage device and like SISO Shift register it acts as a delay element.
- For parallel input it requires 1 clock pulse and for parallel output it requires 0 clock.



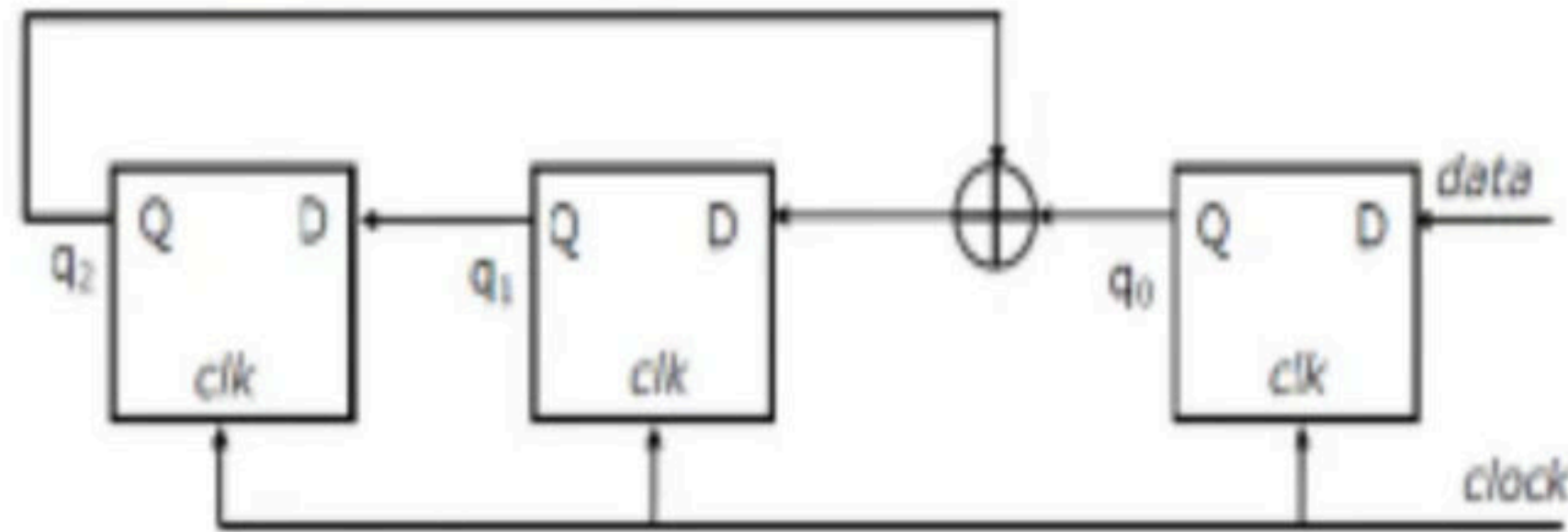
	No of clock (write)	No of clock (Read)	total
SISO	n	$n-1$	$2n-1$
SIPO	n	0	n
PISO	1	$n-1$	n
PIPO	1	0	1

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Q Consider a 32-bit shift register which uses a clock of 1 GHz. If register is operated in SISO mode, find total time required to perform loading and reading?

- a)** 1 ns **b)** 31 ns **c)** 32 ns **d)** 63 ns

Q Consider the circuit in the diagram. The $\oplus$ operator represents Ex-OR. The D flipflops are initialized to zeroes (cleared). **(GATE-2006) (2 Marks)**



The following data: 100110000 is supplied to the “data” terminal in nine clock cycles. After that the values of $q_2q_1q_0$ are:

(A) 000

(B) 001

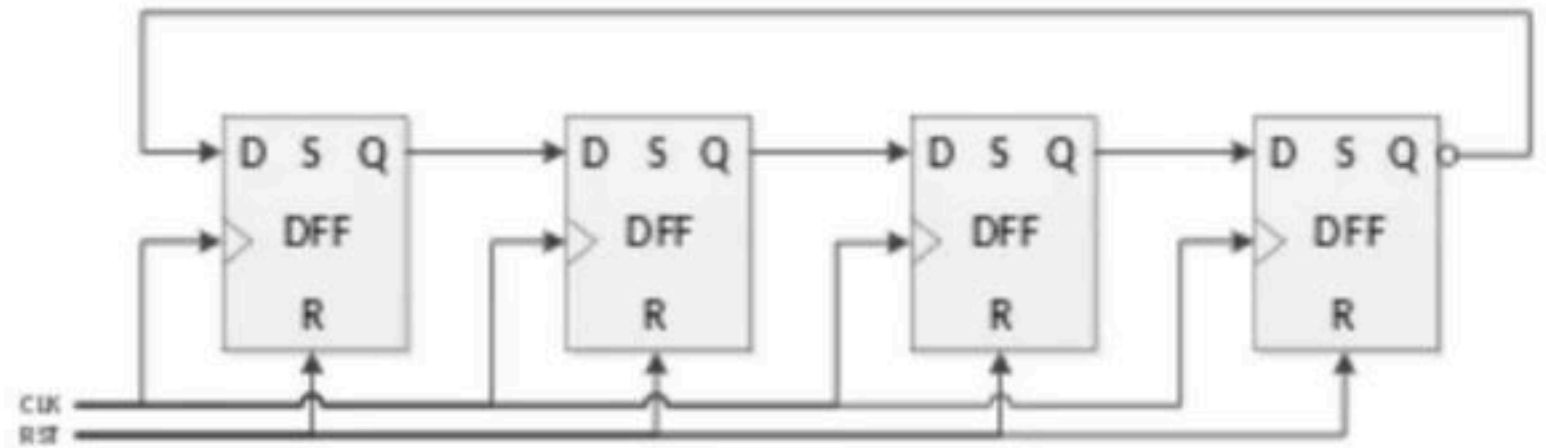
(C) 010

(D) 101

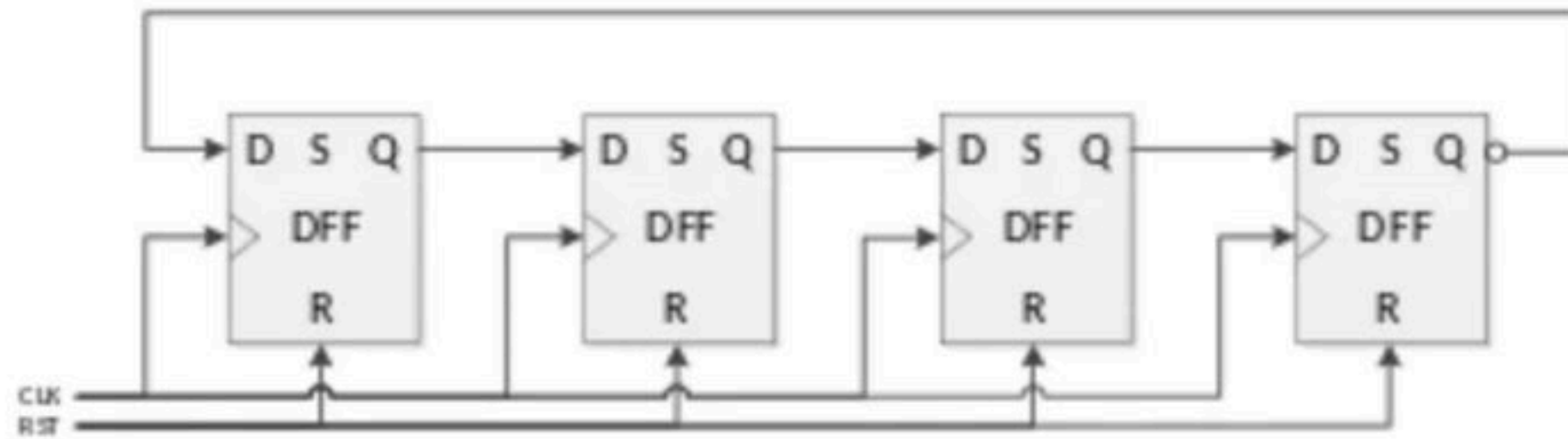
Ring Counter/ Straight Ring Counter

4-Bit Ring Counter

- A ring counter is a circular shift register with only one flip-flop being set at any particular time; all others are cleared.
- The single bit is shifted from one flip-flop to the next to produce the sequence of timing signals.
- Output of the last flip-flop is connected to the input of the first flip-flop in case of ring counter.
- ***No. of states in Ring counter = No. of flip-flop used***



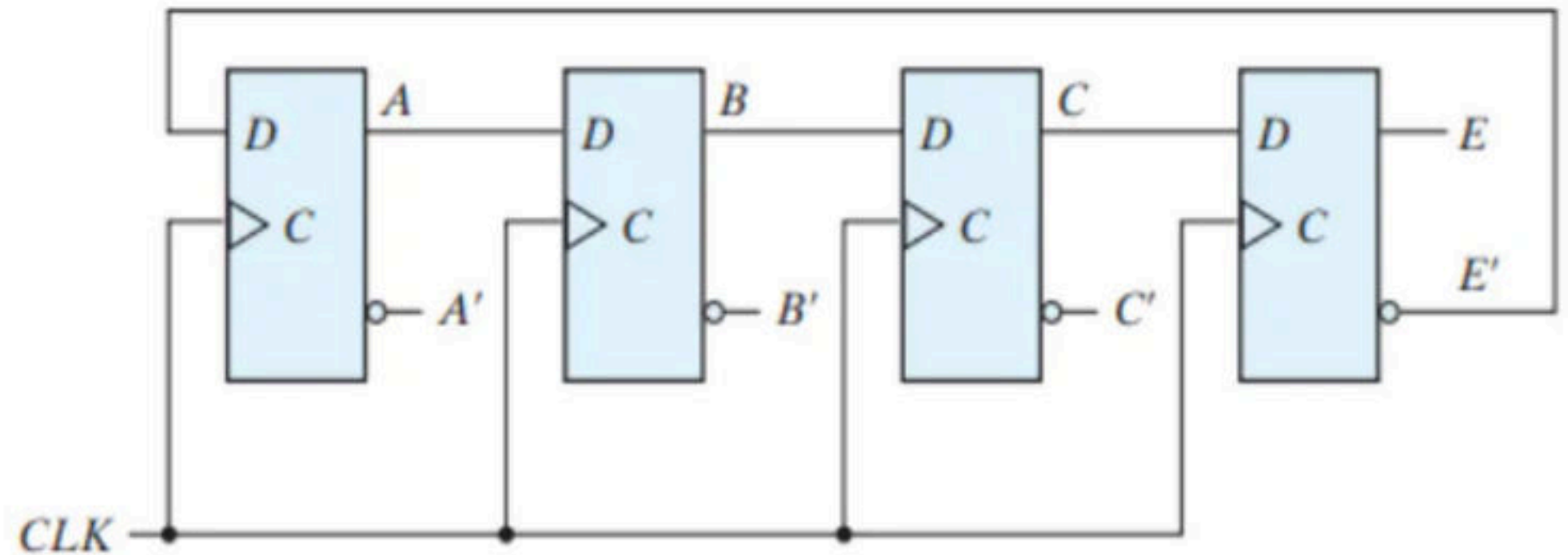
- Number of unused states in Ring Counter is $2^n - n$

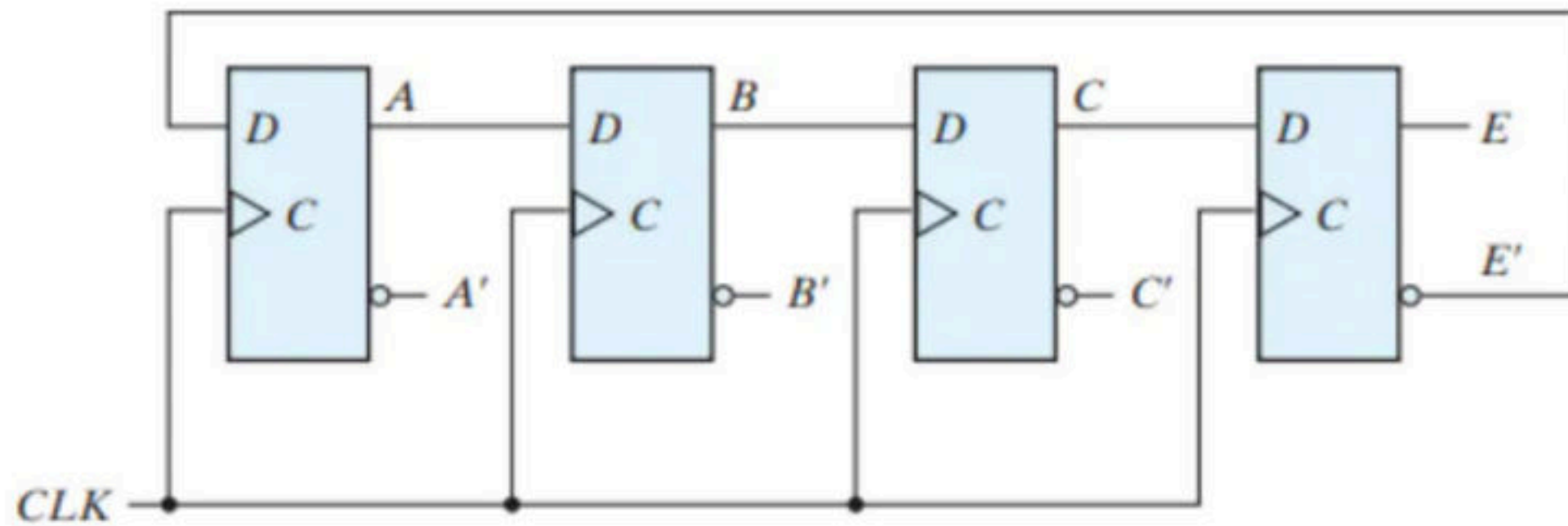


CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
0	1	0	0	0

Johnson Counter/ twisted ring counter/ switch-tail ring counter/ walking ring counter

- A k -bit ring counter circulates a single bit among the flip-flops to provide k distinguishable states.
- The number of states can be doubled if the shift register is connected as a switch-tail ring counter. A switch-tail ring counter is a circular shift register with the complemented output of the last flip-flop connected to the input of the first flip-flop.





CLK	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
0	0	0	0	0

Q Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is **(GATE-2015) (2 Marks)**

(A) 0, 1, 3, 7, 15, 14, 12, 8, 0

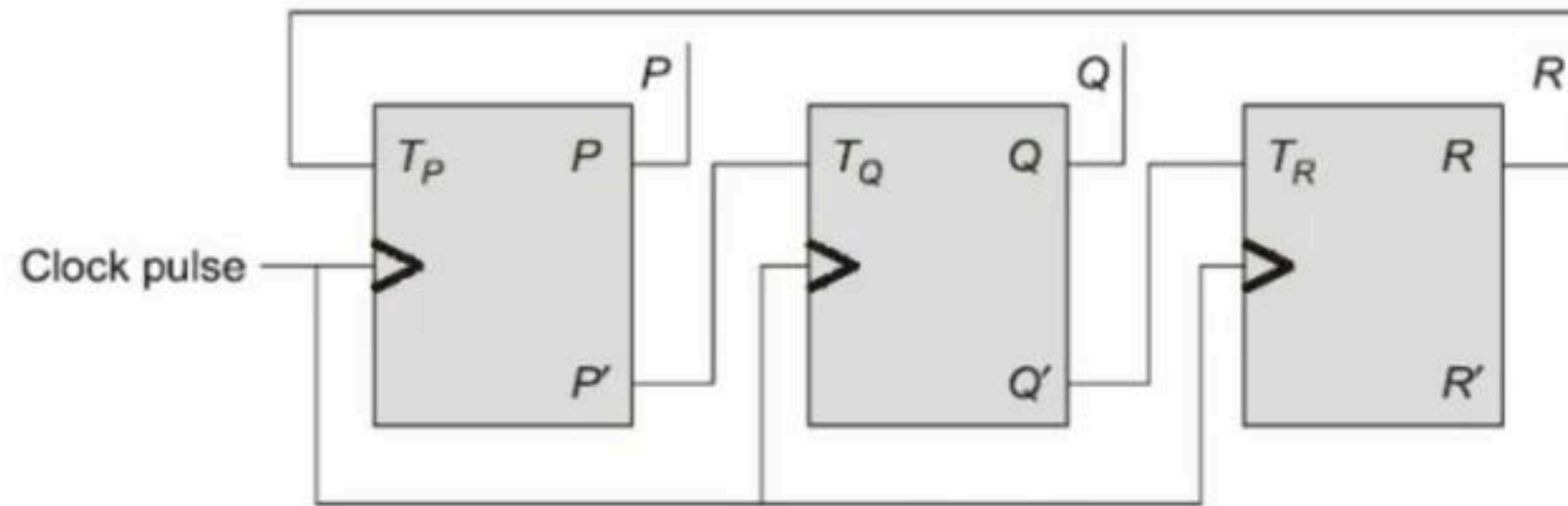
(B) 0, 1, 3, 5, 7, 9, 11, 13, 15, 0

(C) 0, 2, 4, 6, 8, 10, 12, 14, 0

(D) 0, 8, 12, 14, 15, 7, 3, 1, 0

(GATE-2021)

Q.45 Consider a 3-bit counter, designed using T flip-flops, as shown below:



Assuming the initial state of the counter given by PQR as 000. What are the next three states?

(a) 011, 101, 111

(b) 001, 010, 000

(c) 001, 010, 111

(d) 011, 101, 000

Ans. (d)